



ELEC 5690 – Advanced Topics in Artificial Intelligence for Medical Image Analysis

Lecture 06: Medical Images (Multi-modal data and Dermoscopy)

Oct 6, 2026



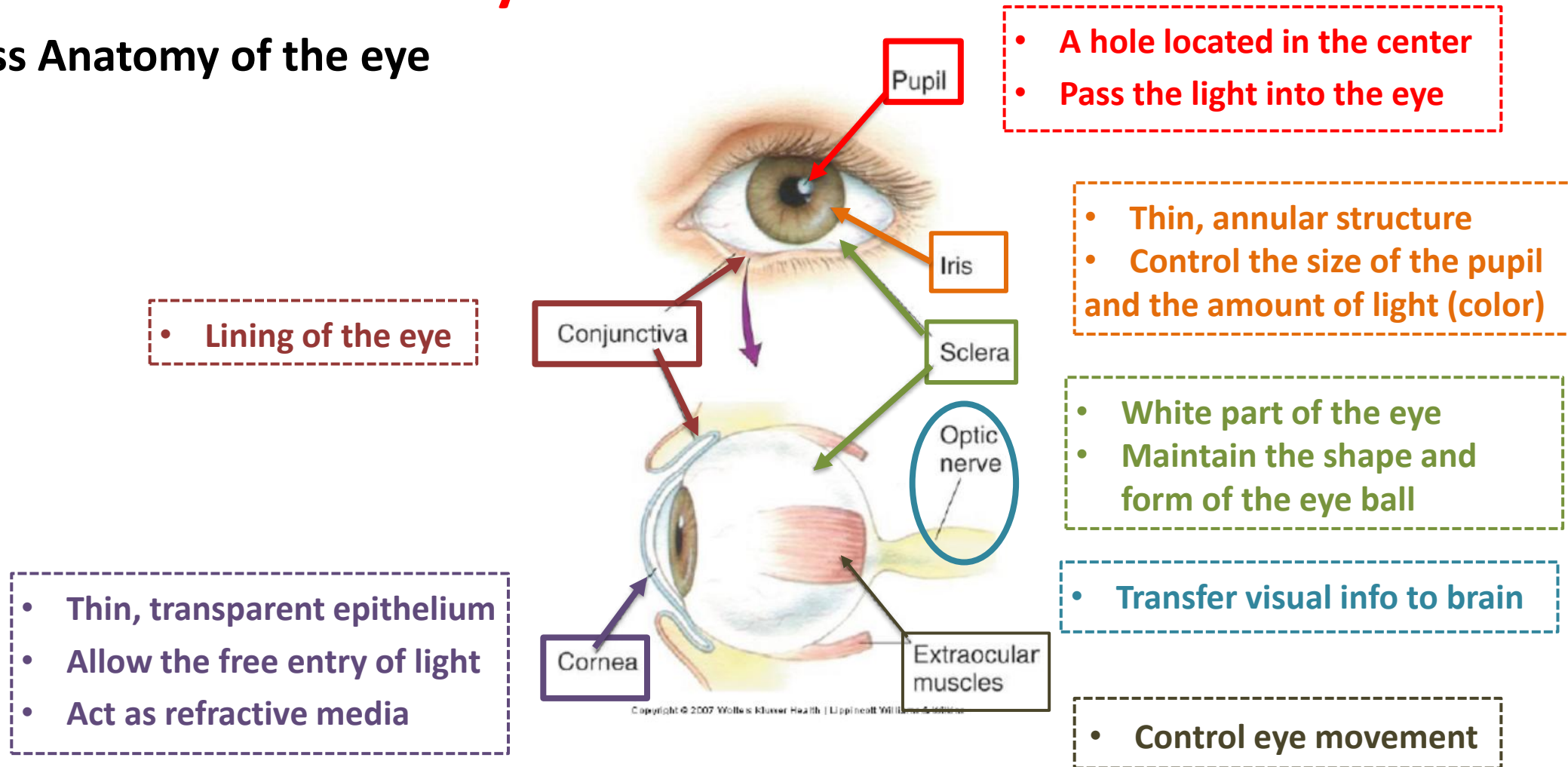
- Last time
- Multimodal data
- Introduction to dermoscopy
- Introduction to ultrasound
- Deep learning applications

1. Last time: introduction to eye physiology



The structure of the eye

Gross Anatomy of the eye



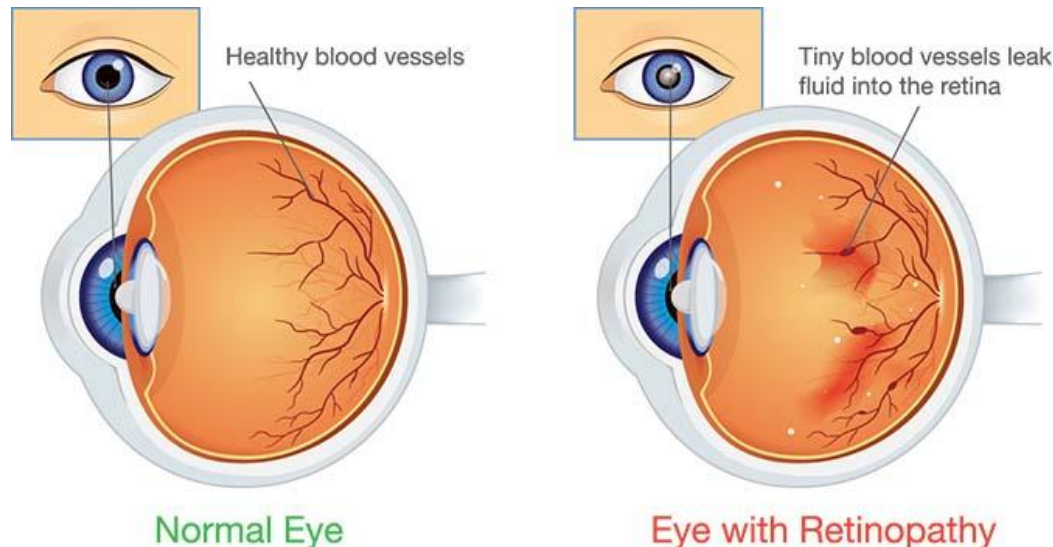
1. Last time: introduction to eye physiology



Common eye diseases

Diabetic retinopathy

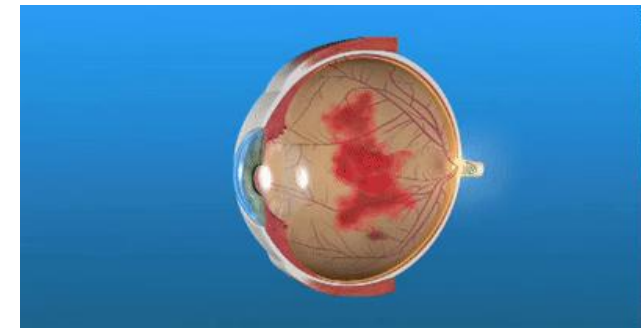
- **diabetes complication** that affects eyes
- high blood sugar levels cause damage to **blood vessels** in the retina



NORMAL

Diabetic Retinopathy

Impaired vision, dark or empty areas



Vitrectomy surgery for diabetic retinopathy

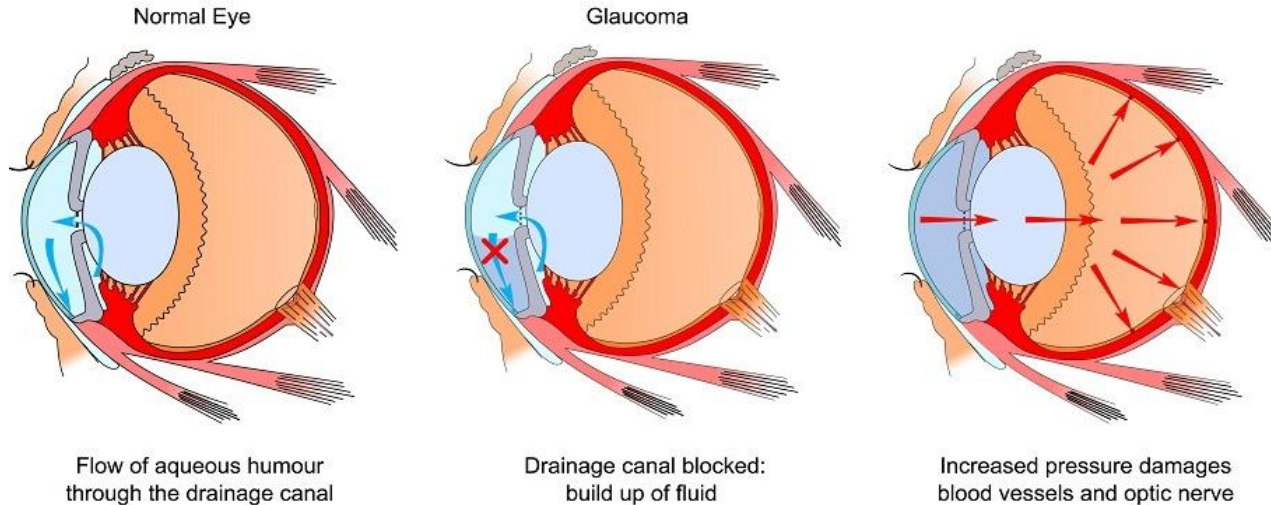
1. Last time: introduction to eye physiology



Common eye diseases

Glaucoma

- one of the leading causes of **blindness** for age above 60
- abnormally **high pressure** in the eye
- if damage to optic nerve, may cause permanent vision loss or even blindness



Prescription eye drops or Glaucoma lasers

1. Last time: introduction to eye physiology



Common eye diseases

Age-related macular degeneration

- it happens when the **macula** is damaged
- very **common**
- often happens in **elder** people > 50
- **blur the sharp**, central vision you need for activities like reading and driving

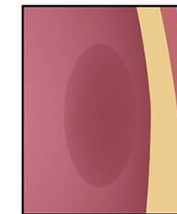
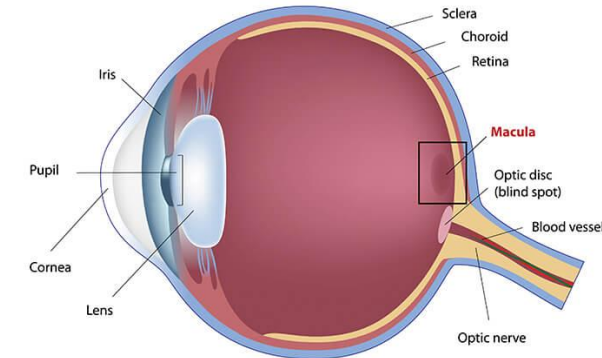


Normal Vision

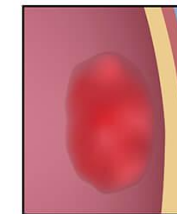


The same scene affected by age-related macular degeneration

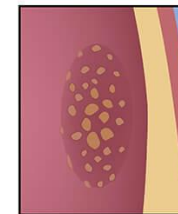
Macular Degeneration



Normal



"Wet" Macular Degeneration



"Dry" Macular Degeneration

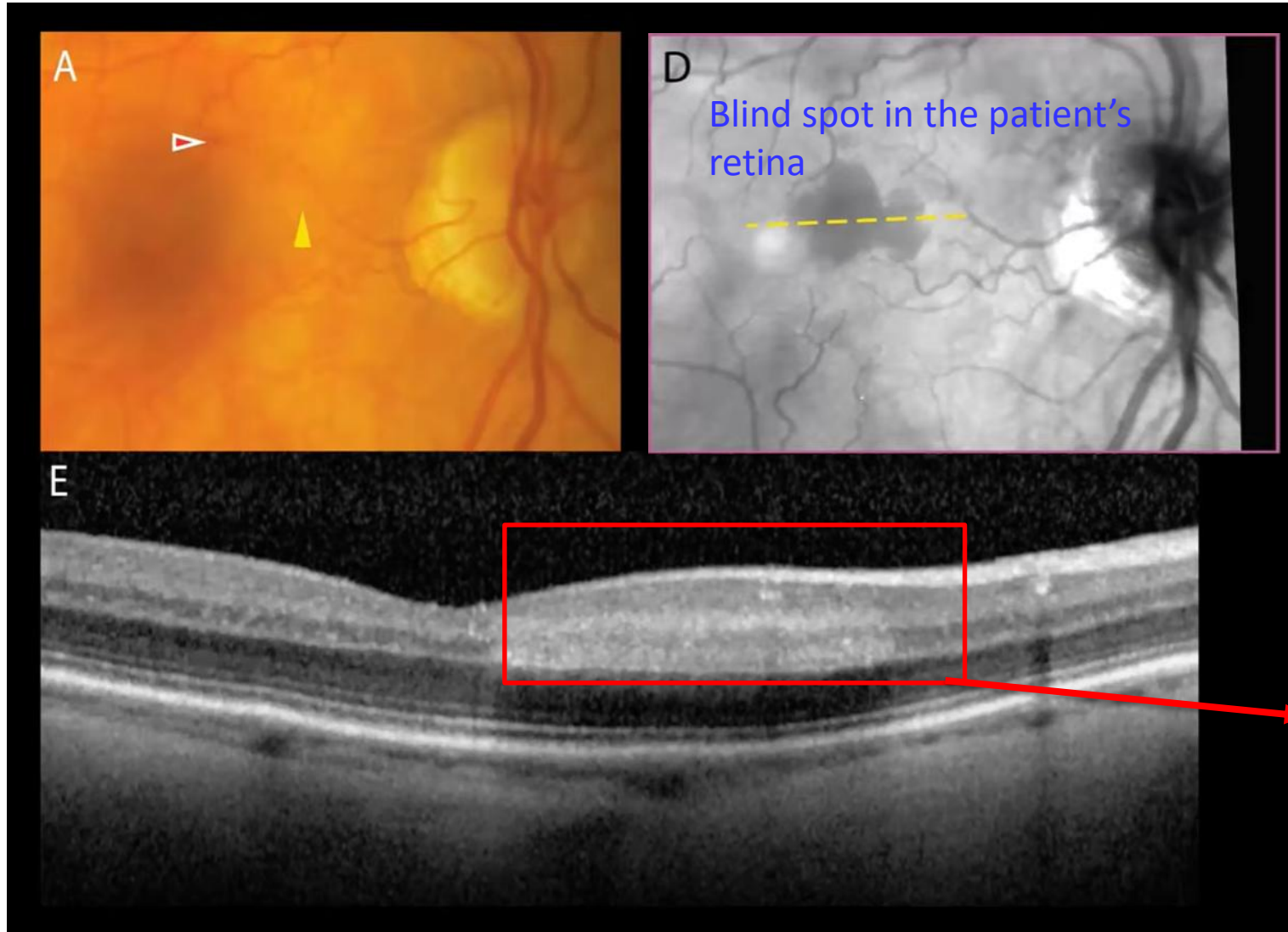
Treatment:
medications called anti-VEGF drugs

1. Last time: Fundus images and OCT



color photograph of a patient, who has an interruption of blood flow on the retina

However, color photo can not contribute a lot in understanding this case.



infrared photography

reflects one of very deep layers on behind of retina.

In addition to standard fundus photography, serial IR imaging may be an important adjunct to detecting outer retinal pathology with patients with peripheral retinoschisis.

A band of tissue within the retina which has increased brightness.

Also causes a shadow behind it.

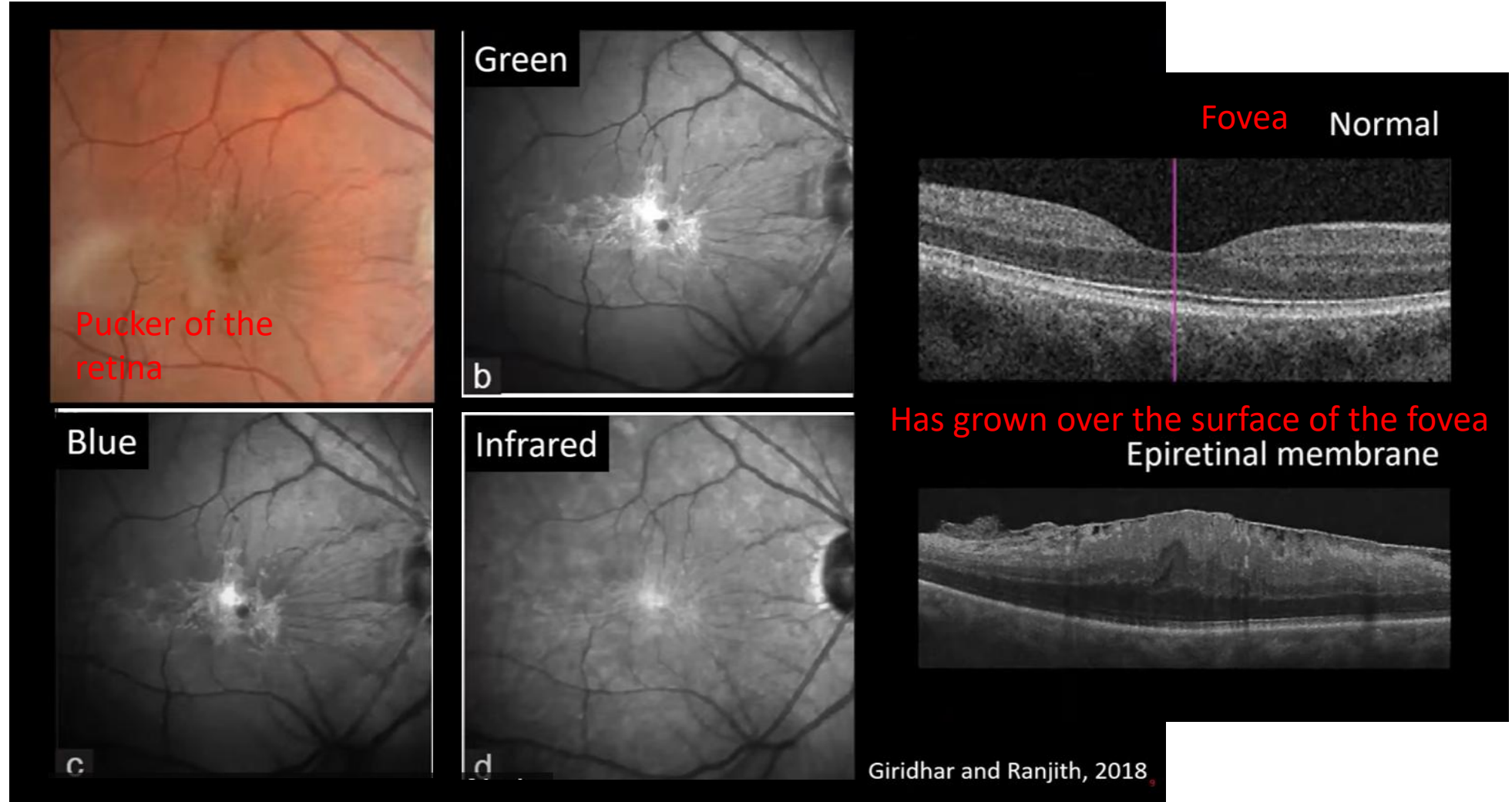
1. Last time: Fundus images and OCT



A patient with
epiretinal
membrane.

A layer of scar tissue
which has gradually
grown the surface of
the retina.

Blue flash in the
camera that
delineates the
edges of the
membrane.



1. Last time: Deep learning applications



Diabetic Retinopathy Detection

The attention map is a heatmap that indicates which pixels play more important roles in making the image-level decision.

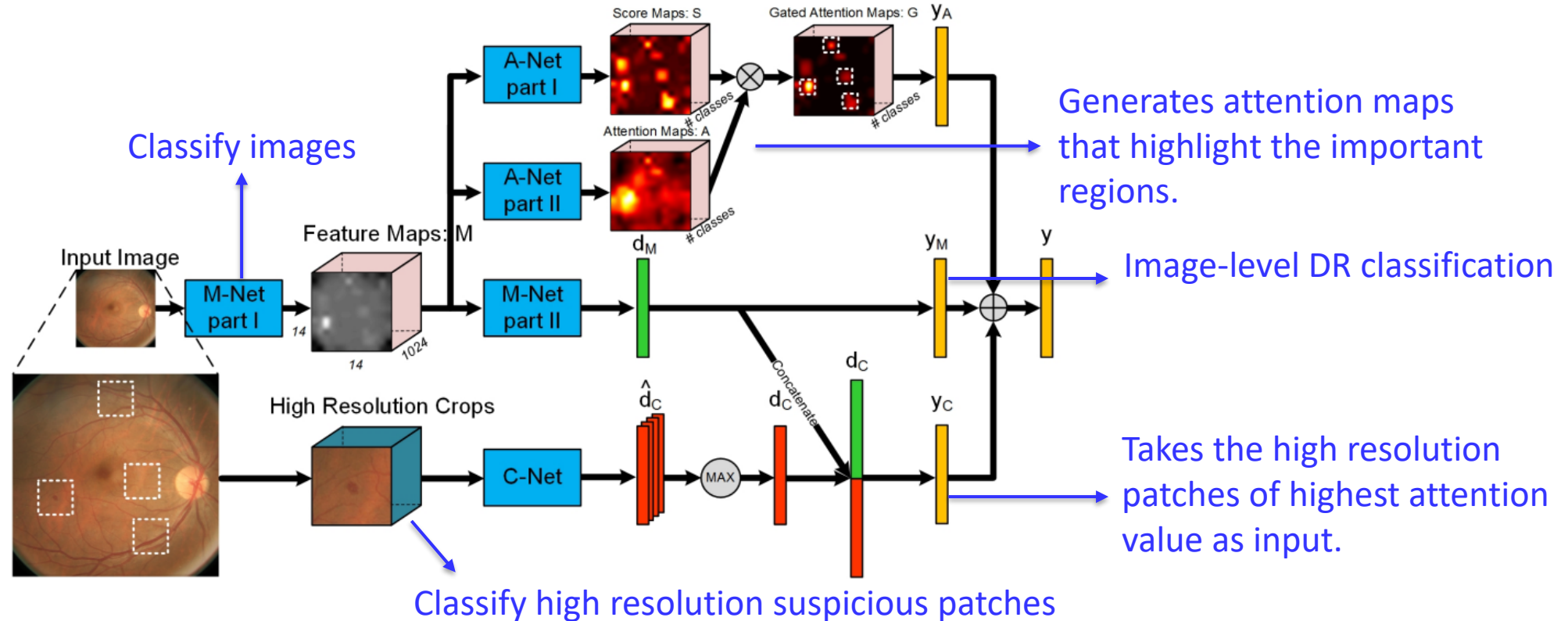


Fig. 1. An overview of Zoom-in-Net. It consists of three sub-networks. M-Net and C-Net classify the image and high resolution suspicious patches, respectively, while A-Net generates the gated attention maps for localizing suspicious regions and mining lesions.

2. Multimodal data



Next topic: Multimodal Data

2. Multimodal data



Multimodal data

Oncologists often use the terms "low-grade" and "high-grade" to categorize malignant gliomas based on how quickly the tumors spread.

Can be very similar, e.g. different image acquisition variants

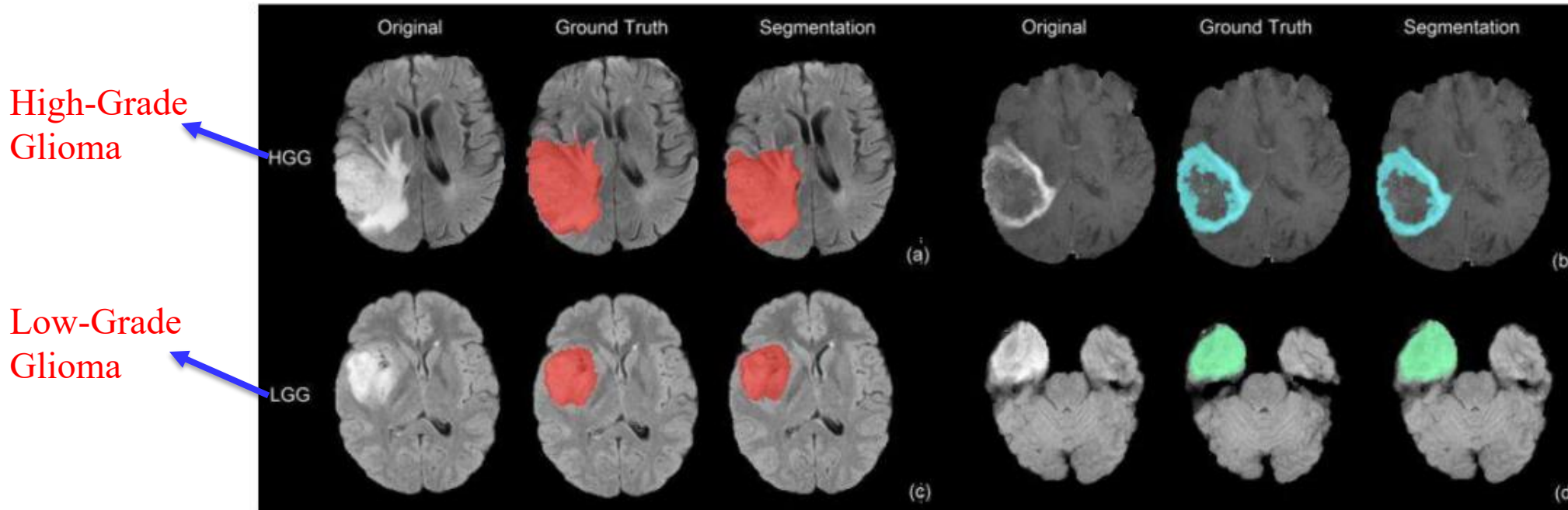


Figure credit: Dong et al. MIUA, 2017.

Low-grade gliomas grow very slowly but are still malignant and can progress to high-grade gliomas if left untreated.

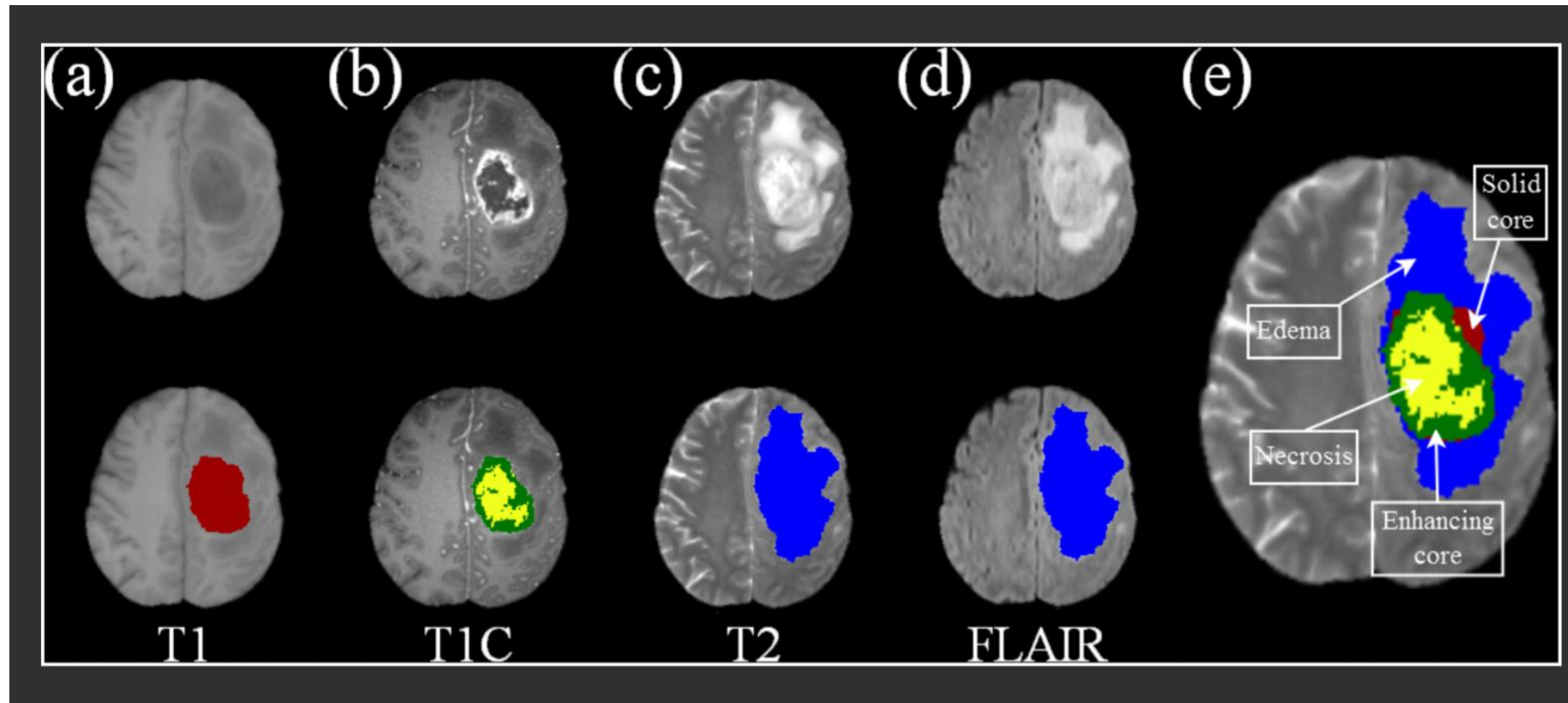
2. Multimodal data



Multimodal MRI in BRAST 2015 datasets

TR refers to the amount of time between successive pulse sequences applied to the same slice.

TE is the time between the delivery of RF pulse and the receipt of the echo signal.



T1-weighted (T1)

are produced by using **short** TE (time to echo) and TR (Repetition time) times.

T1-weighted imaging with gadolinium enhancing contrast (T1c)

Gad enhanced images are especially useful in looking at vascular structures and breakdown in the blood-brain barrier

T2-weighted (T2)

are produced by using **longer** TE and TR times.

FLAIR

Fluid Attenuated Inversion Recovery

Similar to T2-weighted imaging, except that TE and TR times are **very long**.

2. Multimodal data



Multimodal data

Or very different, e.g. different types of clinical data

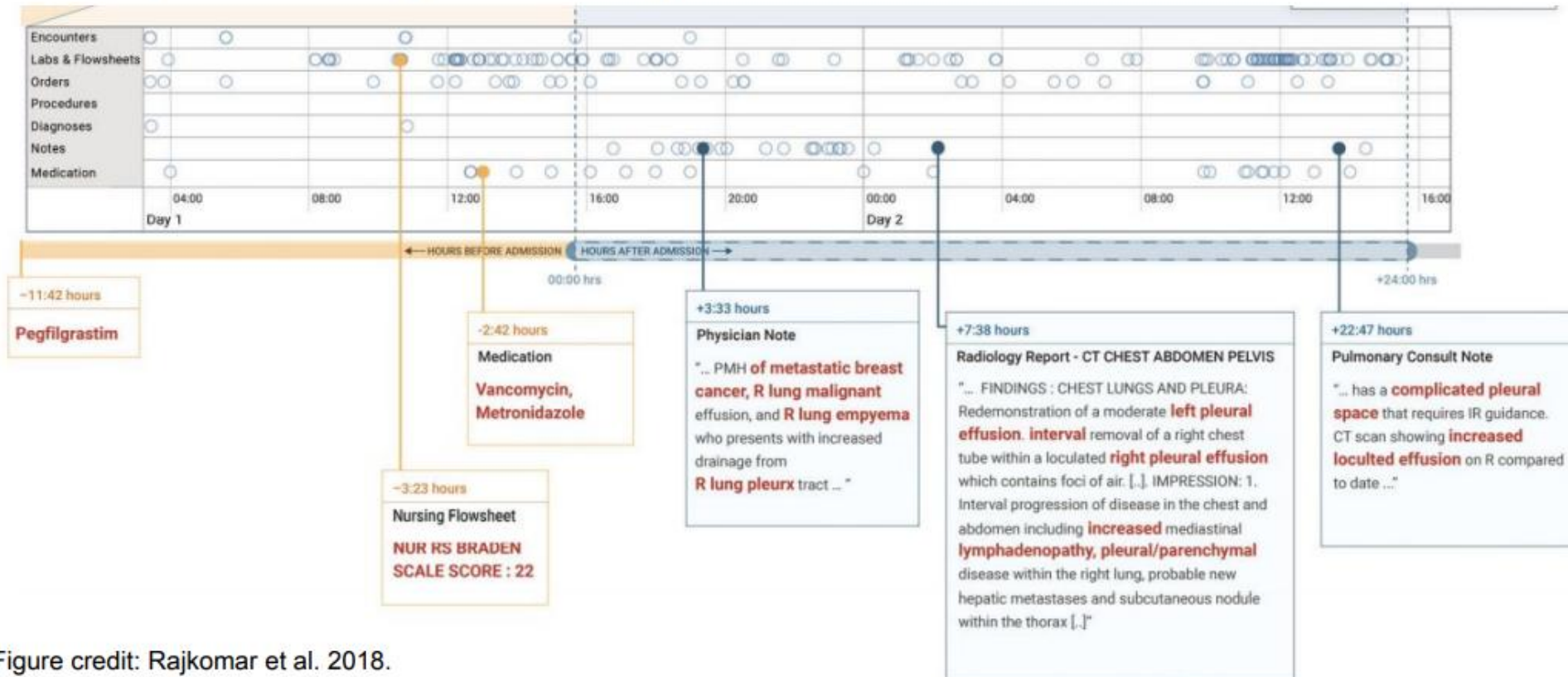


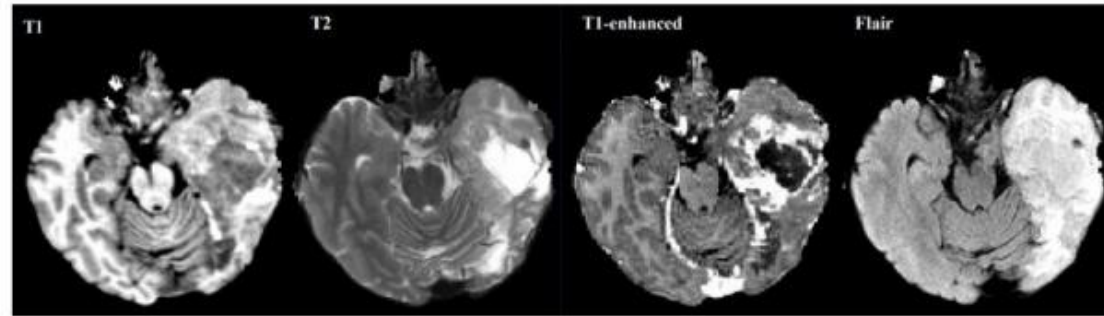
Figure credit: Rajkomar et al. 2018.

2. Multimodal data



Similar data: can fuse at input

- Havaei et al.: brain tumor segmentation from multimodal MR images

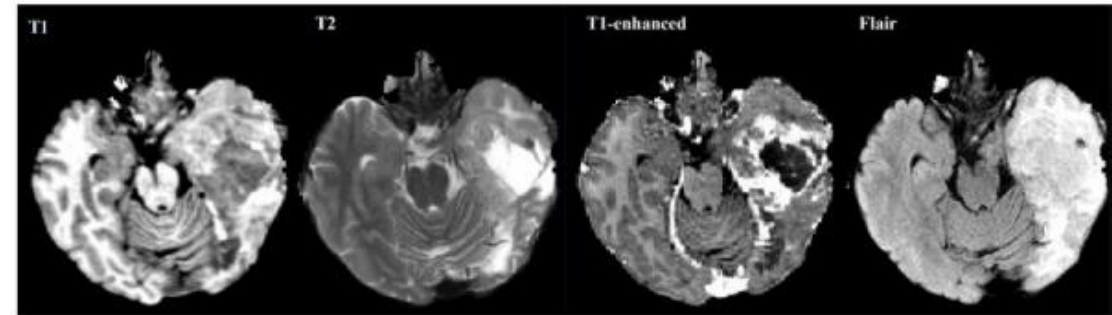


2. Multimodal data

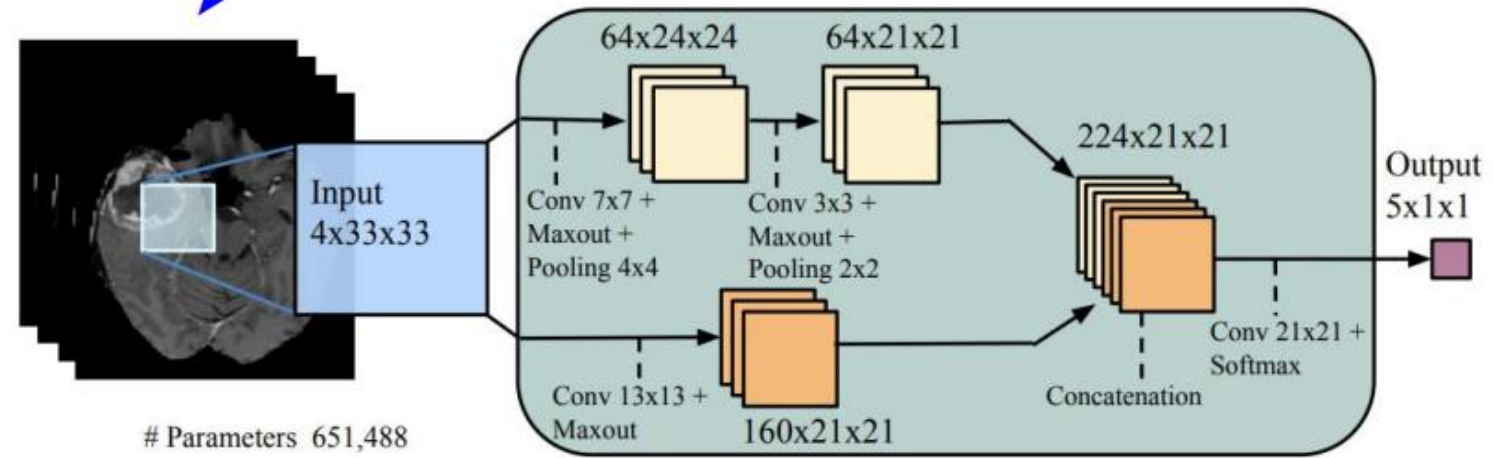


Similar data: can fuse at input

- Havaei et al.: brain tumor segmentation from multimodal MR images



Stack modalities such that each channel of input is a different modality.



2. Multimodal data

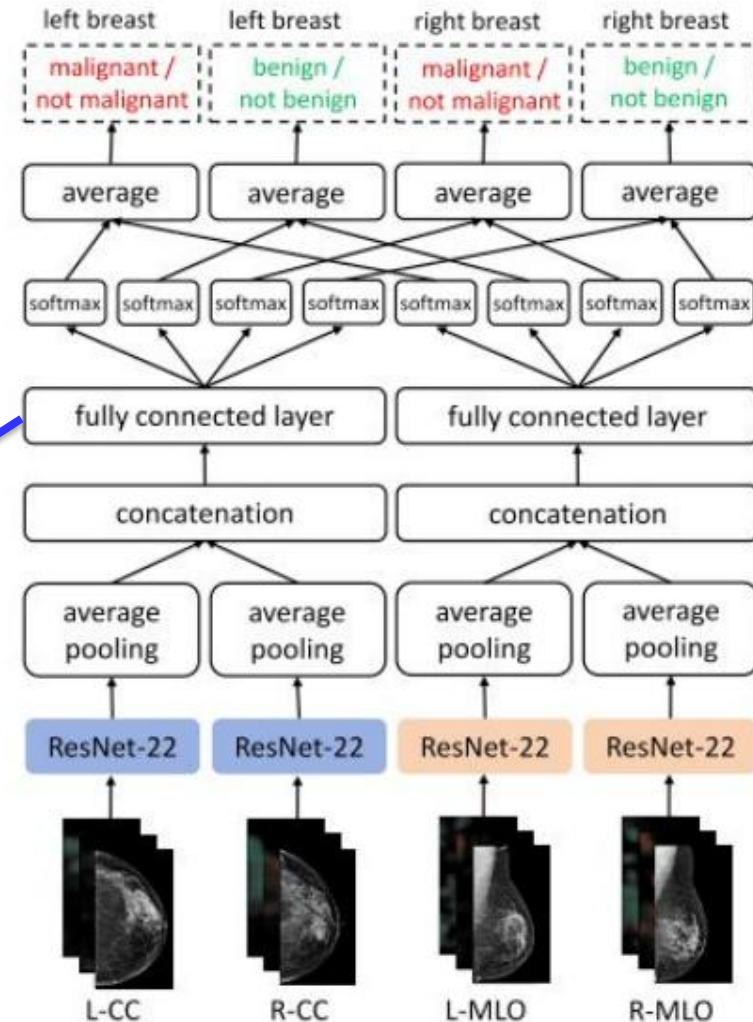


More different data:
may want some layers of
modality-specific processing

Wu et al. 2019:

- Binary classification of breast malignant and benign findings
- Model based on ResNet architecture
- Multi-view network (different views can be considered different modalities)

Make separate
predictions for
CC and MLO,
and then
averaged for
prediction.



Wu et al. Deep Neural Networks Improve Radiologists' Performance in Breast Cancer Screening. IEEE Trans Med Imaging, 2019.

L-CC (left craniocaudal), R-CC (right craniocaudal), L-MLO (left mediolateral oblique), R-MLO (right mediolateral oblique)

2. Multimodal data

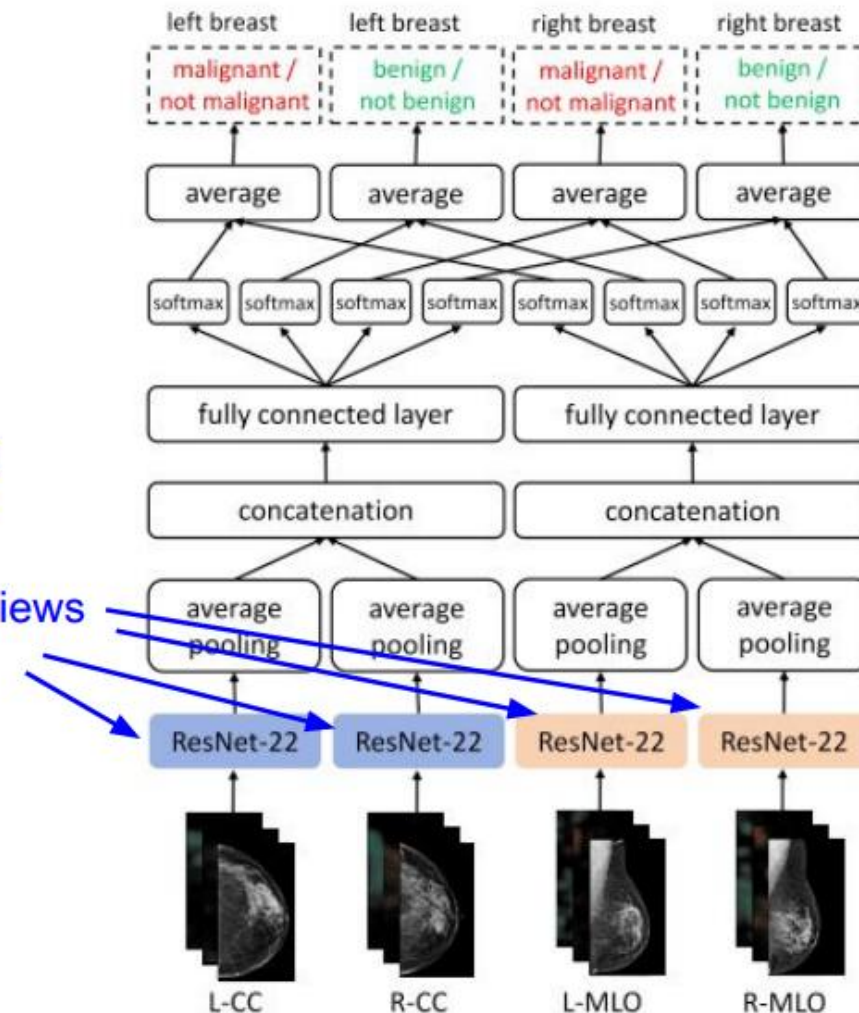


More different data:
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Separate initial
processing for
different
mammogram views



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2. Multimodal data

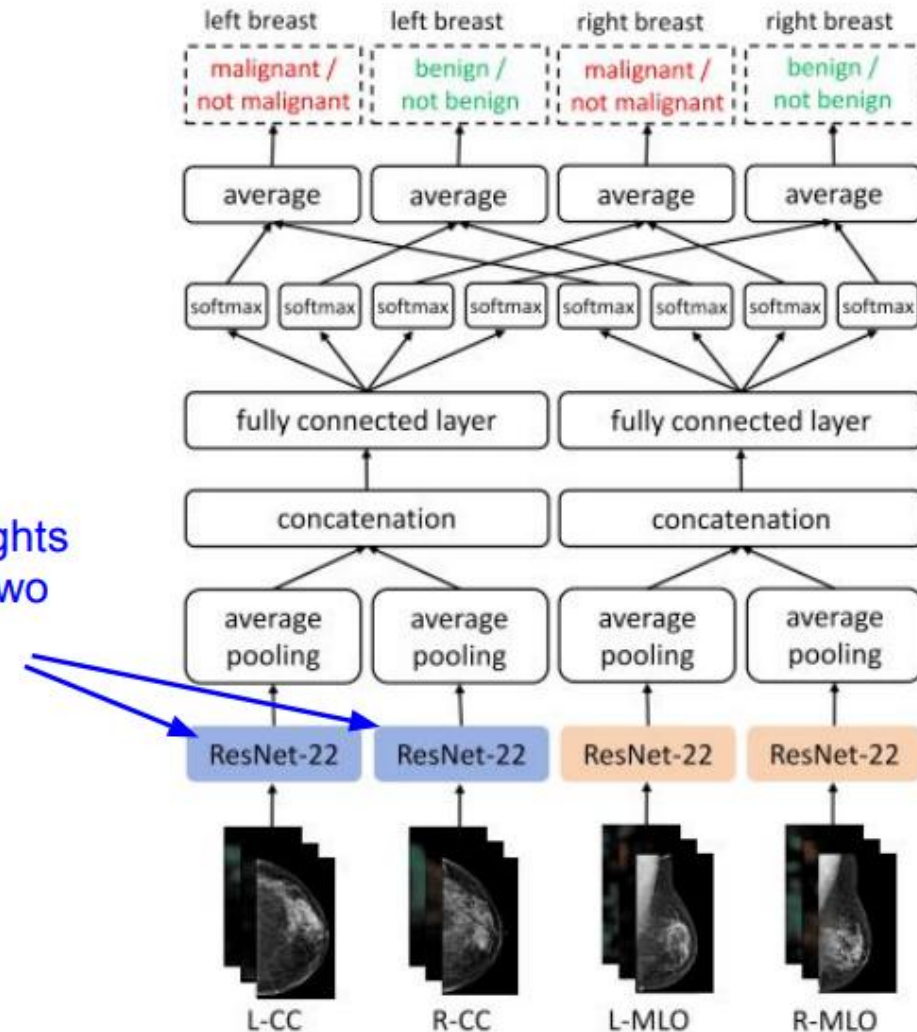


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Shared weights
across the two
networks



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2. Multimodal data

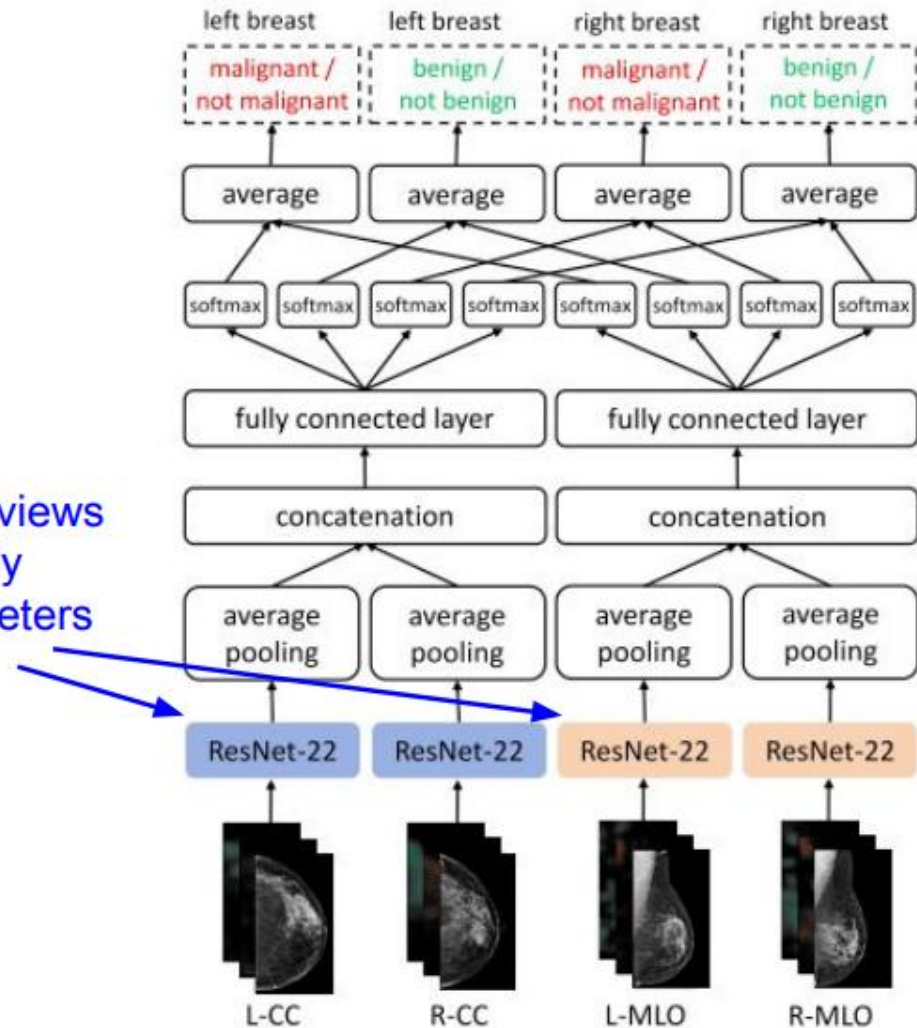


More different data:
may want some layers of
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Wu et al. 2019:

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More different views
have separately
learned parameters



2. Multimodal data

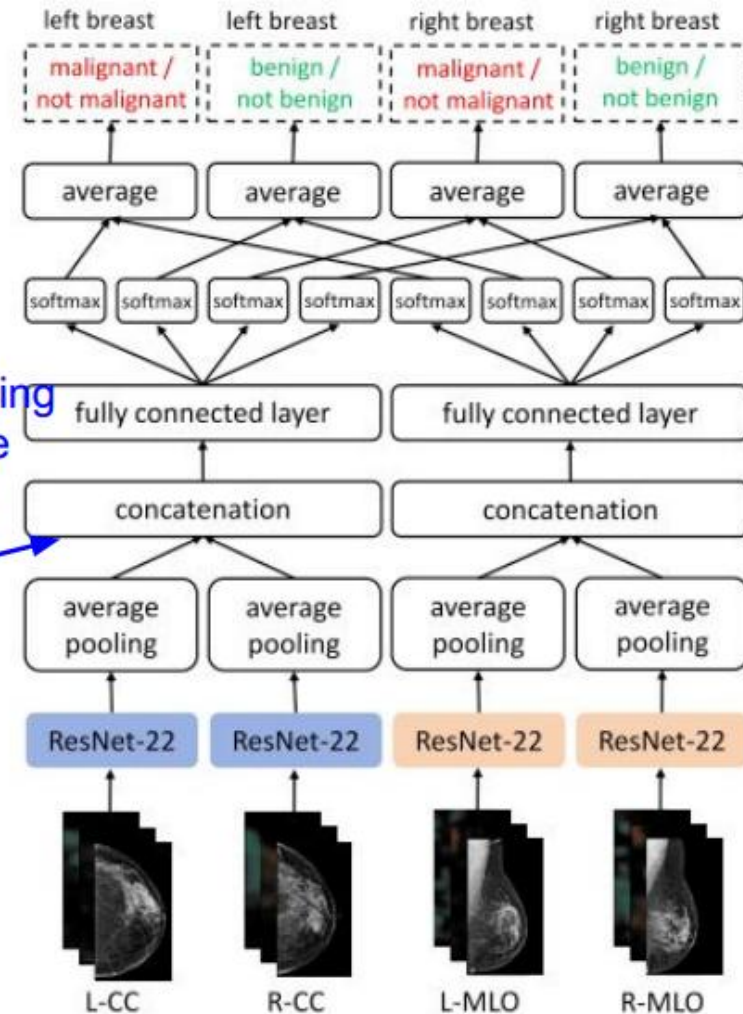


More different data:
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Multimodal fusion at intermediate part of processing (very common): concatenate outputs of modality-specific processing into one feature vector.



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2. Multimodal data

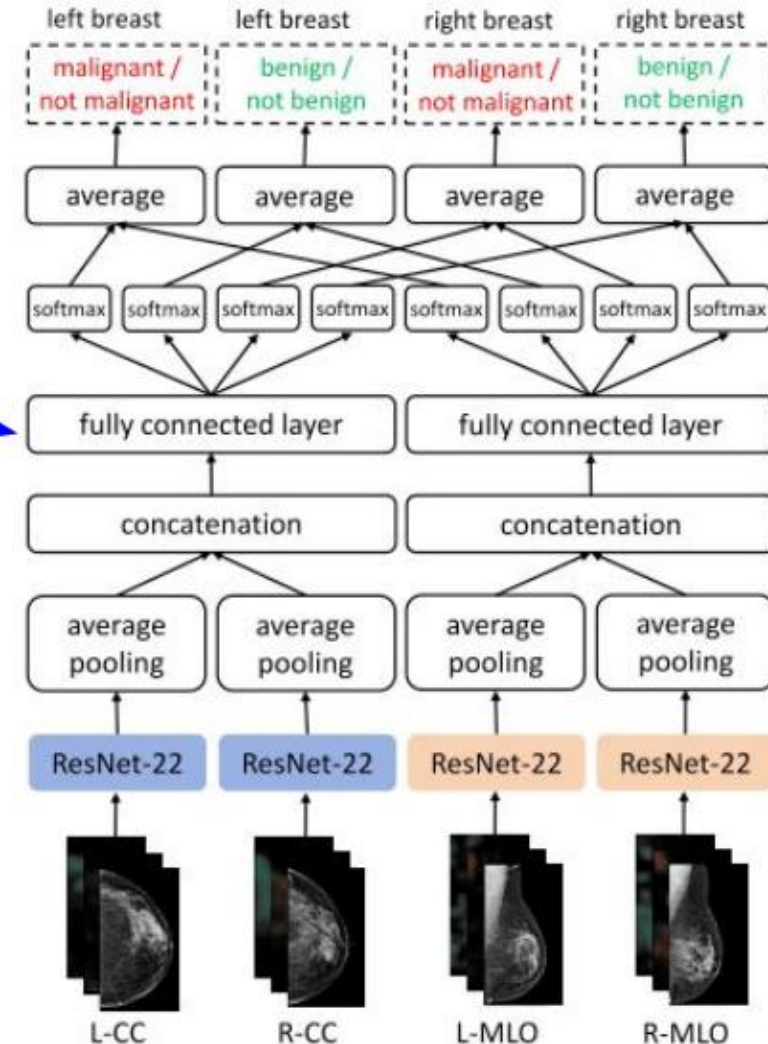


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may want some layers of
modality-specific processing

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Fully connected layer (or several) afterwards.
Concatenated feature vector no longer contains spatial relationships suitable for conv layers.



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2. Multimodal data



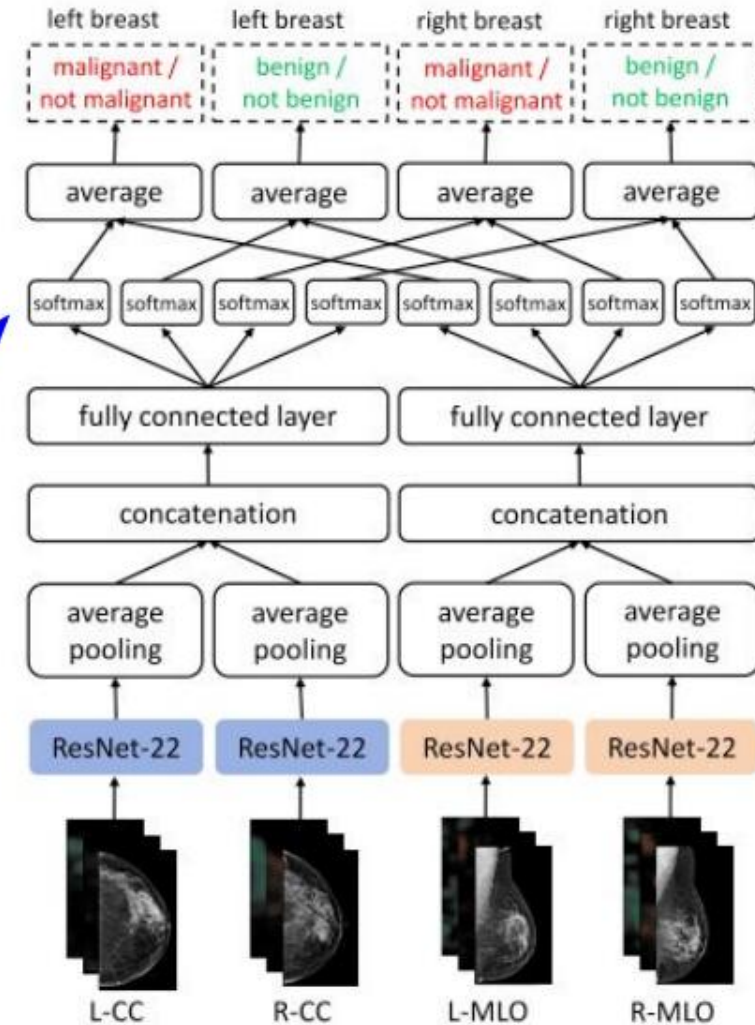
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Predict all
4 binary
outputs
from each
view



2. Multimodal data

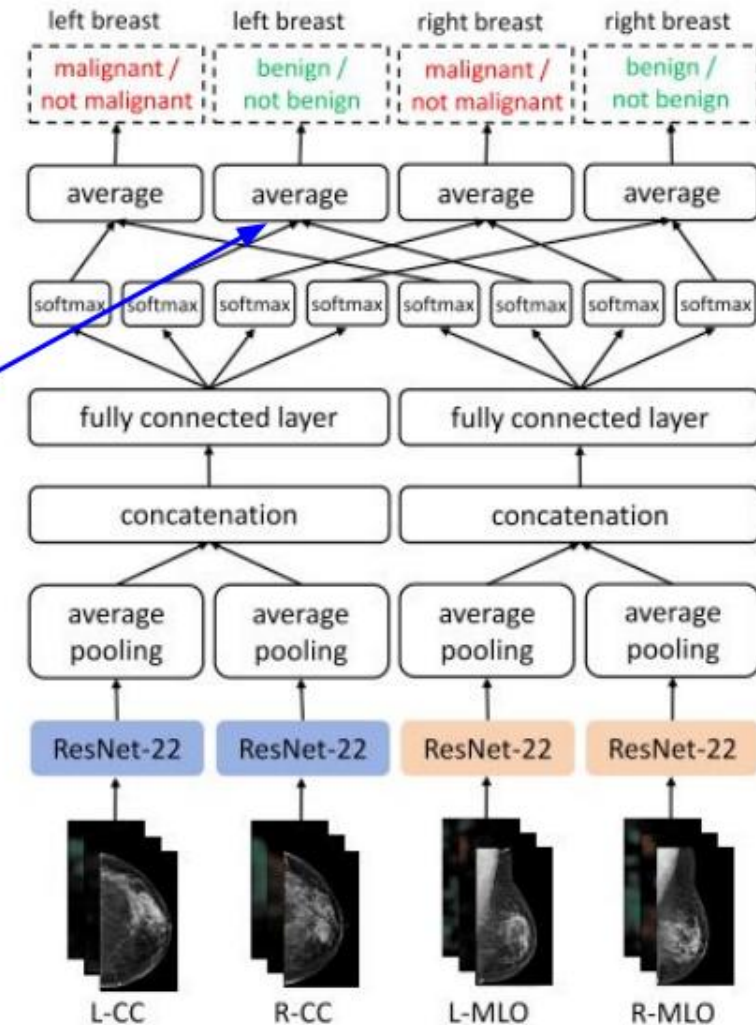


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Wu et al. 2019:

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This model also uses a second type of fusion for the CC vs. MLO views: late fusion of predictions through averaging.



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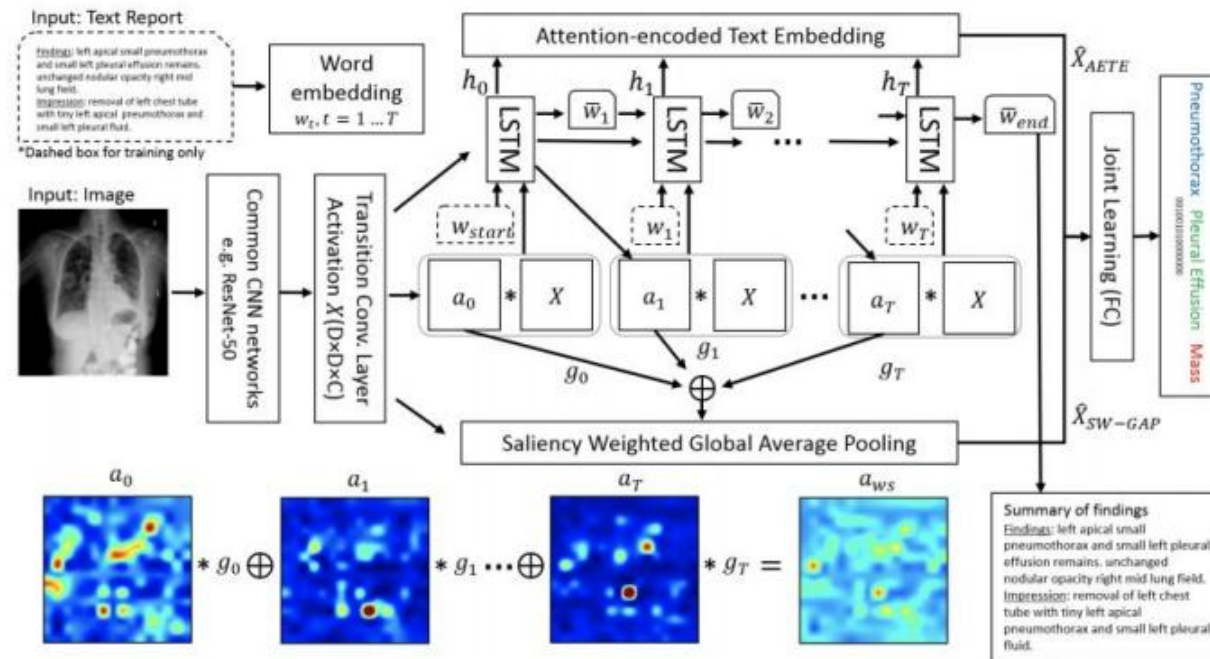
2. Multimodal data



A recurrent network approach for combining multimodal data

Wang et al. 2018:

- Jointly process chest x-rays and associated reports to produce disease labels that can be used to produce auto-annotation disease labels



Wang et al. TieNet: Text-Image Embedding Network for Common Thorax Disease Classification and Reporting in Chest X-rays. CVPR, 2018.

2. Multimodal data

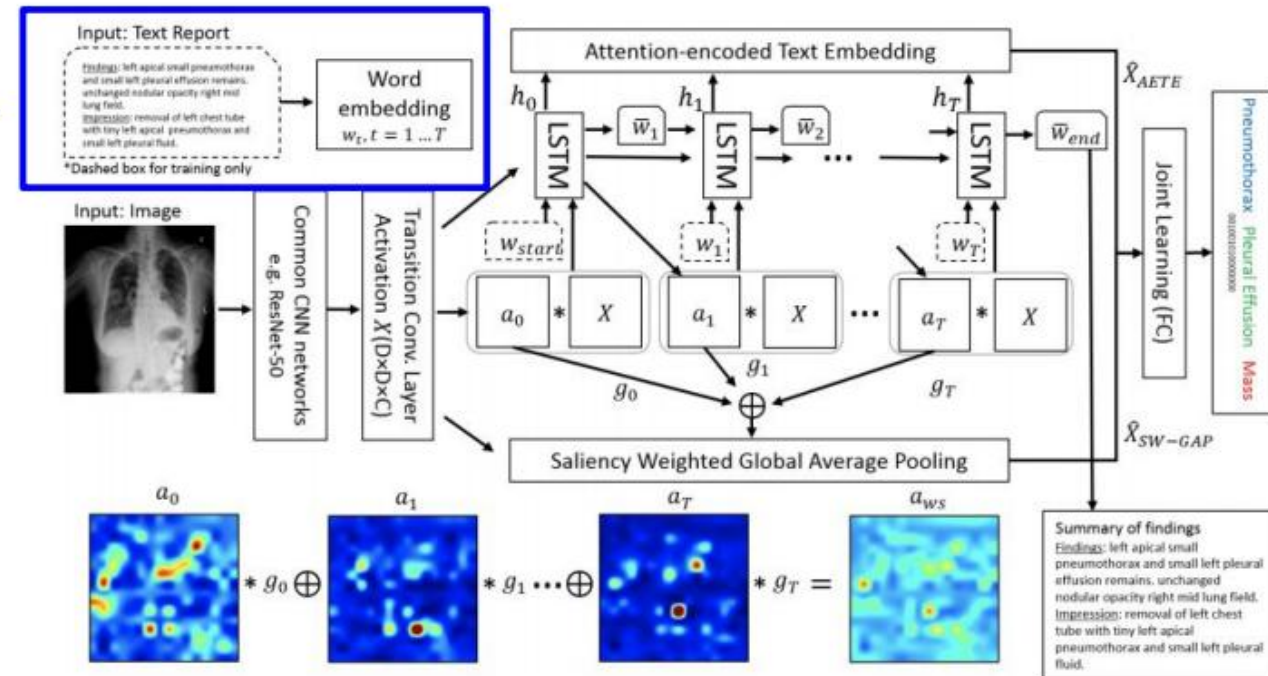


A recurrent network approach for combining multimodal data

Wang et al. 2018:

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Use NLP approaches to generate word embedding representations of words in text



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2. Multimodal data

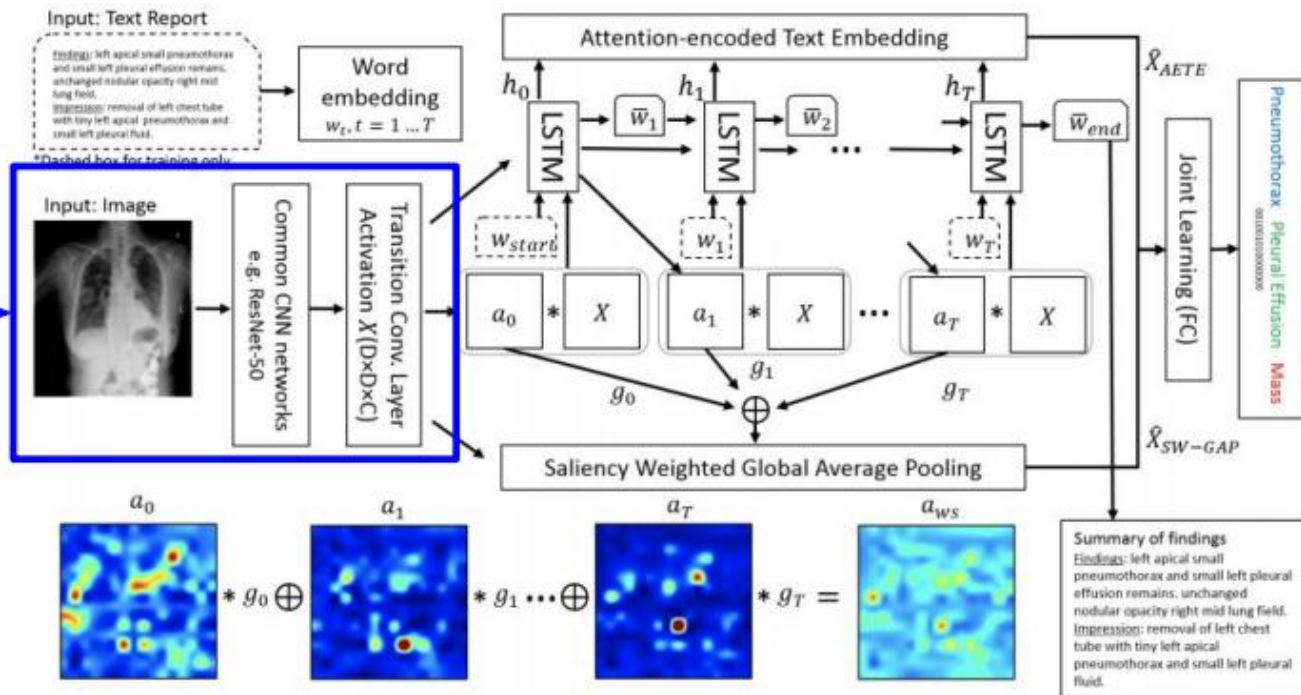


A recurrent network approach for combining multimodal data

Wang et al. 2018:

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Use common CNN networks to generate feature representation of image data



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2. Multimodal data

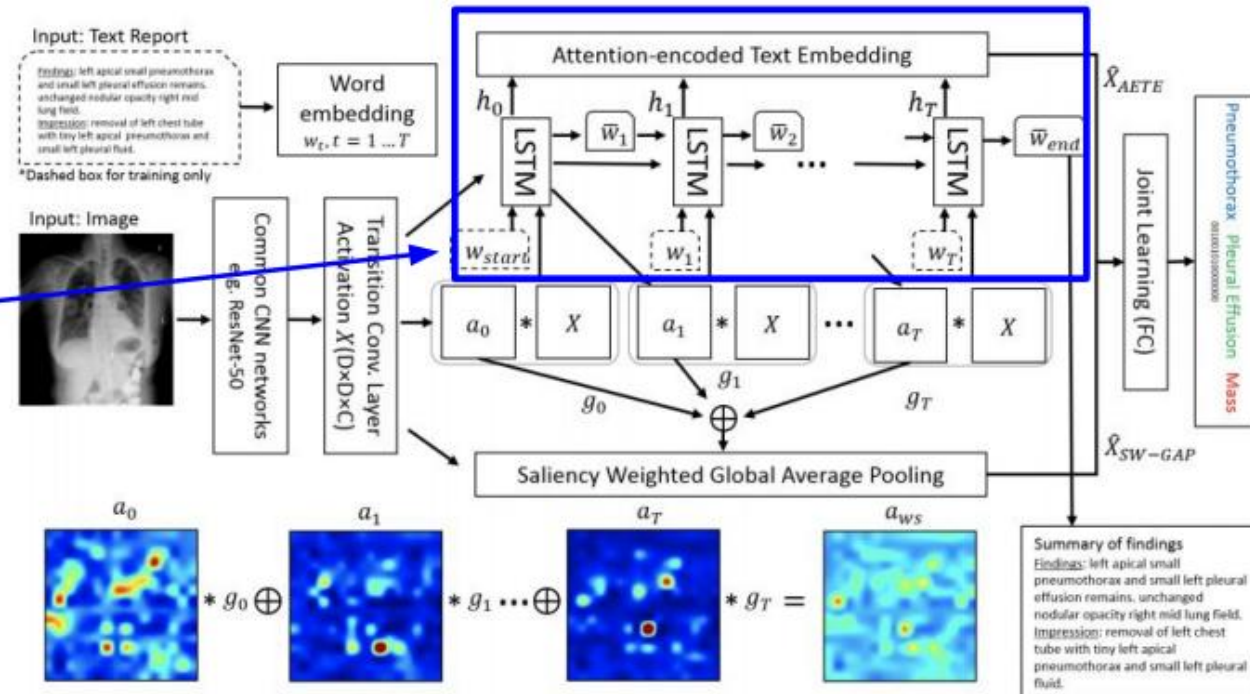


A recurrent network approach for combining multimodal data

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Use LSTM to process sequence of text data embedding representations



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2. Multimodal data



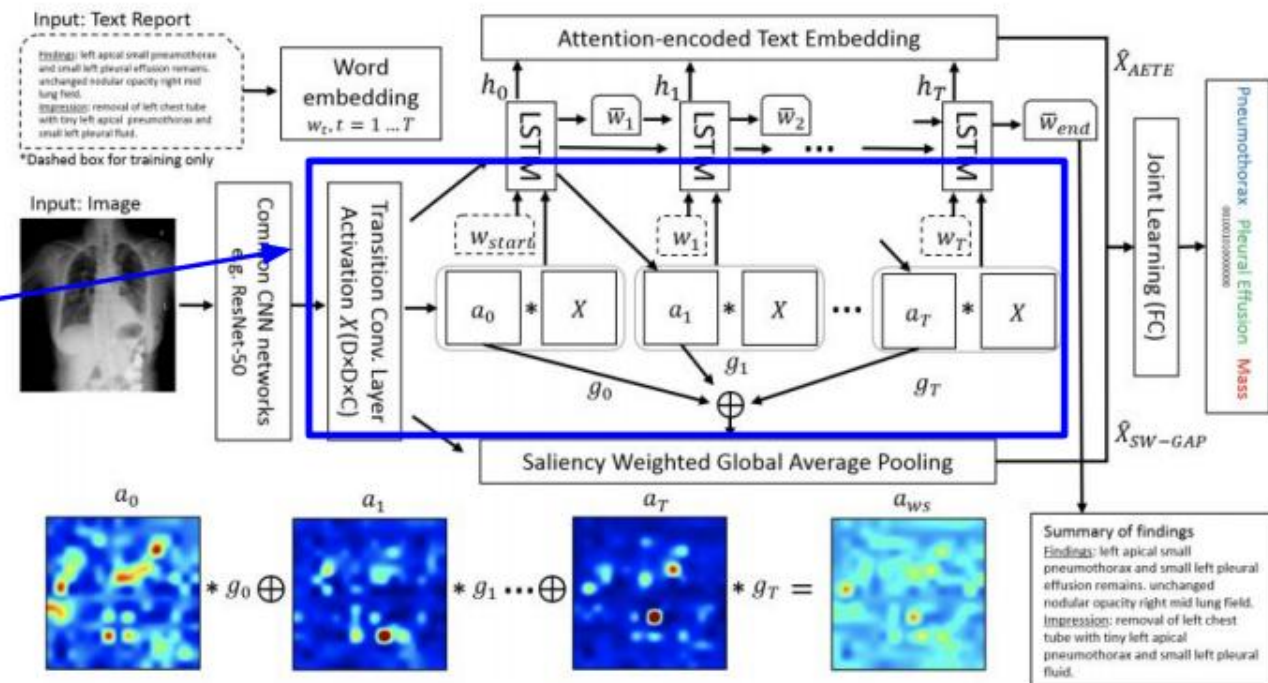
A recurrent network approach for combining multimodal data

Wang et al. 2018:

- Jointly process chest x-rays and associated reports to produce disease labels that can be used to produce auto-annotation disease labels

We produce a sequence of saliency values for each word. These saliency values are used to weight and select the spatial attention maps.

Image data is an additional input to the LSTM at each time step (with soft-attention weighting)



Wang et al. TieNet: Text-Image Embedding Network for Common Thorax Disease Classification and Reporting in Chest X-rays. CVPR, 2018.

2. Multimodal data

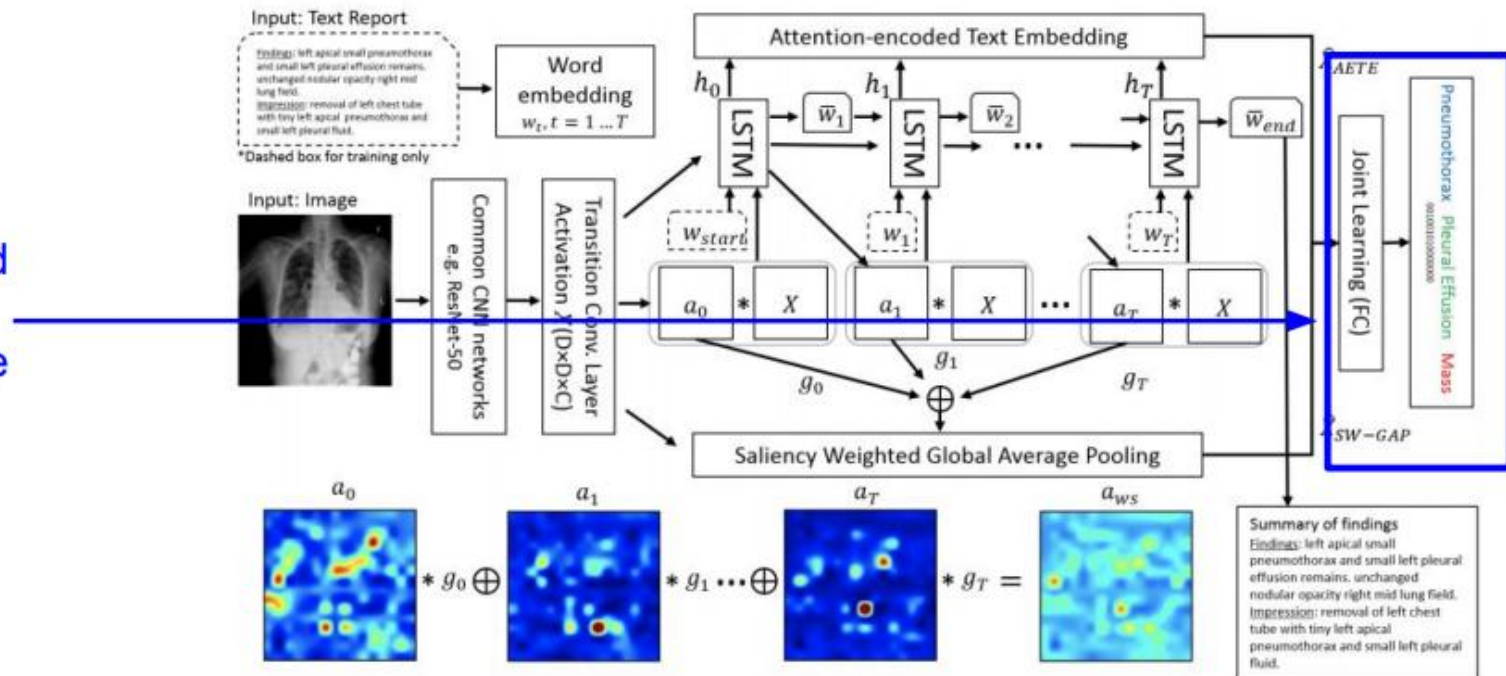


A recurrent network approach for combining multimodal data

Wang et al. 2018:

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Final fully-connected layer fusion and prediction of disease labels



Wang et al. TieNet: Text-Image Embedding Network for Common Thorax Disease Classification and Reporting in Chest X-rays. CVPR, 2018.

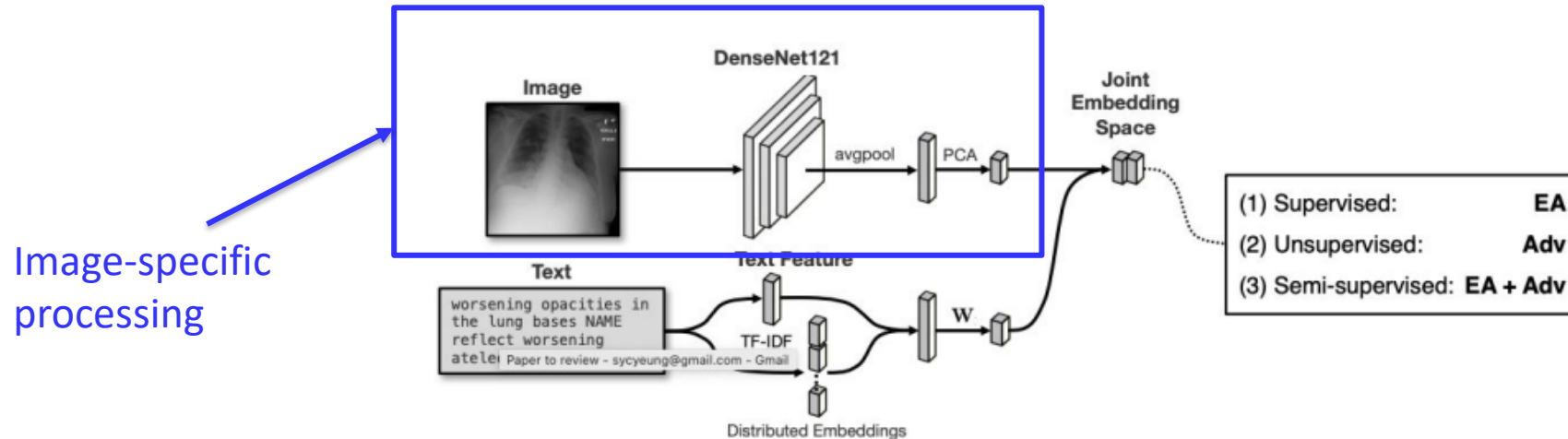
2. Multimodal data



Another direction of research: learning multimodal embedding spaces

Hsu et al. 2018:

- Learn mapping from images and text to vectors in the same embedding space, such that images are embedded closer to their corresponding reports than other reports, and vice versa.
- Can be used for e.g. cross-domain retrieval



Hsu et al. Unsupervised Multimodal Representation Learning across Medical Images and Reports. NeurIPS ML4H, 2018.

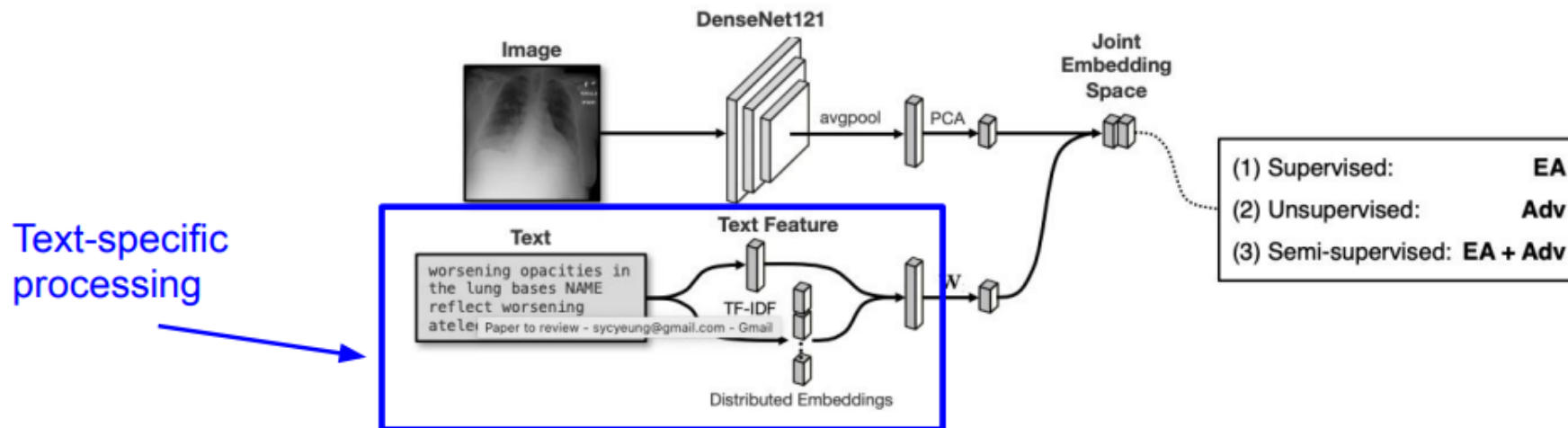
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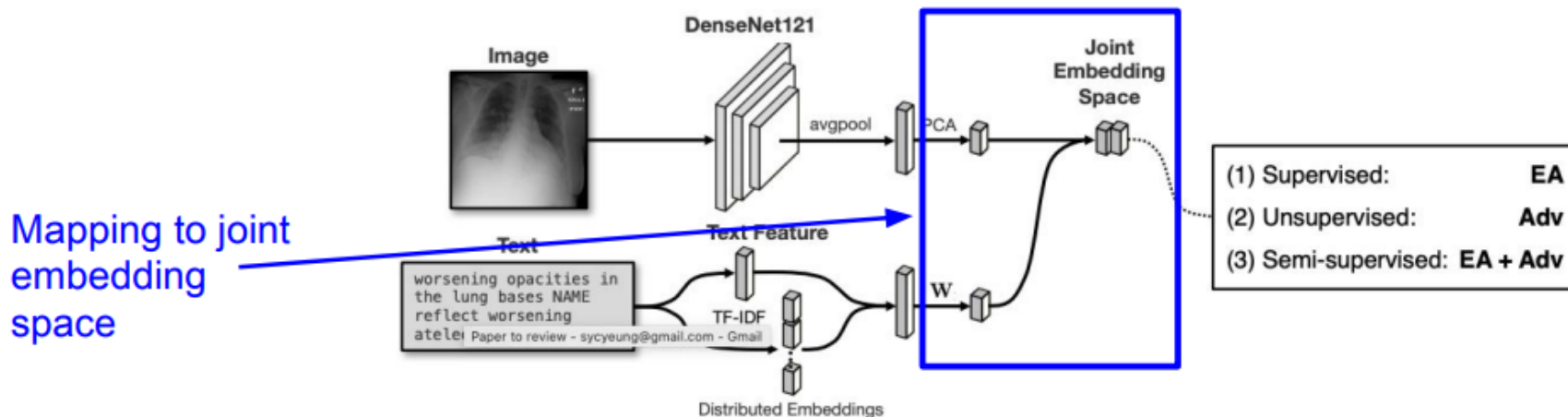
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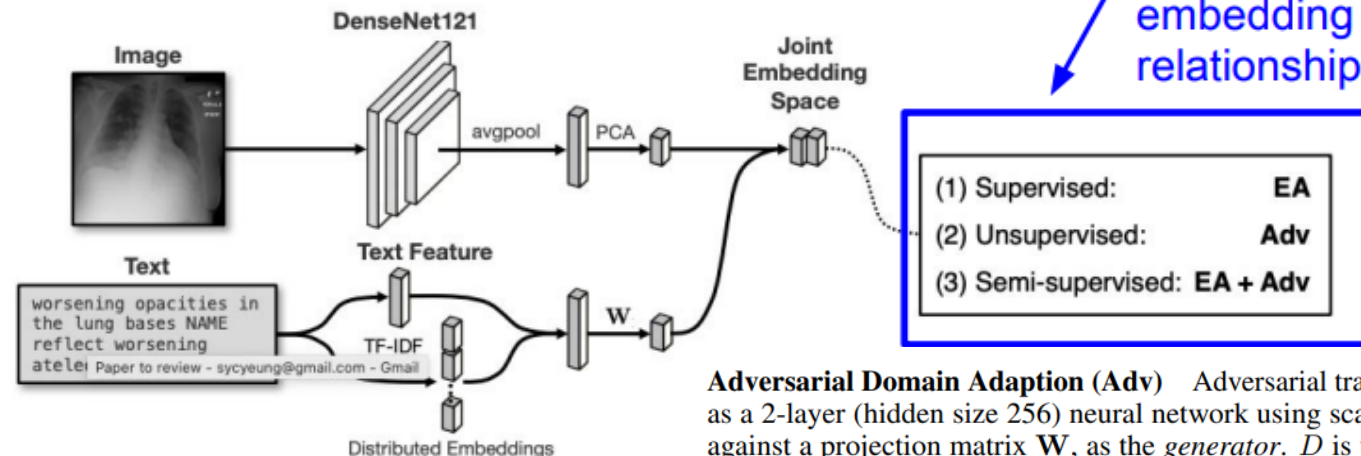


Another direction of research: learning multimodal embedding spaces

Embedding Alignment (EA) Here, we find a linear transformation between two sets of matched points $\mathbf{X} \in \mathbb{R}^{d_X \times n}$ and $\mathbf{Y} \in \mathbb{R}^{d_Y \times n}$ by minimizing $\mathcal{L}_{EA}(\mathbf{X}, \mathbf{Y}) = \|\mathbf{W}^\top \mathbf{X} - \mathbf{Y}\|_F^2$.

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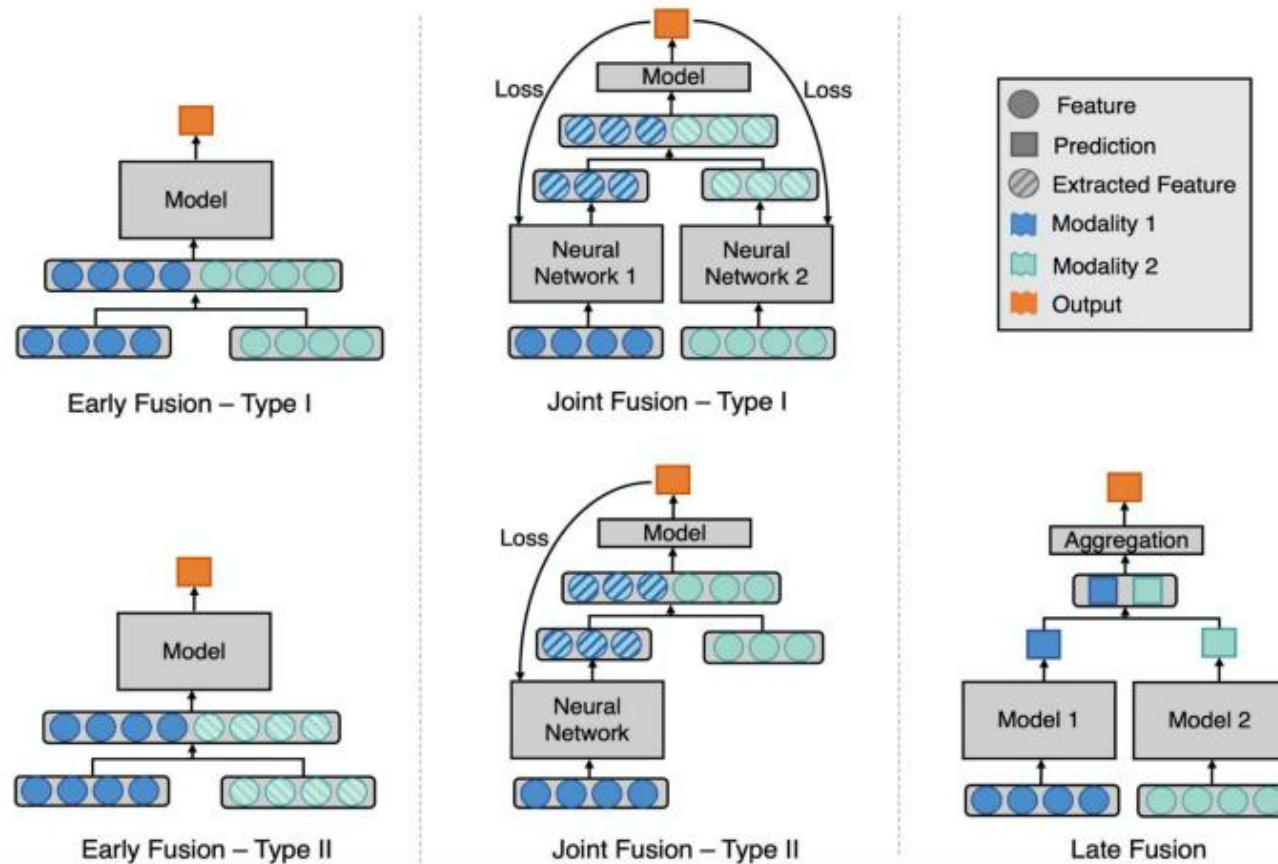


Adversarial Domain Adaption (Adv) Adversarial training pits a *discriminator*, D , implemented as a 2-layer (hidden size 256) neural network using scaled exponential linear units (SELUs) [15], against a projection matrix $\mathbf{W}$, as the *generator*. D is trained to classify points in the joint space according to source modality, and $\mathbf{W}$ is trained adversarially to fool D . Alternatively, D minimizes $\mathcal{L}_{Adv}^D(\mathbf{X}, \mathbf{Y}) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim p(\mathbf{X}, \mathbf{Y})} [-\log D(\mathbf{W}^\top \mathbf{x}) - \log(1 - D(\mathbf{y}))]$ when $\mathbf{W}$ minimizes $\mathcal{L}_{Adv}^W(\mathbf{X}, \mathbf{Y}) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim p(\mathbf{X}, \mathbf{Y})} [-\log(1 - D(\mathbf{W}^\top \mathbf{x}))]$.

2. Multimodal data



Categorizations of multimodal models

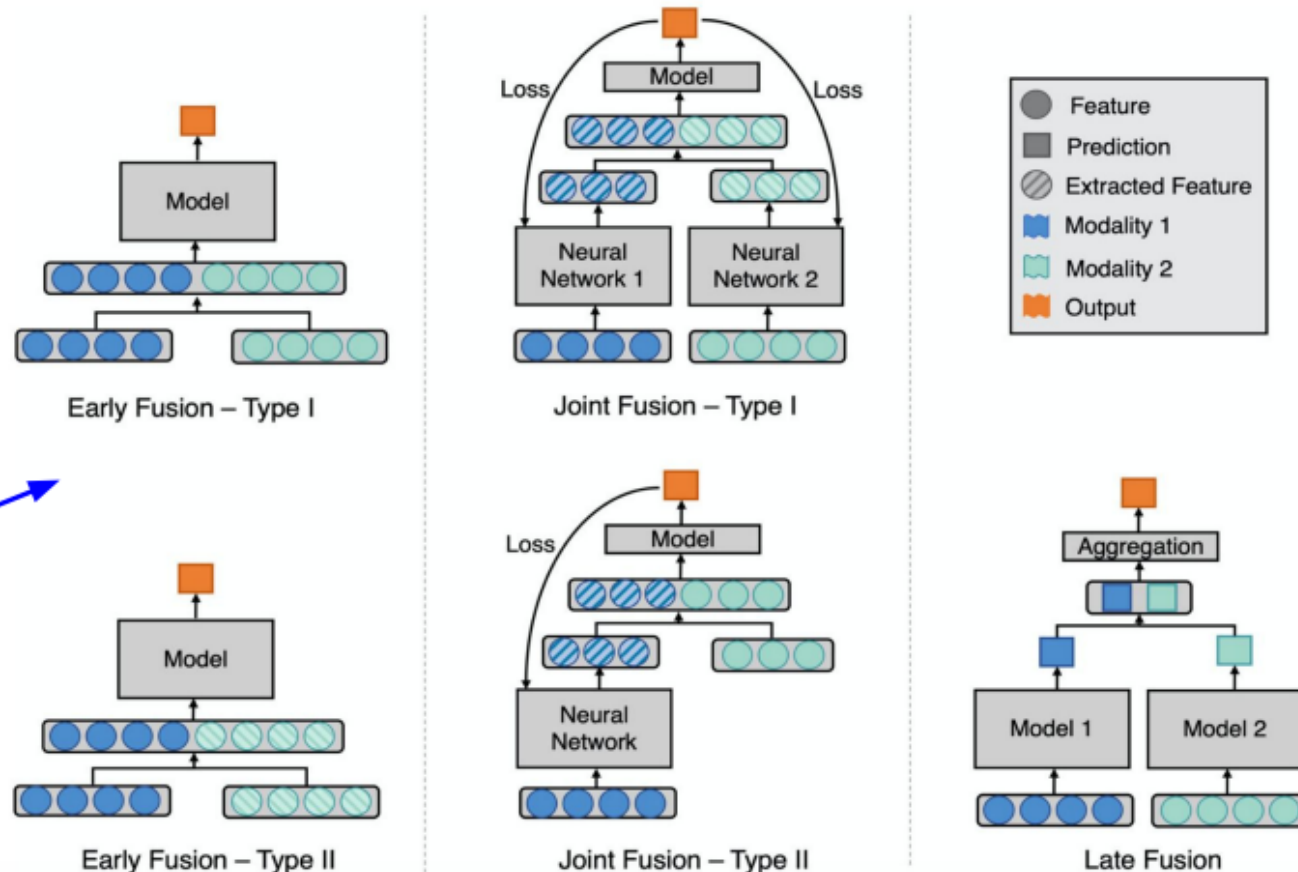


2. Multimodal data



Categorizations of multimodal models

Early fusion:
concatenate /
combine data
before any model
processing.
Includes using
extracted features
as input, if model
gradients are not
backpropagated to
update feature
extractor
parameters

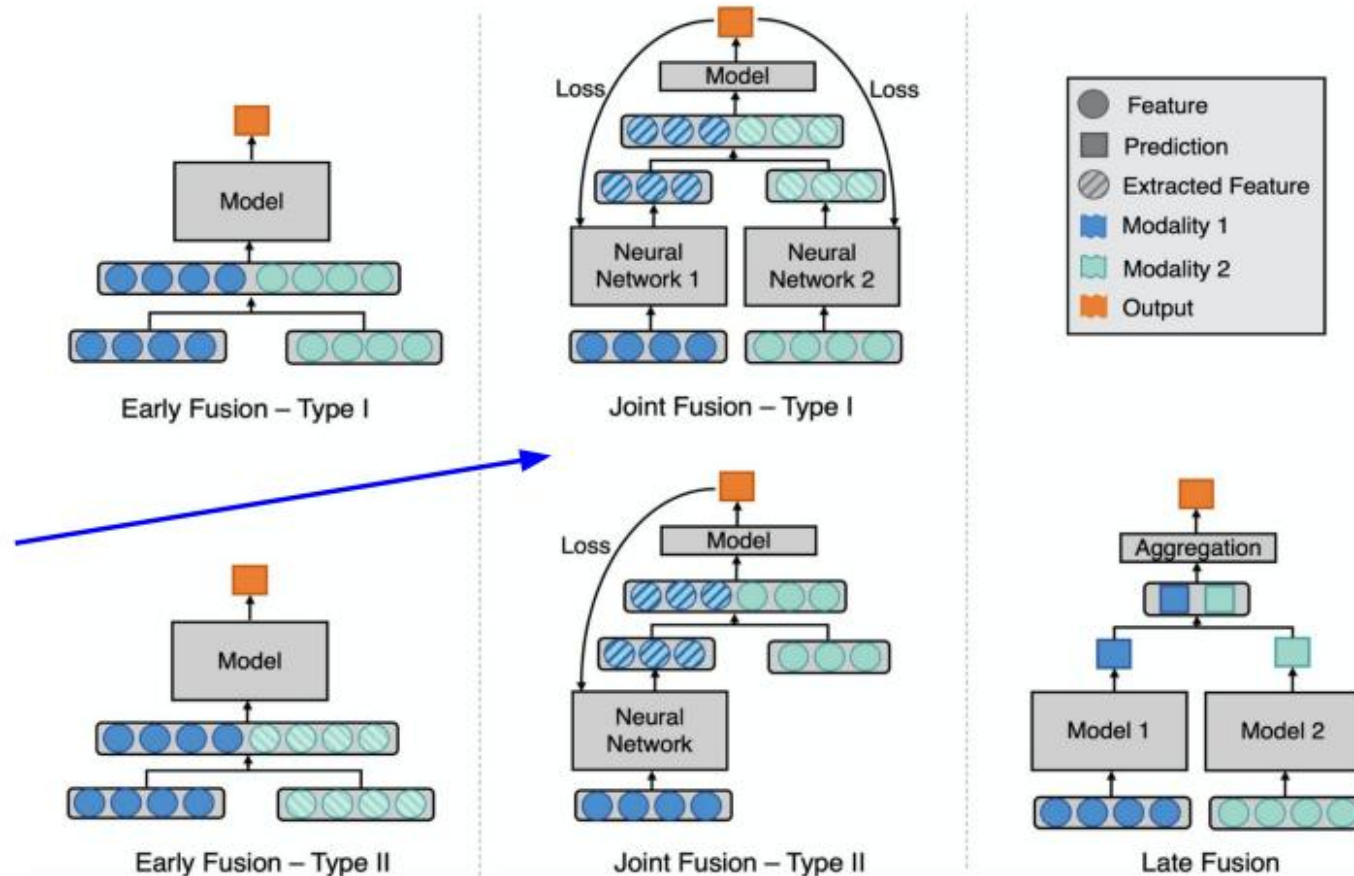


2. Multimodal data



Categorizations of multimodal models

Joint fusion: Both modality-specific components (with learnable parameters) and combined-modality components within the model, that are updated during model training

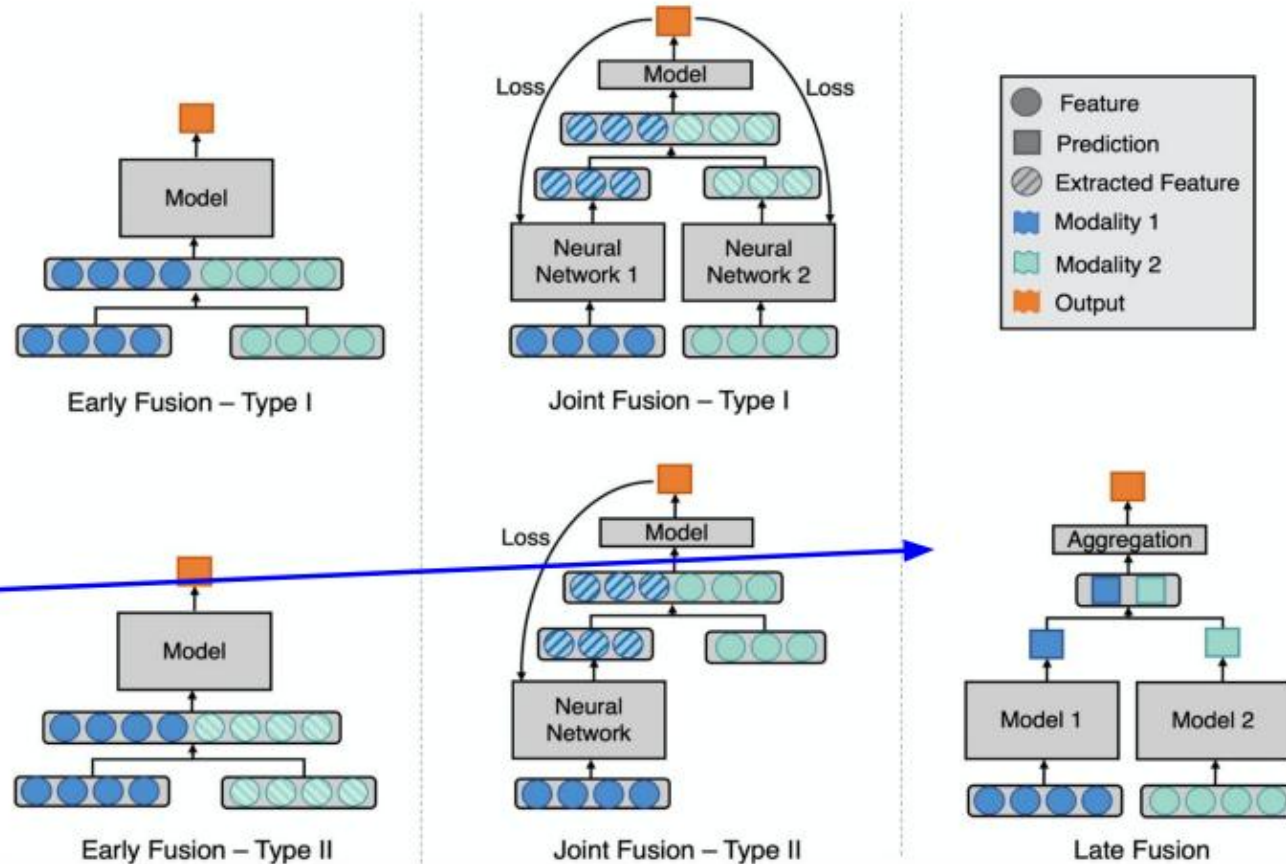


2. Multimodal data



Categorizations of multimodal models

Late fusion:
Main learnable
model
components are
only model
specific.
Individual
modality outputs
are then
aggregated.

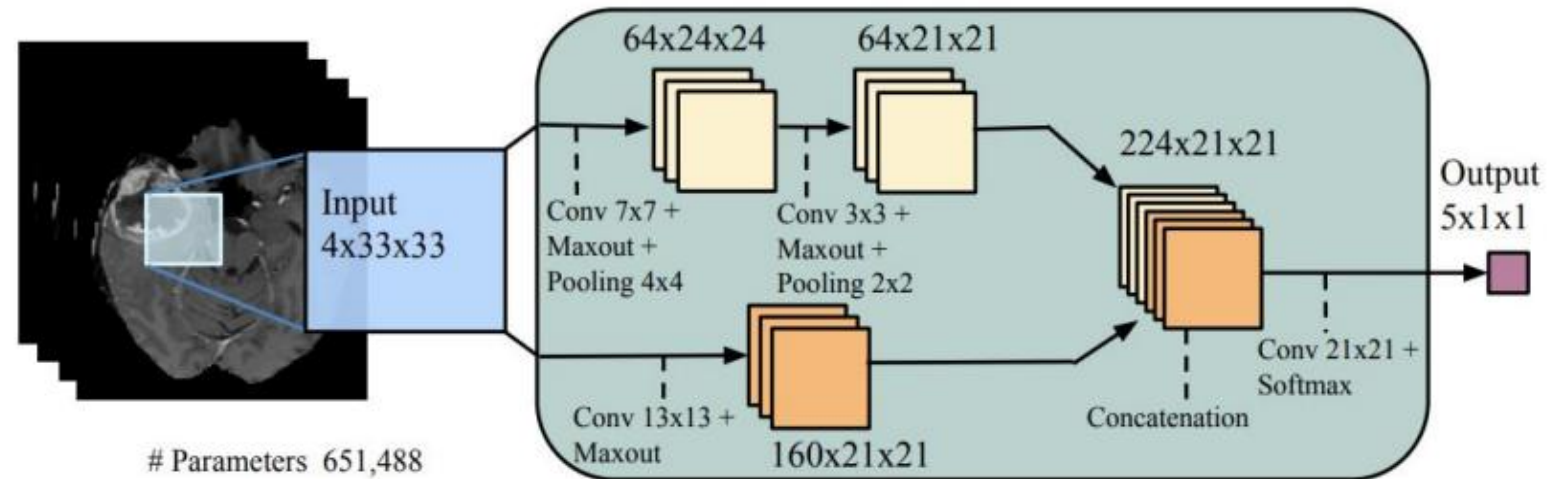
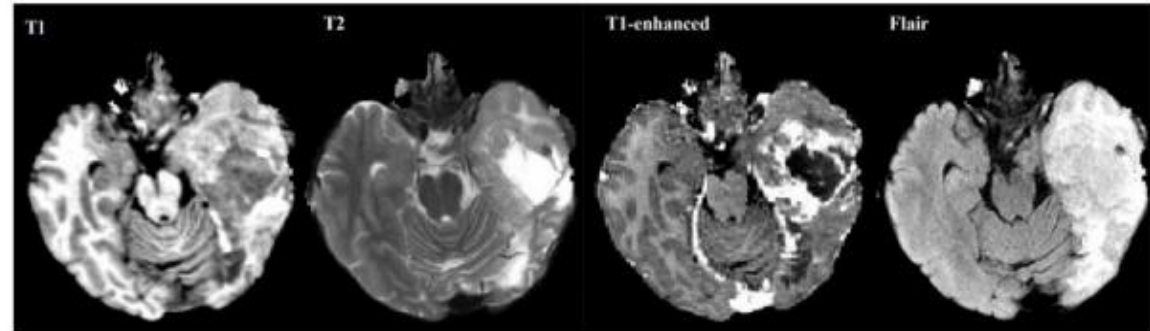


2. Multimodal data



Q: What kind of fusion was this model?

- Havaei et al.: brain tumor segmentation from multimodal MR images



2. Multimodal data

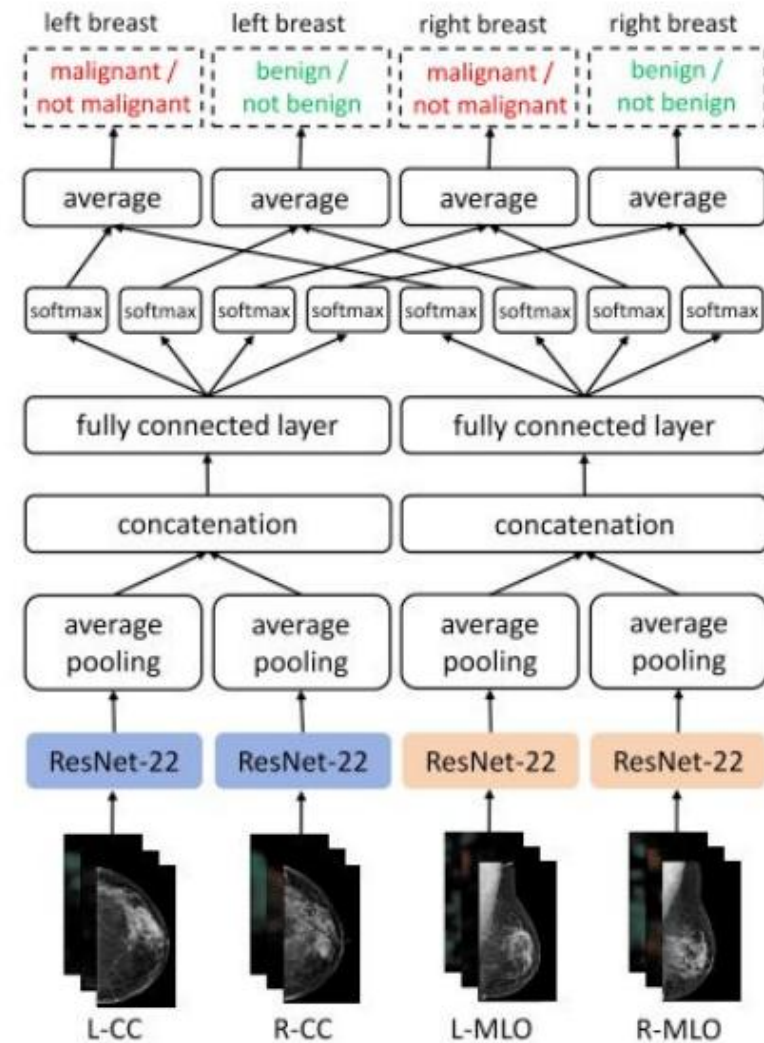


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Wu et al. 2019:

- Binary classification of breast malignant and benign findings
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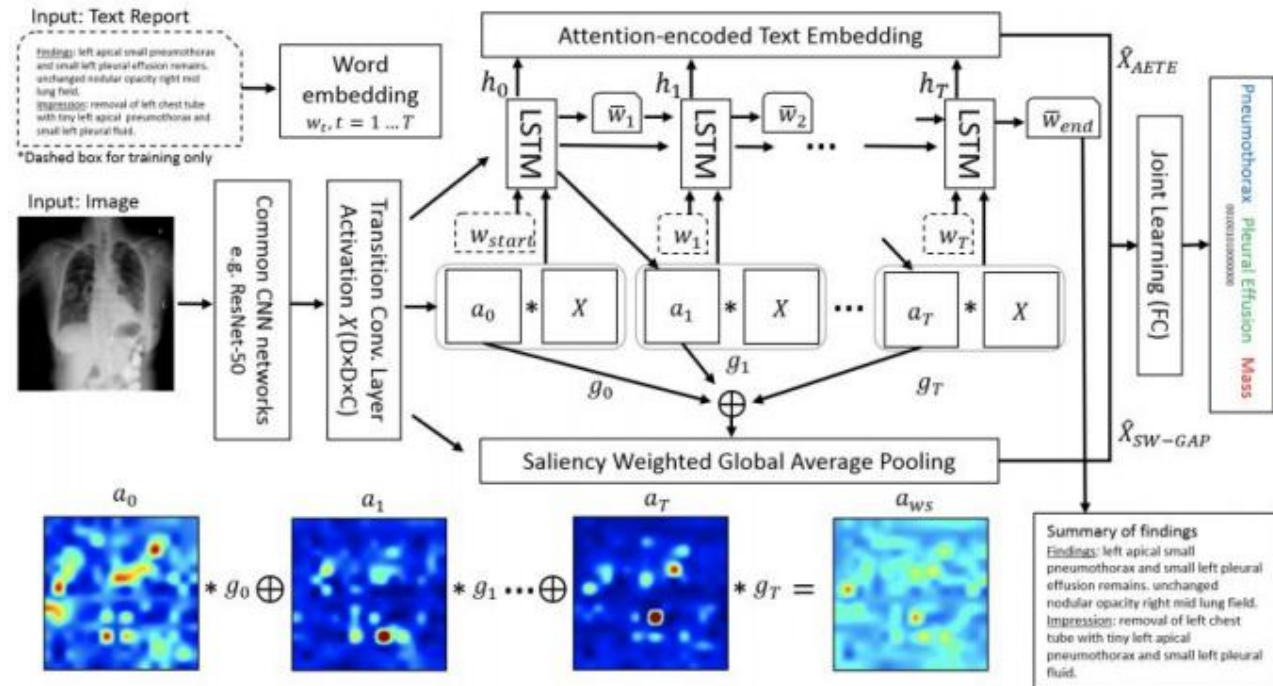
2. Multimodal data



Q: What kind of fusion was this model?

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- Jointly process chest x-rays and associated reports to produce disease labels that can be used to produce auto-annotation disease labels



Wang et al. TieNet: Text-Image Embedding Network for Common Thorax Disease Classification and Reporting in Chest X-rays. CVPR, 2018.

3. Introduction to dermoscopy



Introduction to dermoscopy

3. Introduction to dermoscopy



- General Basics
- Pattern
- Interpretation Algorithm
- Nevus Cell Nevus
- Malignant Melanoma
- Non-Melanoma Skin Cancer
- Basal Cell Carcinoma
- Squamous Cell Carcinoma and Aktinic Keratosis
- Further Skin Changes



3. Introduction to dermoscopy



Dermatoscopy Images



3. Introduction to dermoscopy



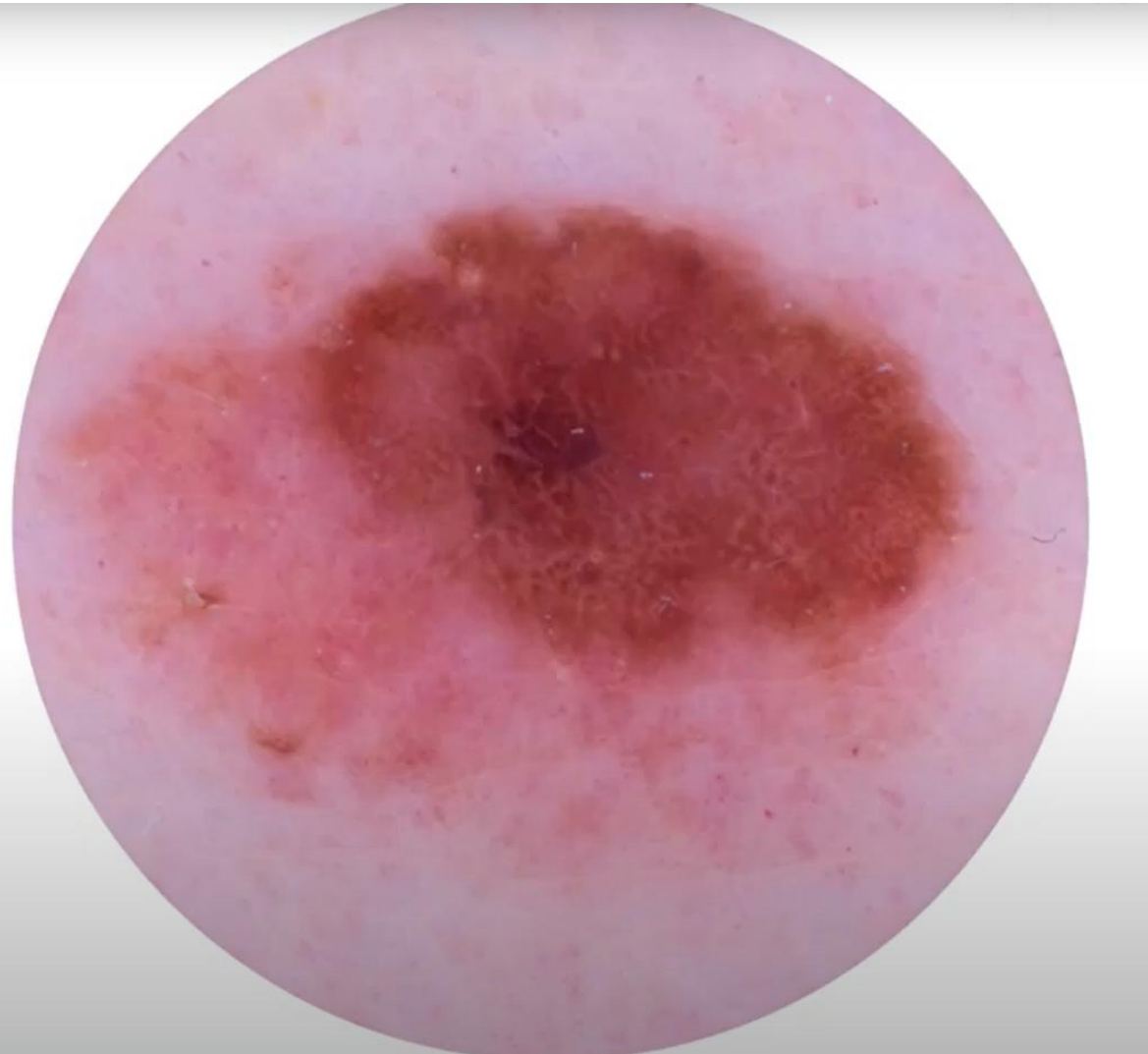
Learning Goals

- Physical basic
- Structure of the skin
- Interpretation of the color

3. Introduction to dermoscopy



Example 1

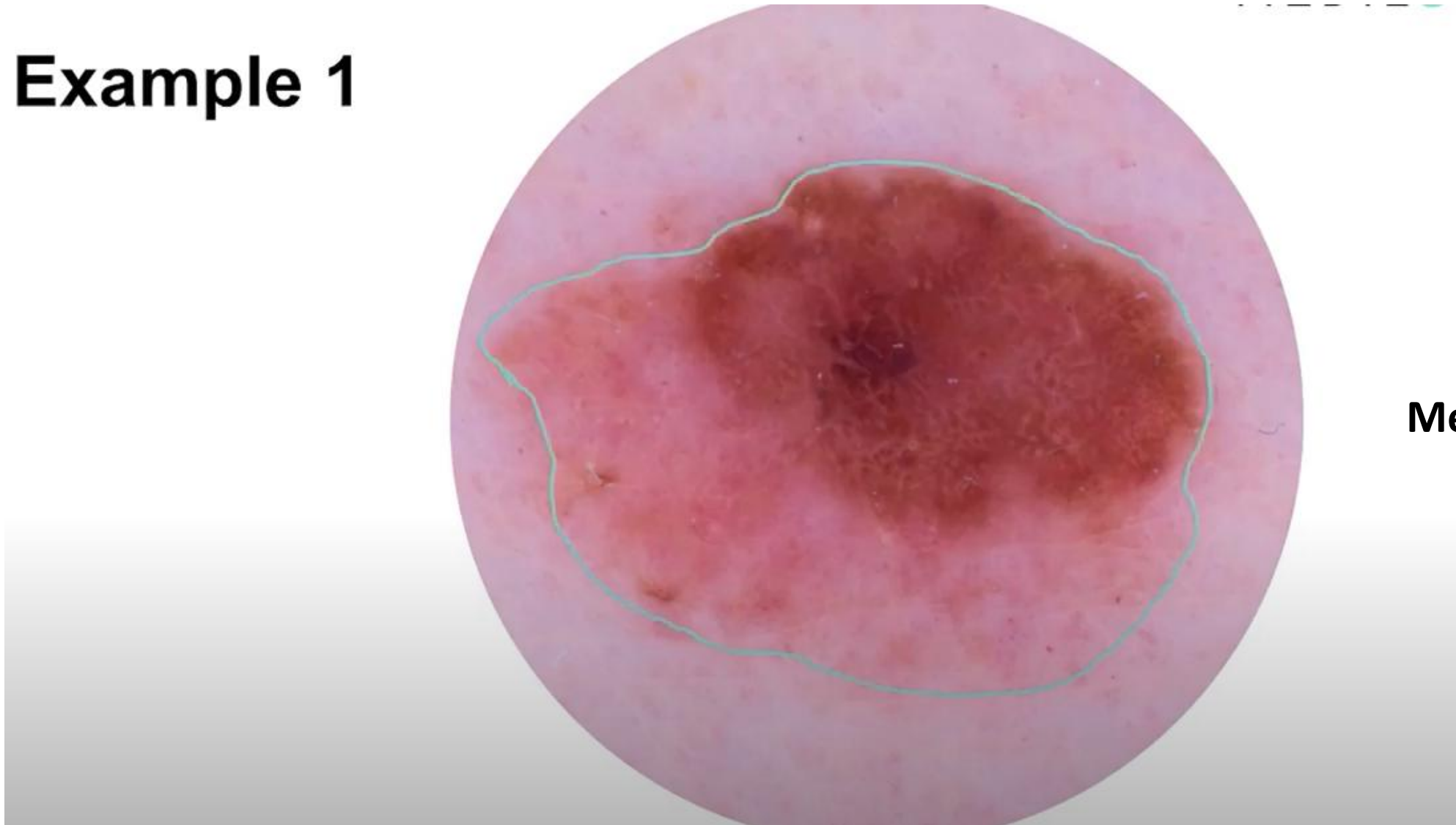


Melanoma

3. Introduction to dermoscopy



Example 1

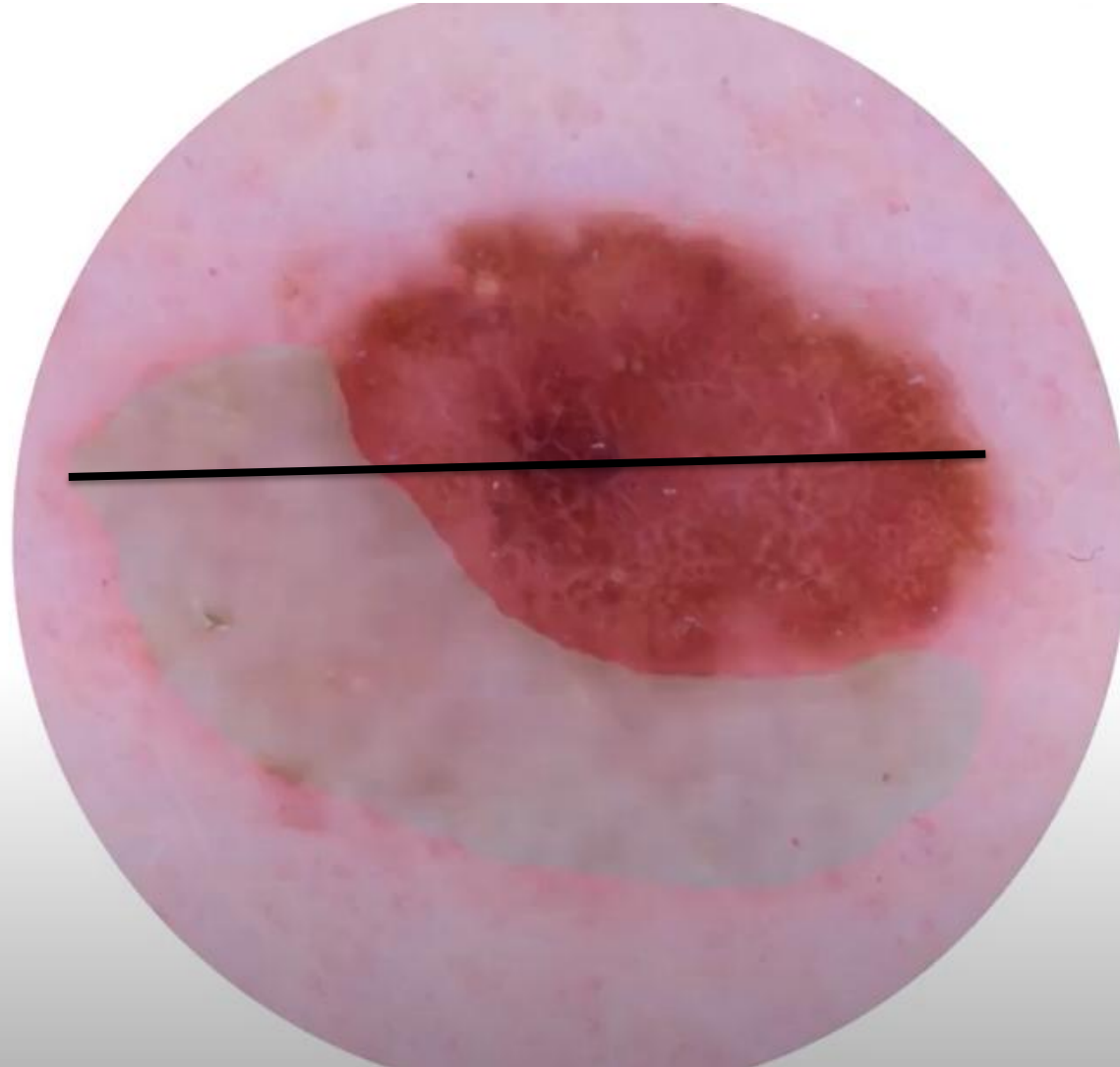


Melanoma

3. Introduction to dermoscopy



Example 1

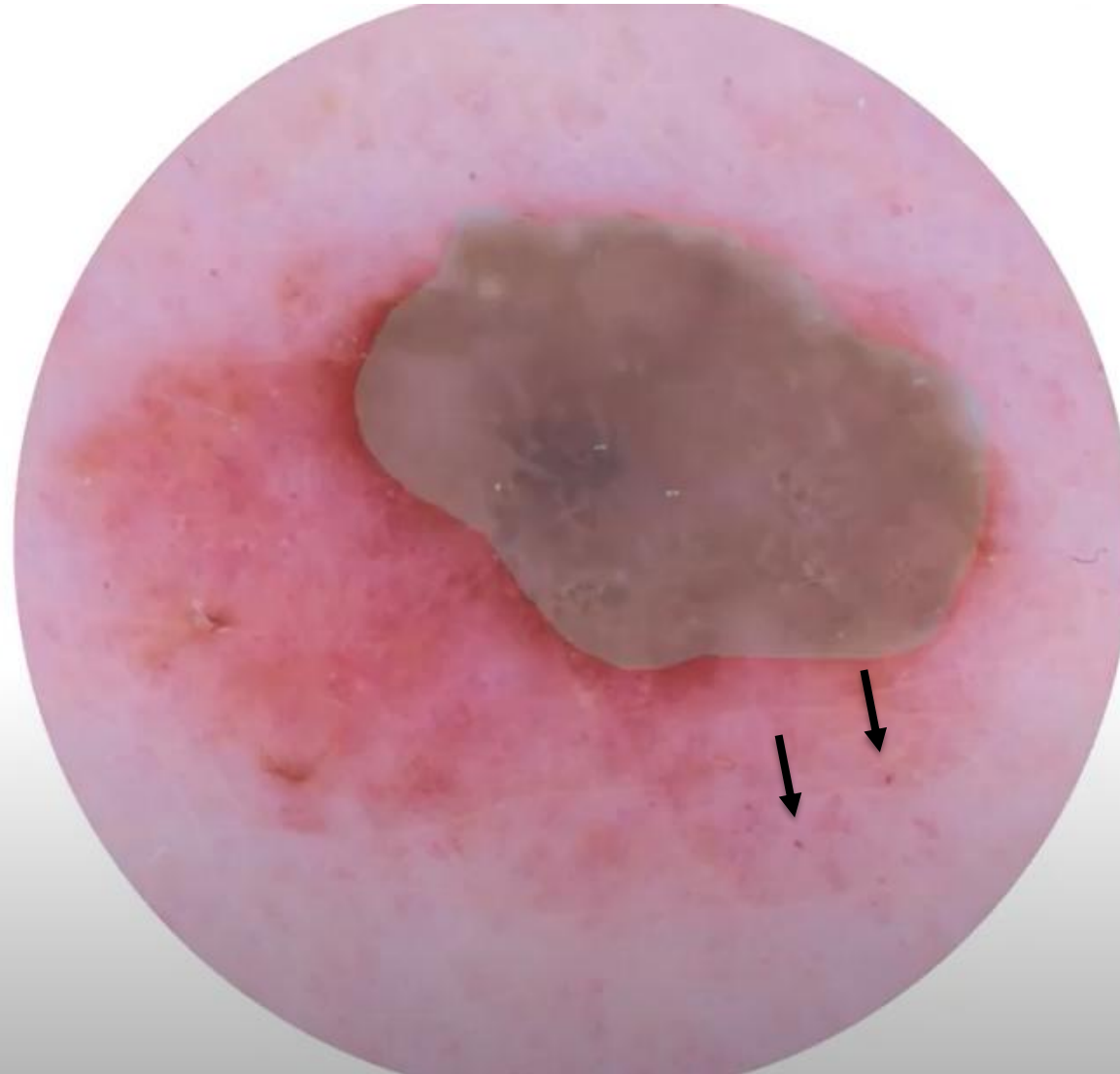


Melanoma

3. Introduction to dermoscopy



Example 1



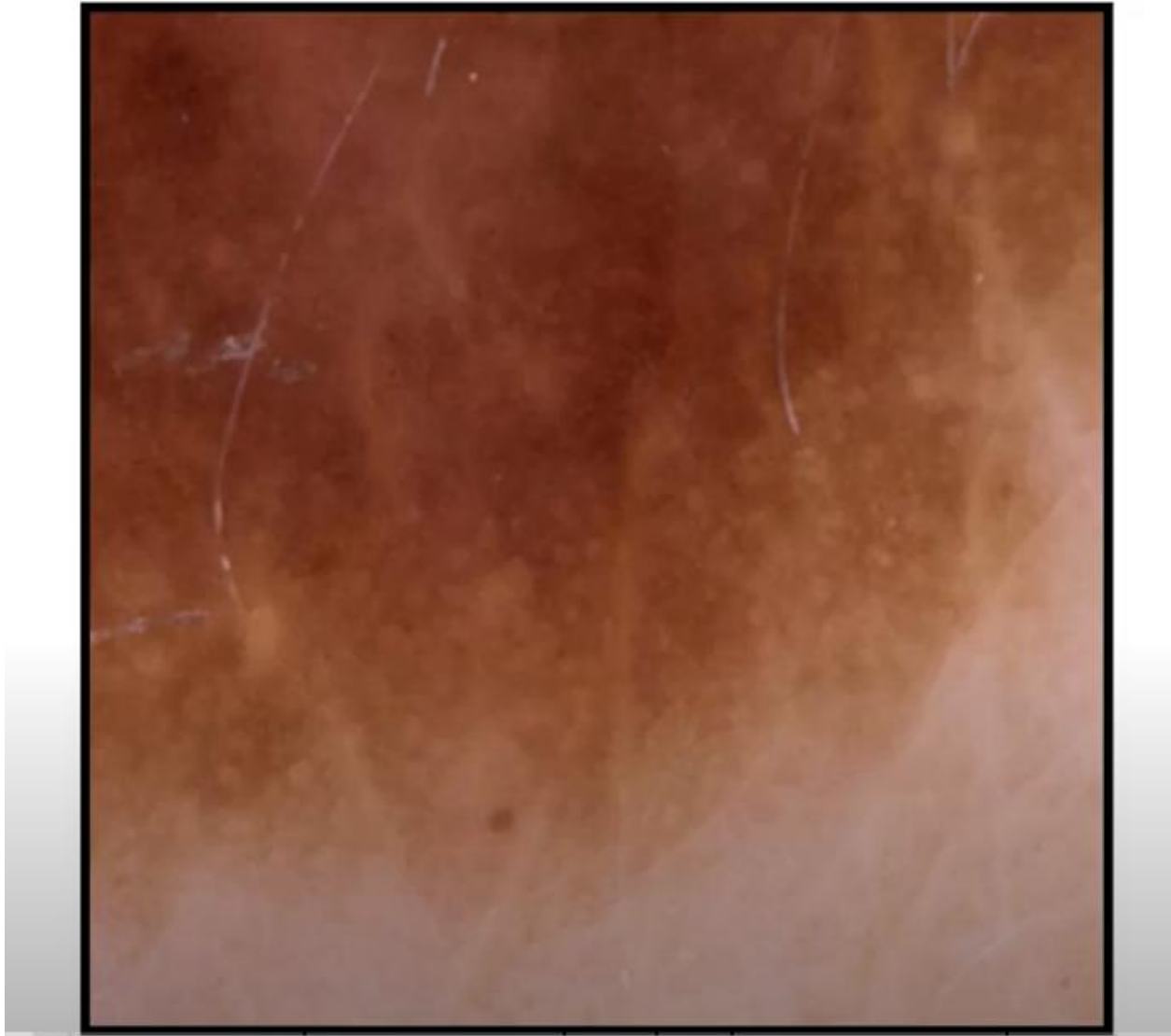
Melanoma

3. Introduction to dermoscopy



nevus cell

3. Introduction to dermoscopy



3. Introduction to dermoscopy



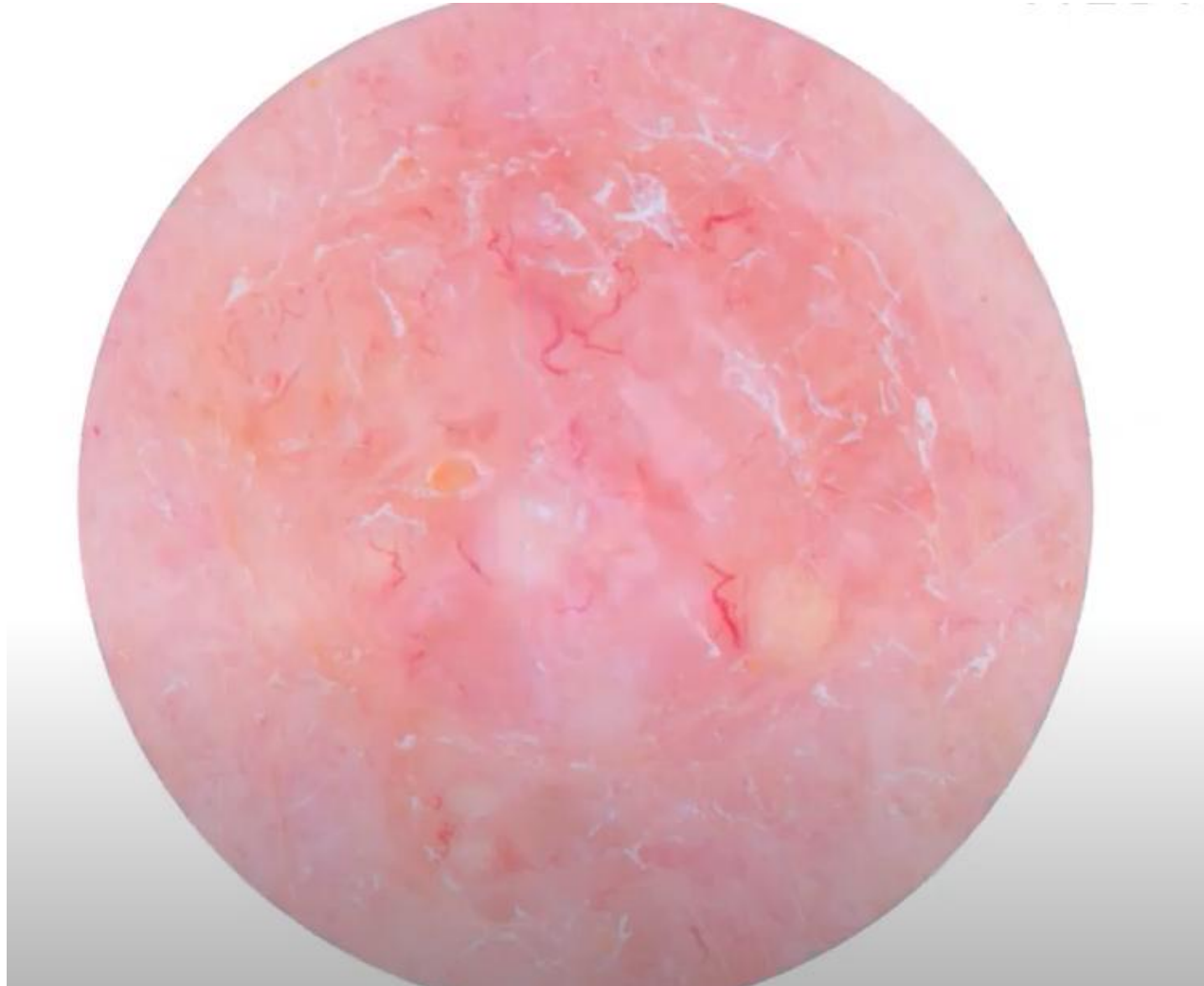
Interpretation Algorithm

Structured image interpretation: ABCD-rule

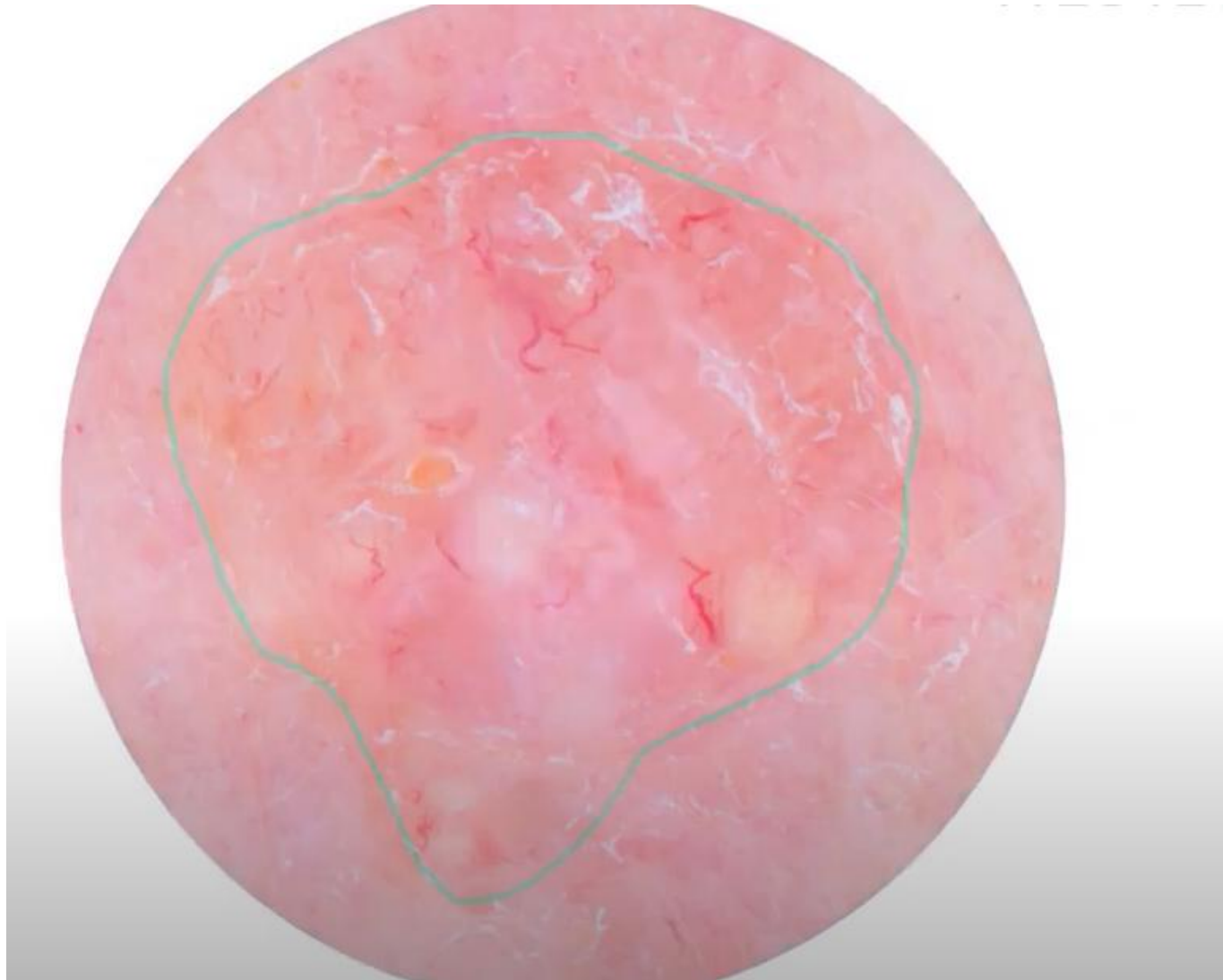
Benign nevus:

- Symmetrical, sharply defined
- Uniform color
- Diameter < 5 mm
- Not raised, similar to other nevi on the body
- Often no vessels available

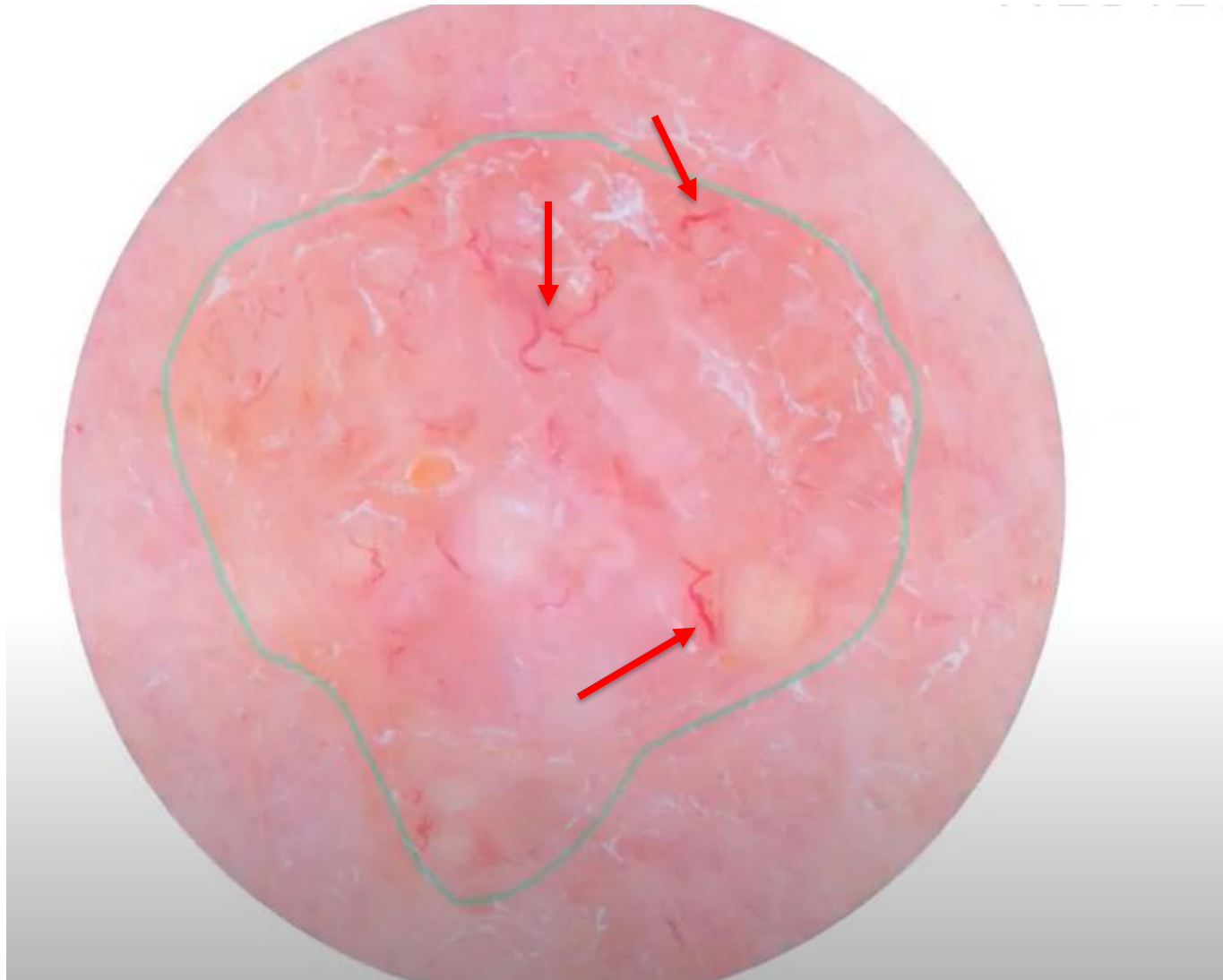
3. Introduction to dermoscopy



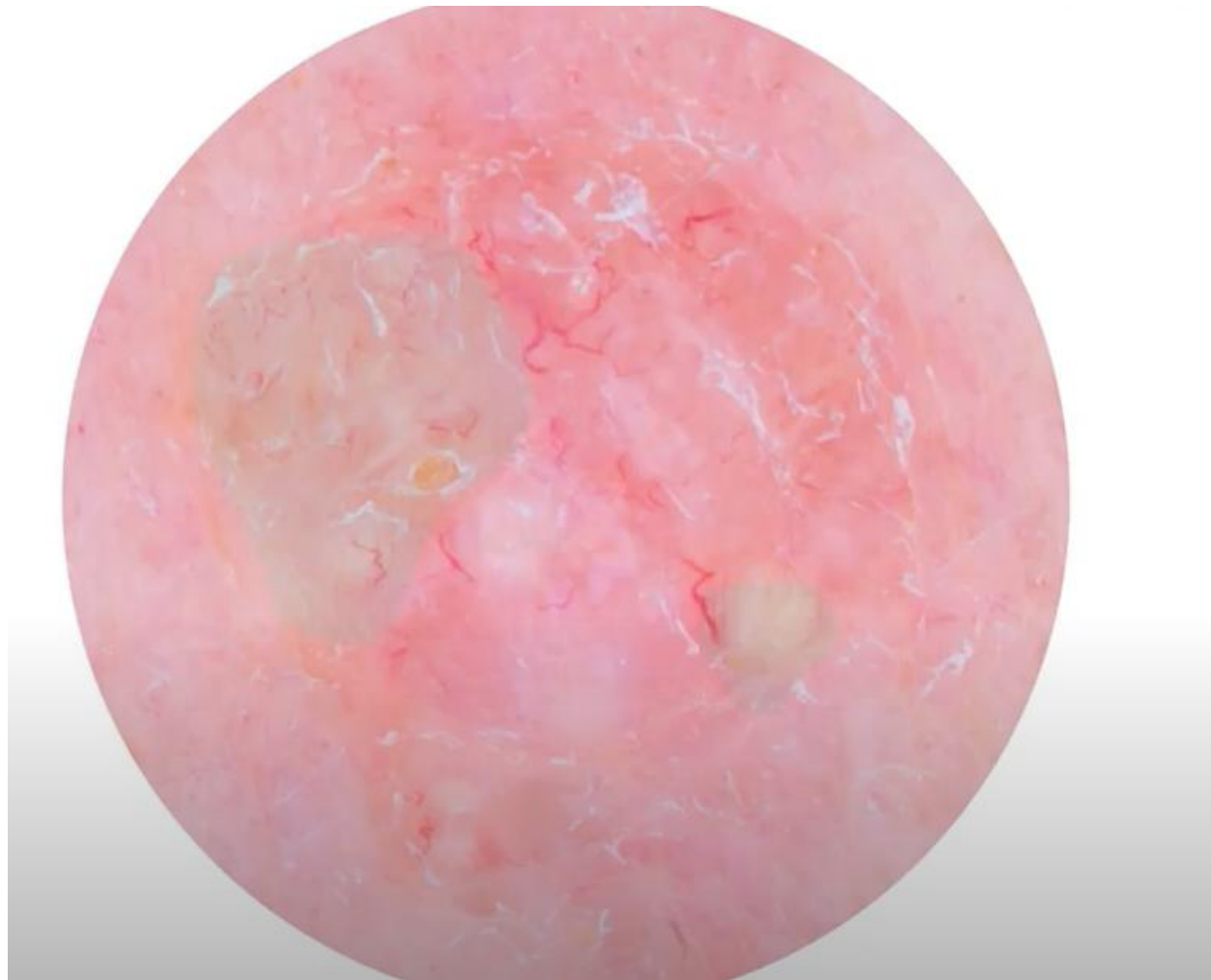
3. Introduction to dermoscopy



3. Introduction to dermoscopy



3. Introduction to dermoscopy

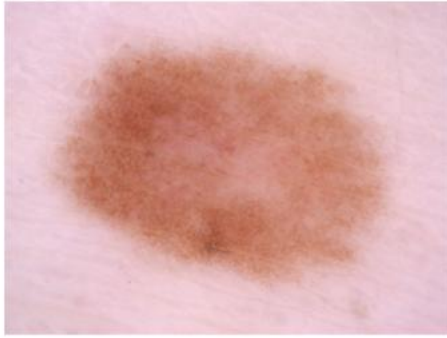


Basal cell carcinoma

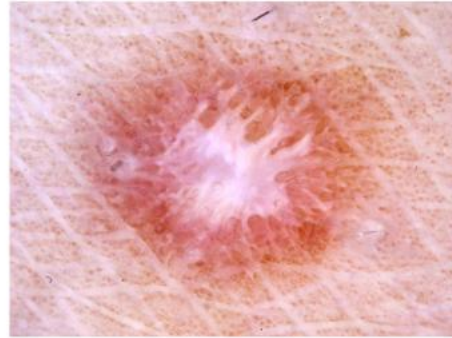
3. Introduction to dermoscopy



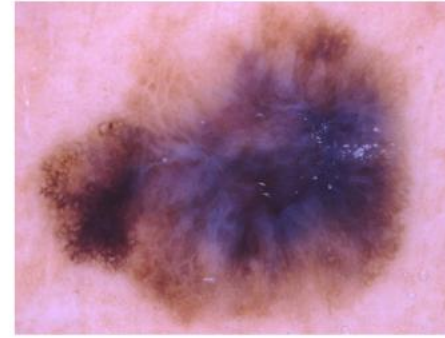
Nevus



Dermatofibroma



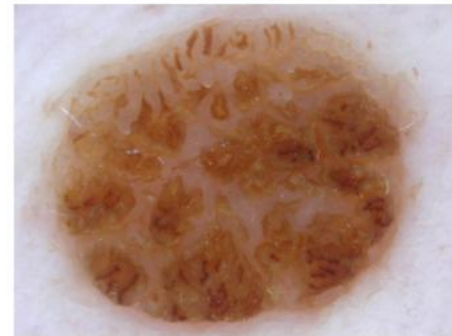
Melanoma



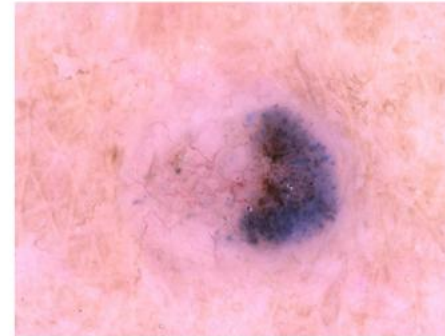
Pigmented
Bowen's



Pigmented Benign
Keratosis



Basal Cell
Carcinoma



Vascular



3. Introduction to dermoscopy



ISICCHALLENGE

Data

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Challenges 

Leaderboards 

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ISIC 2018 Challenge - Task 3: Lesion Diagnosis [Closed]

<https://challenge.isic-archive.com/landing/2018/47/>

4. Deep learning for dermoscopy images



Deep learning applications on dermoscopy images

4. Deep learning for dermoscopy images

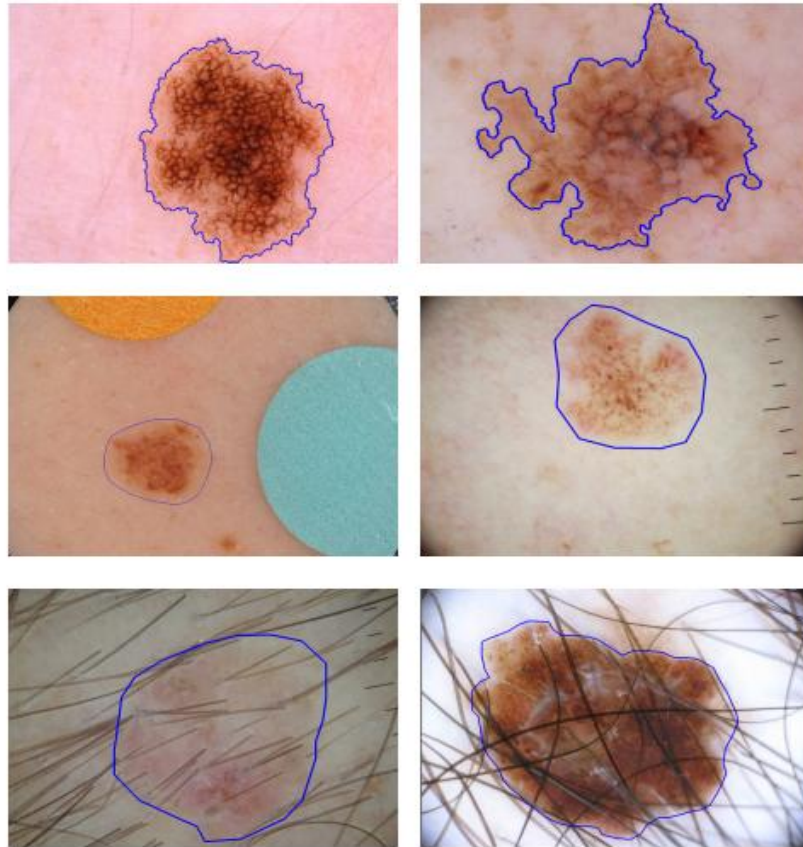


Fig. 1. Automated melanoma recognition from dermoscopy images is a very challenging task. The main challenges include (from [top](#) to [bottom](#)): high degree of visual similarity between melanoma and non-melanoma lesions, relatively low contrast between skin lesions and normal skin regions, and artifacts in images. The left column images show non-melanomas and right column images show melanomas. Blue contours indicate the skin lesions.

Automated Melanoma Recognition in Dermoscopy Images via Very Deep Residual Networks

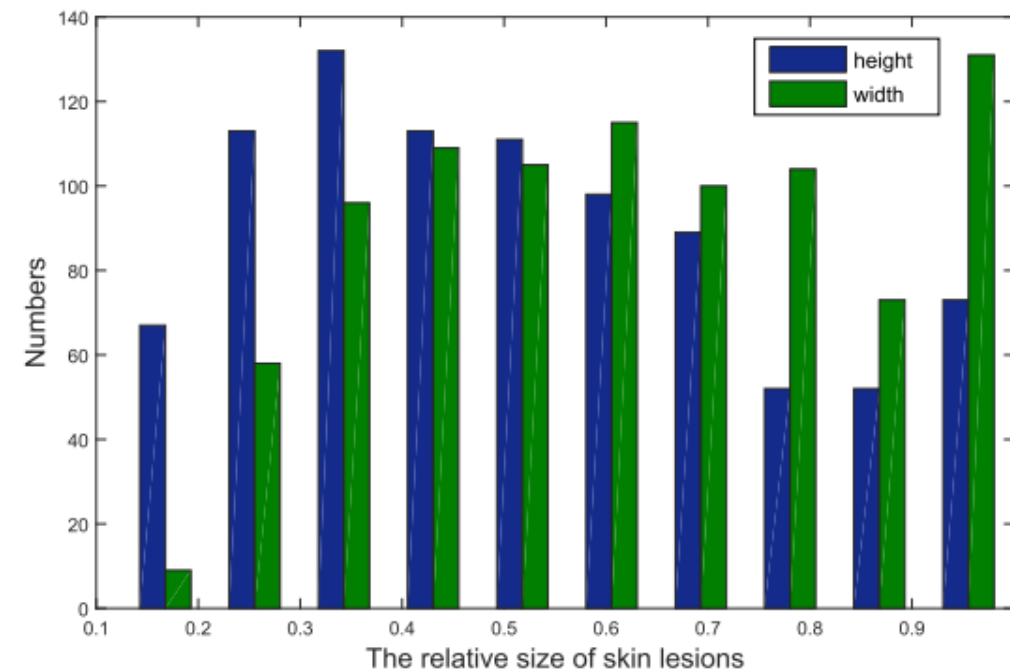


Fig. 3. Distribution of the relative size of skin lesions in our training dermoscopy images.

4. Deep learning for dermoscopy images

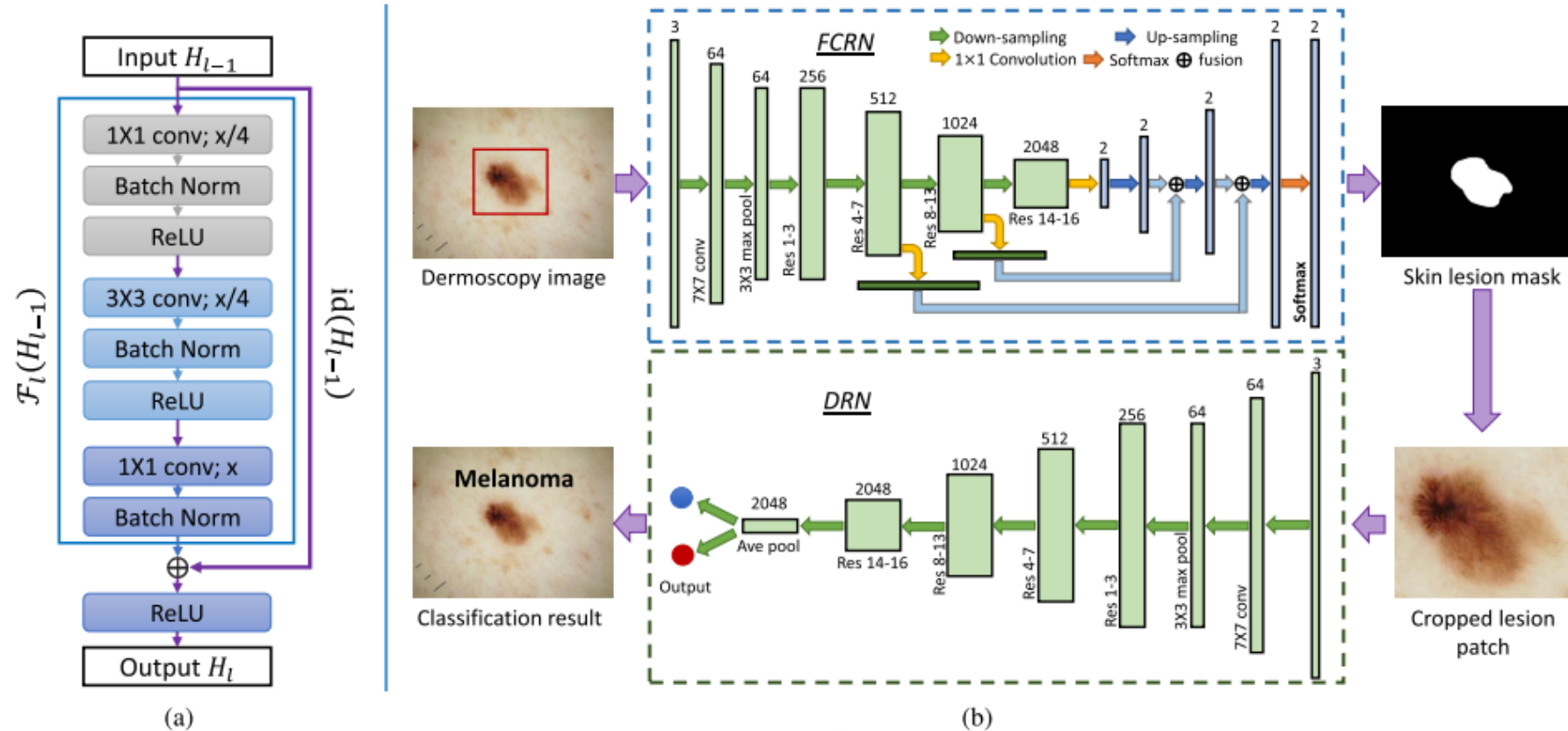


Fig. 2. The flowchart of the proposed framework. (a) Illustration of the ℓ^{th} residual block. x is the number of feature maps of output. We use 1×1 convolutional layers to reduce (and restore) the dimensions. (b) Architectures of fully convolutional residual network (FCRN) for skin lesion segmentation and very deep residual network (DRN) for skin lesion classification. Both of the two networks are trained using the residual learning technique. The numbers above residual blocks represent the number of feature maps. The down-sampling operator is performed by 7×7 convolutional layer (stride is 2), max pooling layer (stride is 2) and convolutional layers (stride is 2) appearing in the first convolutional layers of block 4, 8 and 14.

4. Deep learning for dermoscopy images



TABLE I

ARCHITECTURES OF DOWN-SAMPLING PATH IN
FCRN-38, -50 AND -101

38-layer	50-layer	101-layer
7×7, 64, stride 2		
3×3 max pool, stride 2		
ResBlock 1-3		
ResBlock 4-7	ResBlock 4-7	ResBlock 4-7
ResBlock 8-9	ResBlock 8-13	ResBlock 8-30
ResBlock 10-12	ResBlock 14-16	ResBlock 31-33

TABLE II

COMPARISON OF ARCHITECTURES WITH DIFFERENT DEPTHS

Networks	AC	DI	JA	SE	SP
VGG-16	0.903	0.794	0.707	0.796	0.945
GoogleNet	0.916	0.848	0.776	0.901	0.916
FCRN-38	0.929	0.856	0.785	0.882	0.932
FCRN-50	0.949	0.897	0.829	0.911	0.957
FCRN-101	0.937	0.872	0.803	0.903	0.935

TABLE III

COMPARISON OF FCRNs WITH DIFFERENT SCHEMES OF
MULTI-SCALE CONTEXTUAL FEATURE INTEGRATION

Network	AC	DI	JA	SE	SP
FCRN8	0.949	0.897	0.829	0.911	0.957
FCRN16	0.930	0.855	0.785	0.882	0.934
FCRN32	0.923	0.848	0.773	0.899	0.914

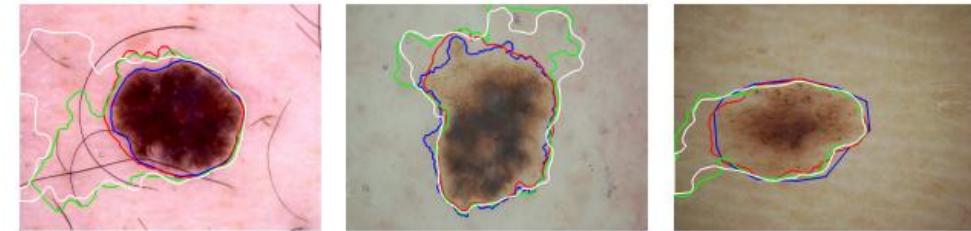


Fig. 4. The segmentation results of FCRN8, FCRN16 and FCRN32. The blue, red, green and white contours indicate the segmentation results of ground truth, FCRN8, FCRN16 and FCRN32, respectively.

4. Deep learning for dermoscopy images



TABLE IV

RESULTS OF SKIN LESION SEGMENTATION CHALLENGE ON ISBI 2016

Method	AC	DI	JA	SE	SP
EXB	0.953	0.910	0.843	0.910	0.965
CUMED (ours)	0.949	0.897	0.829	0.911	0.957
Mahmudur	0.952	0.895	0.822	0.880	0.969
SFU-mial	0.944	0.885	0.811	0.915	0.955
TMUteam	0.946	0.888	0.810	0.832	0.987
UiT-Seg	0.939	0.881	0.806	0.863	0.974
IHPC-CS	0.938	0.879	0.799	0.910	0.941
UNIST	0.940	0.867	0.797	0.876	0.954
Jose Luis	0.934	0.869	0.791	0.870	0.978
Marco romelli	0.936	0.864	0.786	0.883	0.962

Note: There are totally 28 submissions. The top 10 entries are shown here and the ranking (from top to bottom) was made according to the JA.

TABLE V

RESULTS OF CLASSIFICATION WITH AND WITHOUT SEGMENTATION

Method	AC	AUC	AP	SE	SP
Without segmentation	0.828	0.782	0.560	0.427	0.927
With segmentation	0.855	0.783	0.624	0.547	0.931

TABLE VI

COMPARISON OF DIFFERENT NETWORKS ON THE CLASSIFICATION TASK

Method	AC	AUC	AP	SE	SP
VGG-16	0.826	0.826	0.529	0.413	0.928
GoogleNet	0.847	0.801	0.581	0.507	0.931
DRN-50	0.855	0.783	0.624	0.547	0.931

4. Deep learning for dermoscopy images

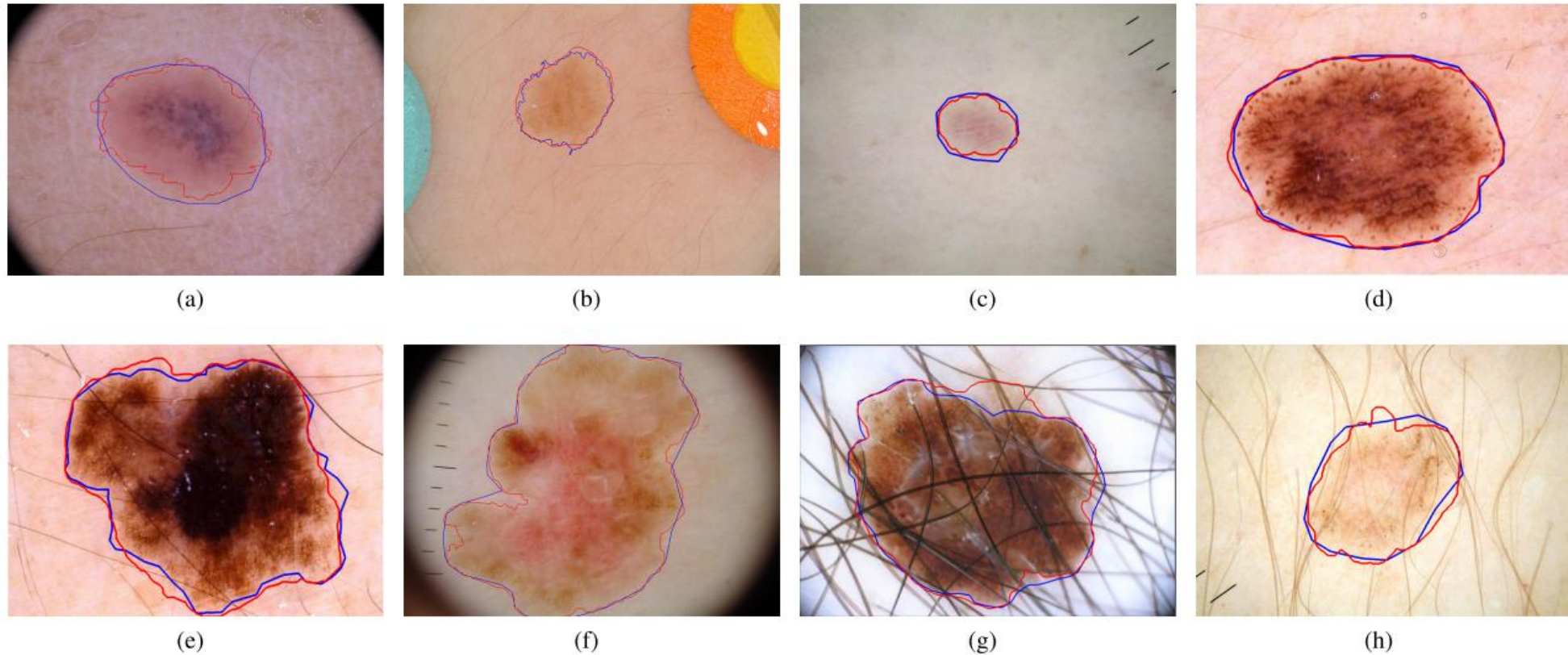


Fig. 5. Examples of some skin lesion segmentation results from test images. The first and second rows are non-melanoma and melanoma lesions, respectively. The red and blue contours indicate the segmentation results of our method and ground truth, respectively.

4. Deep learning for dermoscopy images



TABLE VII

RESULTS OF SKIN LESION CLASSIFICATION FOR DIFFERENT CLASSIFIERS

Method	AC	AUC	AP	SE	SP
Softmax	0.850	0.790	0.609	0.507	0.934
SVM	0.844	0.779	0.616	0.520	0.824
Fusion	0.855	0.783	0.624	0.547	0.931

TABLE VIII

RESULTS OF SKIN LESION CLASSIFICATION CHALLENGE ON ISBI 2016

Method	AC	AUC	AP	SE	SP
CUMED (ours)	0.855	0.804	0.637	0.507	0.941
GTDL	0.813	0.802	0.619	0.573	0.872
BF-TB	0.834	0.826	0.598	0.320	0.961
ThrunLab	0.786	0.796	0.563	0.667	0.816
Jordan Yap	0.844	0.775	0.559	0.240	0.993
Haebeom Lee	0.821	0.793	0.555	0.200	0.974
GT-DL1	0.815	0.813	0.552	0.467	0.901
GT-DL2	0.681	0.800	0.545	0.787	0.655
Sebastien PARIS	0.731	0.793	0.542	0.773	0.720
USYD-BMIT	0.599	0.780	0.537	0.853	0.536

Note: There are totally 25 submissions. The top 10 entries are shown here and the ranking was made according to the AP scores.

4. Deep learning for dermoscopy images

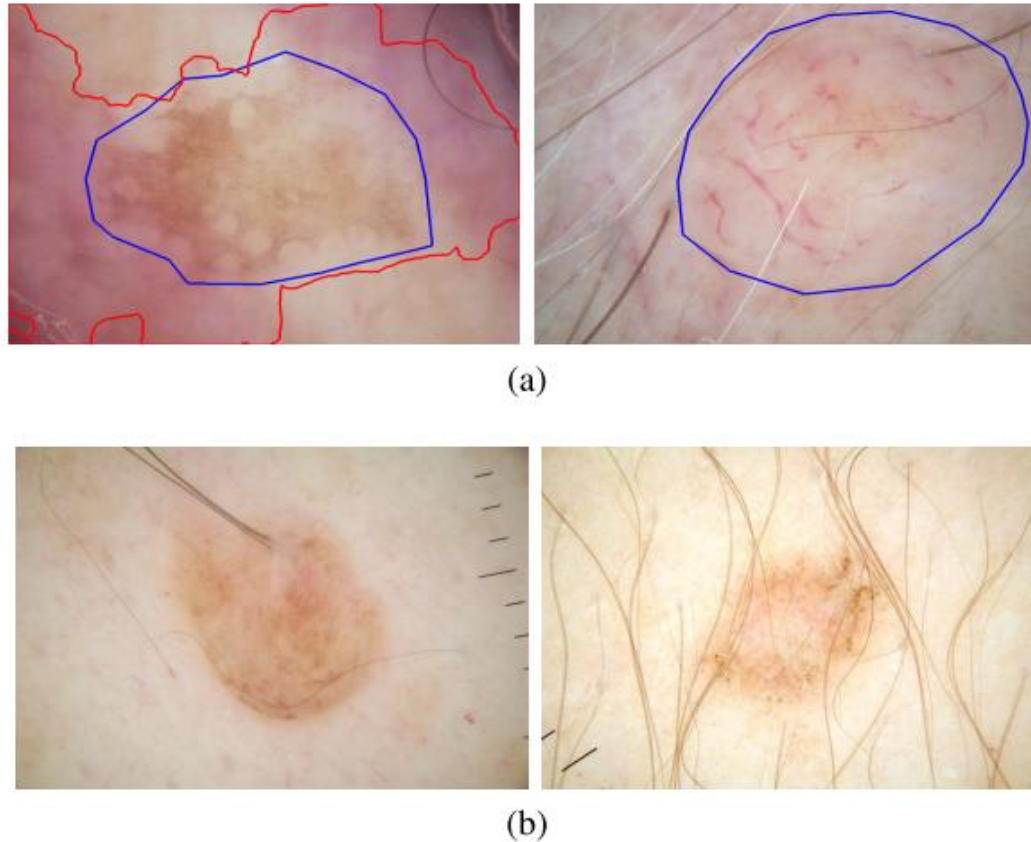


Fig. 6. Some failure cases of our framework: (a) failure cases of segmentation, where red and blue contours indicate the segmentation results of our method and the ground truth, respectively, and (b) melanoma lesions neglected by our framework.

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

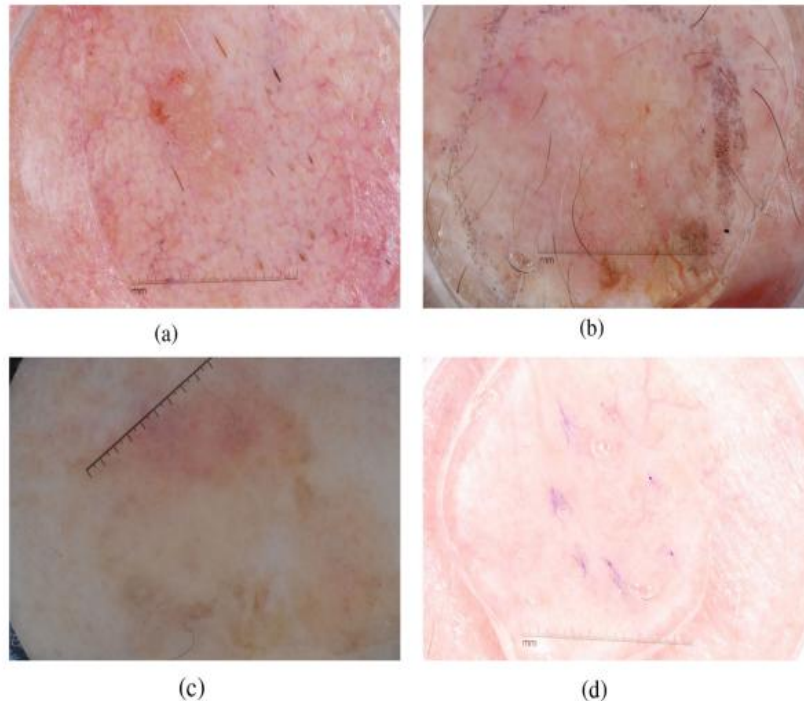


Fig. 1. Illustrative image examples to show difficulties in dermoscopy image analysis. (a) hard to identify with veins; (b) the presence of artifacts; (c) obscure and irregular boundaries; and (d) low contrast in visual impression.

TABLE I

AN OVERVIEW OF SKIN LESION ANALYSIS IN DERMOSCOPY IMAGES AMONG DIFFERENT DEEP LEARNING METHODS

Methods	Task	End-to-end	Loss Function	Model
Esteva 2017 [16]	classification(cls)	YES	cross entropy(CE)	Inception v3
Yu 2017 [17]	cls	NO	CE	FCRN+DRN
Lopez 2017 [20]	cls	NO	CE	VGG16
Vasconcelos 2017 [21]	cls	NO	CE	GoogleNet
Yu Z 2017 [22]	cls	NO	CE	AlexNet+SVM
Li 2018 [18]	segmentation(seg)	YES	CE	ResNet34
Venkatesh 2018 [19]	seg	YES	CE	UNET
Yuan Y 2017 [23]	seg	NO	Jaccard Loss(JL)	CDNN29
Matt Berseth 2017 [24]	seg	NO	CE	UNET
Bi L 2017 [25]	seg	YES	CE	ResNet+FCN
Vesal 2018 [26]	seg	NO	CE	Faster-RCNN
Menegola A 2017 [27]	seg	NO	Dice Coefficient	VGG16
Yang 2017 [28]	cls+seg	NO	CE	GoogleNet
Chen 2018 [29]	cls+seg	NO	CE	ResNet101
Ours	cls+detection+seg	YES	Focal Loss+JL	FPN+RPN+FCN

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

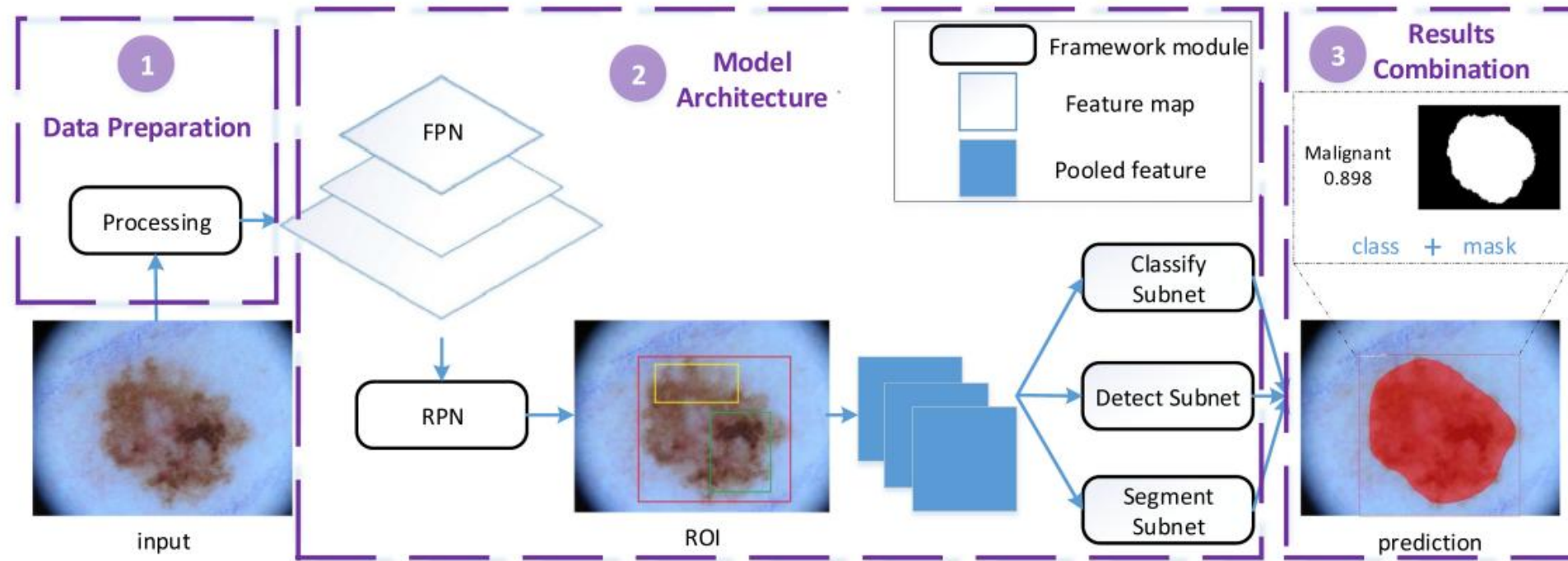


Fig. 2. Flowchart of the proposed skin lesion analysis framework. The dermoscopy images are processed by data preparation and model architecture in sequence. In results combination, predictions of three convolution subnets are fused to produce final results.

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

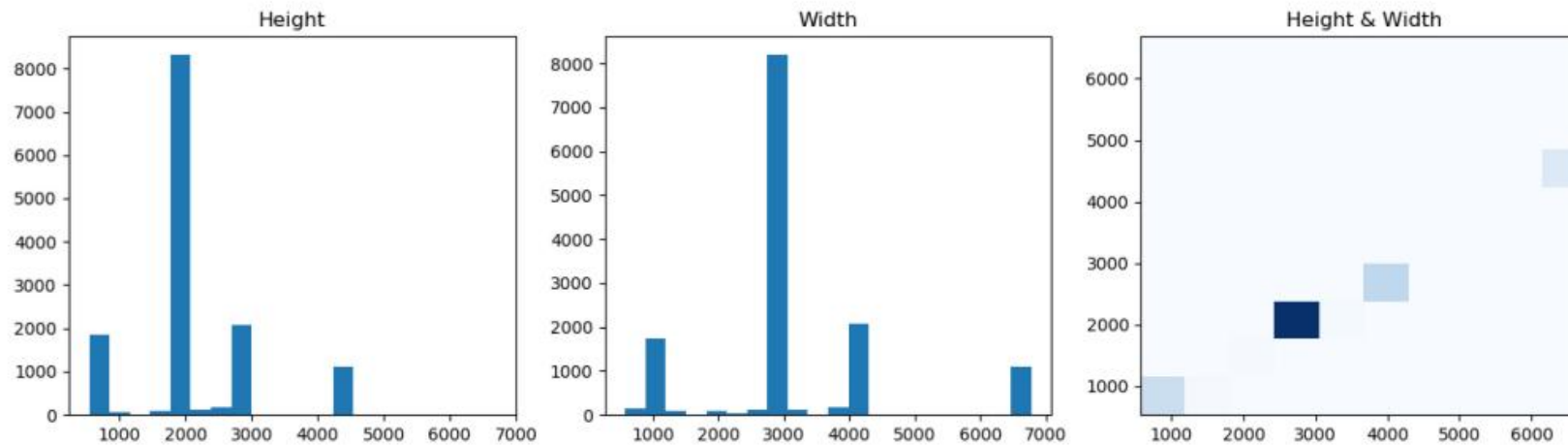


Fig. 3. Statistical information of dermoscopy images regarding the image size. For the left and medium charts, the horizontal axis denotes the number of pixels and the vertical axis stands for occurrence frequency. For the right chart, horizontal and vertical axes represent the width and the height (pixel sizes) of the images respectively, and a darker color means a higher frequency.

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

TABLE II
TRAINING DATASET: STATISTICAL INFORMATION ABOUT DERMOSCOPY IMAGES IN IMAGE SIZE AND COLOR

	mean	median	min	max
#Height	2163.18	2016.00	540.00	6688.00
#Width	3220.65	3024.00	576.00	6780.00
#Color(R,G,B)	(175.65,147.35,129.03)	(174.54, 147.22, 128.17)	(49.74, 27.59, 23.12)	(254.15, 236.11, 230.88)

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

1) *RPN Loss Function Based on the Focal Loss*: RPN classifies the foreground from all anchors and does the bounding box regression to obtain a precise location. Hence, its loss function should include classification and regression losses, and we employ the following loss function:

$$L_{rpn}(p_i, t_i) = \frac{1}{N_{cls}} \sum_i L_{cls}(p_i, p_i^*) + \rho \frac{1}{N_{reg}} \sum_i p_i^* \cdot L_{reg}(t_i, t_i^*). \quad (1)$$

where p_i and t_i stand for the prediction probability of anchor i being foreground and its estimated coordinates respectively, while p_i^* and t_i^* denote the corresponding ground truth information, and N_{cls} and N_{reg} are the number of anchors used for computing the classification loss $L_{cls}(p_i, p_i^*)$ and the regression loss $L_{reg}(t_i, t_i^*)$ respectively. ρ is a balancing parameter to make L_{cls} and L_{reg} terms roughly equally weighted, and we set $\rho = 10$ in our experiments.

Focal loss

Focal Loss for Dense Object Detection

<https://arxiv.org/pdf/1708.02002.pdf>

4. Deep learning for dermoscopy images



Focal Loss for Dense Object Detection <https://arxiv.org/pdf/1708.02002.pdf>

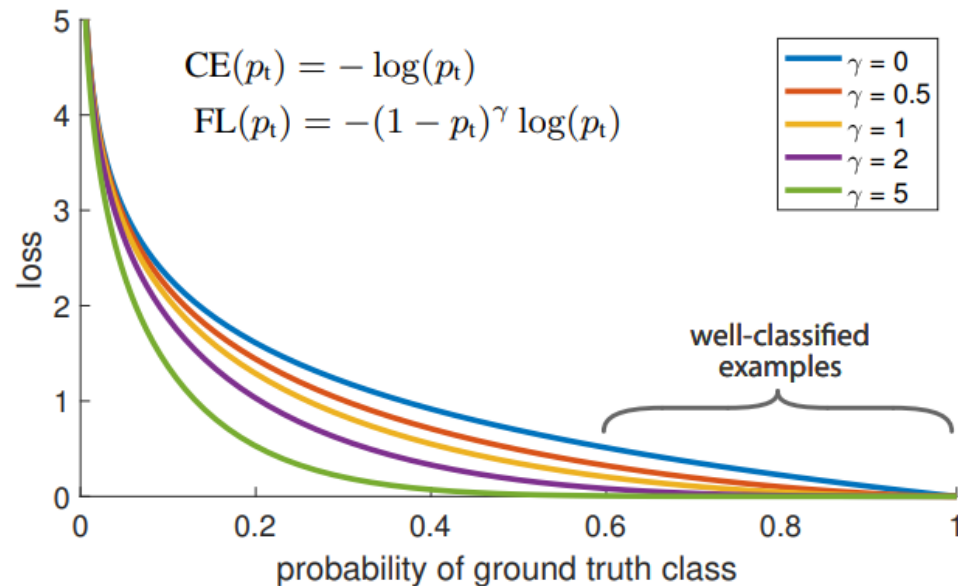


Figure 1. We propose a novel loss we term the *Focal Loss* that adds a factor $(1 - p_t)^\gamma$ to the standard cross entropy criterion. Setting $\gamma > 0$ reduces the relative loss for well-classified examples ($p_t > .5$), putting more focus on hard, misclassified examples. As our experiments will demonstrate, the proposed focal loss enables training highly accurate dense object detectors in the presence of vast numbers of easy background examples.

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

1) *RPN Loss Function Based on the Focal Loss*: RPN classifies the foreground from all anchors and does the bounding box regression to obtain a precise location. Hence, its loss function should include classification and regression losses, and we employ the following loss function:

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Focal loss

TABLE III

ISIC 2017 TEST DATASET: CLASSIFICATION RESULTS FROM DIFFERENT LOSS FUNCTIONS (WITH HYPERPARAMETERS IMAGENET, RESNET101, 1×2)

Type	AC	AUC	SE	SP	AP
Original	0.788	0.754	0.991	0.573	0.796
Dice Loss(DL)	0.790	0.765	0.990	0.664	0.787
Focal Loss(FL)	0.801	0.749	0.986	0.730	0.804
Jaccard Loss(JL)	0.793	0.783	0.991	0.647	0.815
FL+JL	0.813	0.794	0.977	0.731	0.823
FL+JL+DL	0.806	0.790	0.979	0.727	0.818

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

TABLE IV

ISIC 2017 TEST DATASET: CLASSIFICATION RESULTS WITH DIFFERENT NETWORK HYPERPARAMETERS (WITH THE FOCAL LOSS + JAC LOSS)

Initialization	Backbone	Batchsize	AC	AUC	SE	SP	AP
#COCO	ResNet50	1×2	0.851	0.776	0.986	0.802	0.820
#		2×2	0.905	0.708	0.991	0.880	0.806
#COCO	ResNet101	1×2	0.838	0.731	0.981	0.968	0.825
#		2×2	0.918	0.696	0.978	0.893	0.804
#IMAGENET	ResNet50	1×2	0.859	0.769	0.988	0.821	0.803
#		2×2	0.905	0.610	0.987	0.915	0.809
#IMAGENET	ResNet101	1×2	0.813	0.794	0.977	0.731	0.823
#		2×2	0.868	0.760	0.984	0.880	0.816

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

TABLE V

ISBI 2016 TEST DATASET: CLASSIFICATION PERFORMANCE COMPARISONS
WITH STATE-OF-THE-ART METHODS

Methods	AC	AUC	SE	SP	AP
Ours	0.891	0.805	0.996	0.723	0.815
Ours(ONLY ISBI2016)	0.880	0.796	0.994	0.657	0.782
CUMED	0.855	0.783	0.931	0.547	0.624
Jordan Yap	0.844	0.775	0.993	0.240	0.559
BF-TB	0.834	0.826	0.961	0.320	0.598
Haebeom Lee	0.821	0.793	0.974	0.200	0.555
GTDL	0.813	0.802	0.872	0.573	0.619

TABLE VI

ISIC 2017 TEST DATASET: SEGMENTATION RESULTS FROM DIFFERENT
LOSS FUNCTIONS (WITH HYPERPARAMETERS IMAGENET,
RESNET101, 1×2)

Type	JA	DI	AC	SE	SP
Original	0.789	0.868	0.952	0.829	0.983
Dice Loss(DL)	0.794	0.873	0.948	0.830	0.972
Focal Loss(FL)	0.805	0.880	0.950	0.846	0.983
Jaccard Loss(JL)	0.813	0.889	0.946	0.861	0.976
FL+JL	0.799	0.878	0.959	0.831	0.986
FL+JL+DL	0.796	0.875	0.958	0.833	0.980

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

TABLE VII

ISIC 2017 TEST DATASET: SEGMENTATION RESULTS WITH DIFFERENT NETWORK HYPERPARAMETERS (WITH THE FOCAL LOSS + JAC LOSS)

Initialization	Backbone	Batchsize	JA	DI	AC	SE	SP
#COCO	ResNet50	1×2	0.814	0.887	0.956	0.850	0.985
#		2×2	0.843	0.906	0.958	0.879	0.984
#COCO	ResNet101	1×2	0.823	0.895	0.955	0.863	0.981
#		2×2	0.849	0.911	0.956	0.888	0.985
#IMAGENET	ResNet50	1×2	0.818	0.894	0.965	0.848	0.986
#		2×2	0.836	0.905	0.968	0.862	0.985
#IMAGENET	ResNet101	1×2	0.799	0.878	0.959	0.831	0.986
#		2×2	0.828	0.899	0.959	0.862	0.983

4. Deep learning for dermoscopy images



An End-to-End Multi-Task Deep Learning Framework for Skin Lesion Analysis

TABLE VIII

ISIC 2017 TEST DATASET: SEGMENTATION PERFORMANCE COMPARISONS
WITH STATE-OF-THE-ART METHODS

Methods	JA	DI	AC	SE	SP
Ours	0.849	0.911	0.956	0.888	0.985
Ours(ONLY ISIC2017)	0.816	0.883	0.950	0.871	0.982
Venkatesh [19]	0.764	0.856	0.936	0.830	0.976
Xiaomeng Li [18]	0.772	0.856	0.936	0.854	0.972
Yuan Y [23]	0.765	0.849	0.934	0.825	0.975
Matt Berseth [24]	0.762	0.847	0.932	0.820	0.978
Bi L [25]	0.760	0.844	0.934	0.802	0.985
Menegola A [27]	0.754	0.839	0.931	0.817	0.970

TABLE X

ISIC 2017 TEST DATASET: MULTI-TASK ABLATION INVESTIGATION FOR
SEGMENTATION RESULTS (WITH HYPERPARAMETERS IMAGENET,
RESNET101, 1×2 , THE FOCAL LOSS + JAC LOSS)

Type	JA	DI	AC	SE	SP
Seg	0.760	0.827	0.930	0.829	0.964
Cls+Seg	0.782	0.850	0.948	0.823	0.975
Cls+Dec+Seg	0.799	0.878	0.959	0.831	0.986

TABLE IX

ISIC 2017 TEST DATASET: MULTI-TASK ABLATION INVESTIGATION FOR
CLASSIFICATION RESULTS (WITH HYPERPARAMETERS IMAGENET,
RESNET101, 1×2 , THE FOCAL LOSS + JAC LOSS)

Type	AC	AUC	SE	SP	AP
Cls	0.763	0.752	0.968	0.705	0.787
Cls+Seg	0.798	0.786	0.977	0.720	0.815
Cls+Dec+Seg	0.813	0.794	0.977	0.731	0.823

4. Deep learning for dermoscopy images



Deeply Supervised Rotation Equivariant Network for Lesion Segmentation in Dermoscopy Images

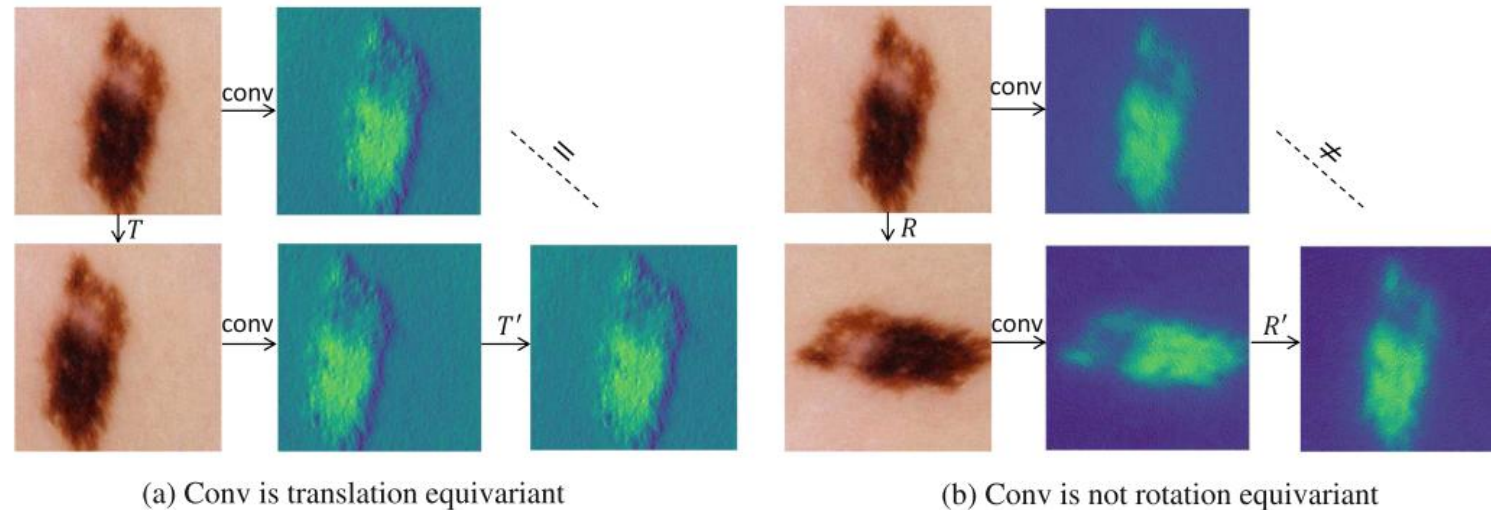


Fig. 1. Convolution layer is translation equivariant (a); but convolution is not rotation equivariant (Zoom in to see the detailed comparison), as shown in (b).

4. Deep learning for dermoscopy images



Deeply Supervised Rotation Equivariant Network for Lesion Segmentation in Dermoscopy Images

2.1 G -convolution

Convolutional kernel

The regular first convolution layer is a function that maps the input to feature maps with K channels $f : \mathbb{Z}^2 \rightarrow \mathbb{R}^K$. The function can be described as Eq. 1.

$$[f * \varphi](x) = \sum_{y \in \mathbb{Z}^2} \sum_k f_k(y) \varphi_k(x - y), \quad (1)$$

where φ_k denotes the convolution kernel.

To encode rotation equivariance in the network, Cohen et al. [6] proposed to conduct convolution on groups, where the group $p4$ consists of all compositions of translations and rotations by 90° about any center of rotation in the grid, and

4. Deep learning for dermoscopy images



Deeply Supervised Rotation Equivariant Network for Lesion Segmentation in Dermoscopy Images

the group $p4m$ additionally includes reflections. Specifically, for the input layer, the $(\mathbb{Z}^2 \rightarrow G)$ convolution is defined as

$$[f * \varphi](g) = \sum_{y \in \mathbb{Z}^2} \sum_k f_k(y) \varphi_k(g^{-1}y), \quad (2)$$

where g is a transformation in the predefined group $p4$ or $p4m$. Then, in the following layers, feature maps and filters are both functions on G and the $(G \rightarrow G)$ convolution can be described as

$$[f * \varphi](g) = \sum_{h \in G} \sum_k f_k(h) \varphi_k(g^{-1}h) \quad (3)$$

4. Deep learning for dermoscopy images

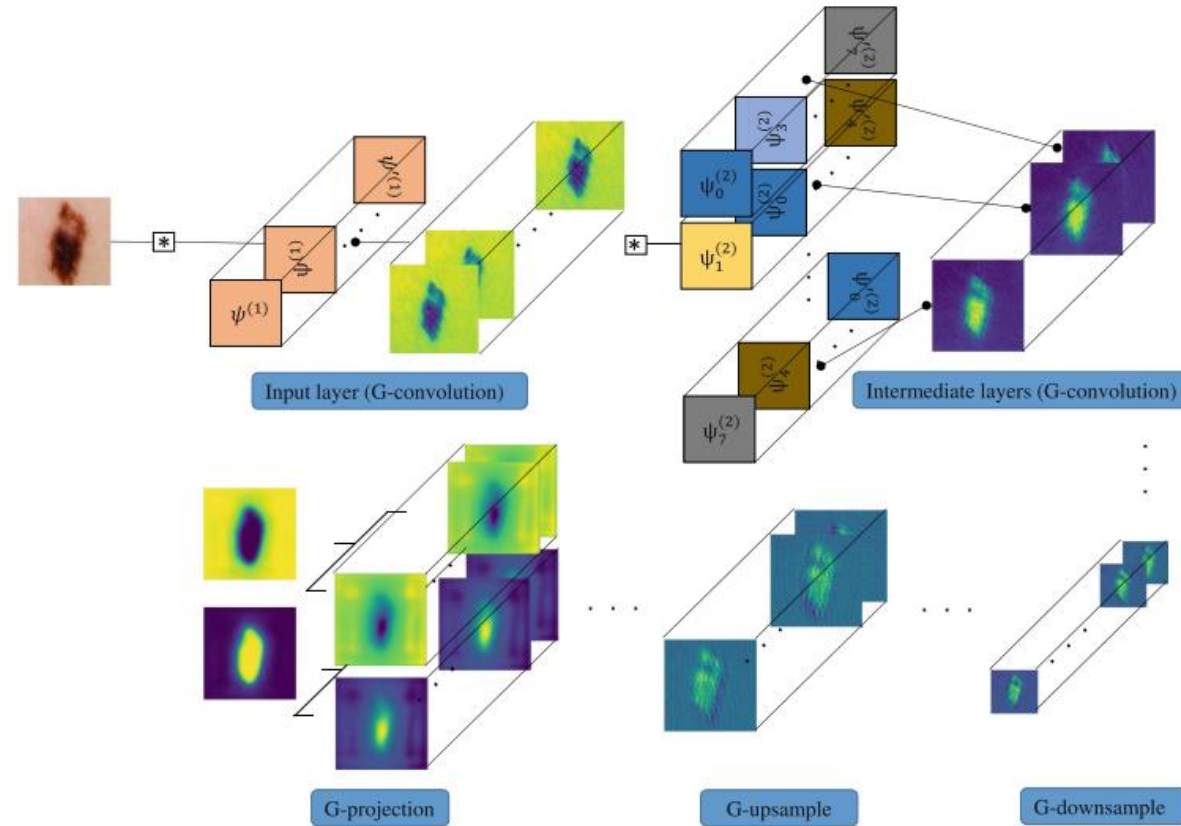
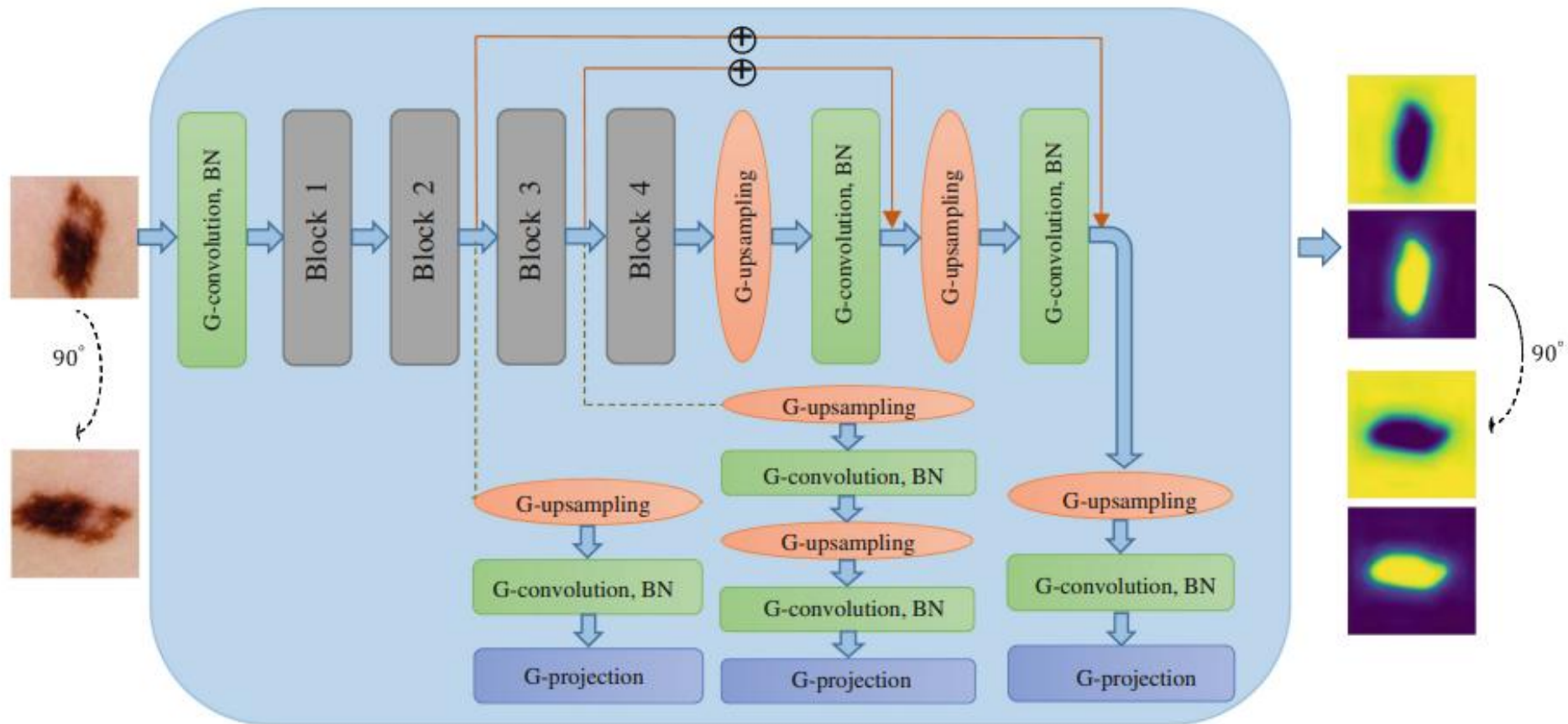


Fig. 2. The illustration of the G -convolution, G -upsampling and G -projection operation. Except the G -projection layer, we only show 1 channel in all other layers to simplify the illustration.

4. Deep learning for dermoscopy images



4. Deep learning for dermoscopy images



Table 1. Ablation study of the deeply supervised rotation equivariant network.

Model	No. of para	Evaluation metrics				
		JA	DI	AC	SE	SP
ResnetFCN34*	22.8M	71.27	80.21	91.39	78.31	96.78
(RE)-ResnetFCN34*	22.8M	74.54	83.27	92.58	81.05	97.59
DS-U-ResnetFCN34*	23.2M	74.38	83.06	92.51	82.52	97.14
(RE)-DS-U-ResnetFCN34* (ours)	23.2M	76.65	85.00	93.27	84.61	96.80
(RE)-DS-U-ResnetFCN34 (ours)	23.5M	77.23	85.60	93.55	85.40	97.15

4. Deep learning for dermoscopy images



Table 2. Comparison with state-of-the-art methods on the ISIC 2017 test dataset.

Team	JA	DI	AC	SE	SP
Our Method	0.772	0.856	0.936	0.854	0.972
Yuan and Lo [17]	0.765	0.849	0.934	0.825	0.975
Berseth [2]	0.762	0.847	0.932	0.820	0.978
Bi et al. [3]	0.760	0.844	0.934	0.802	0.985
RECOD	0.754	0.839	0.931	0.817	0.970
Jer	0.752	0.837	0.930	0.813	0.976
NedMos	0.749	0.839	0.930	0.810	0.981
INESC	0.735	0.824	0.922	0.813	0.968
Shenzhen U (Lee)	0.718	0.810	0.922	0.789	0.975

4. Deep learning for dermoscopy images



More about group equivariant neural networks

<https://tacocohen.wordpress.com/>



Introduction to ultrasound

5. Introduction to ultrasound physics



- How ultrasound is used?
- How in
- What s



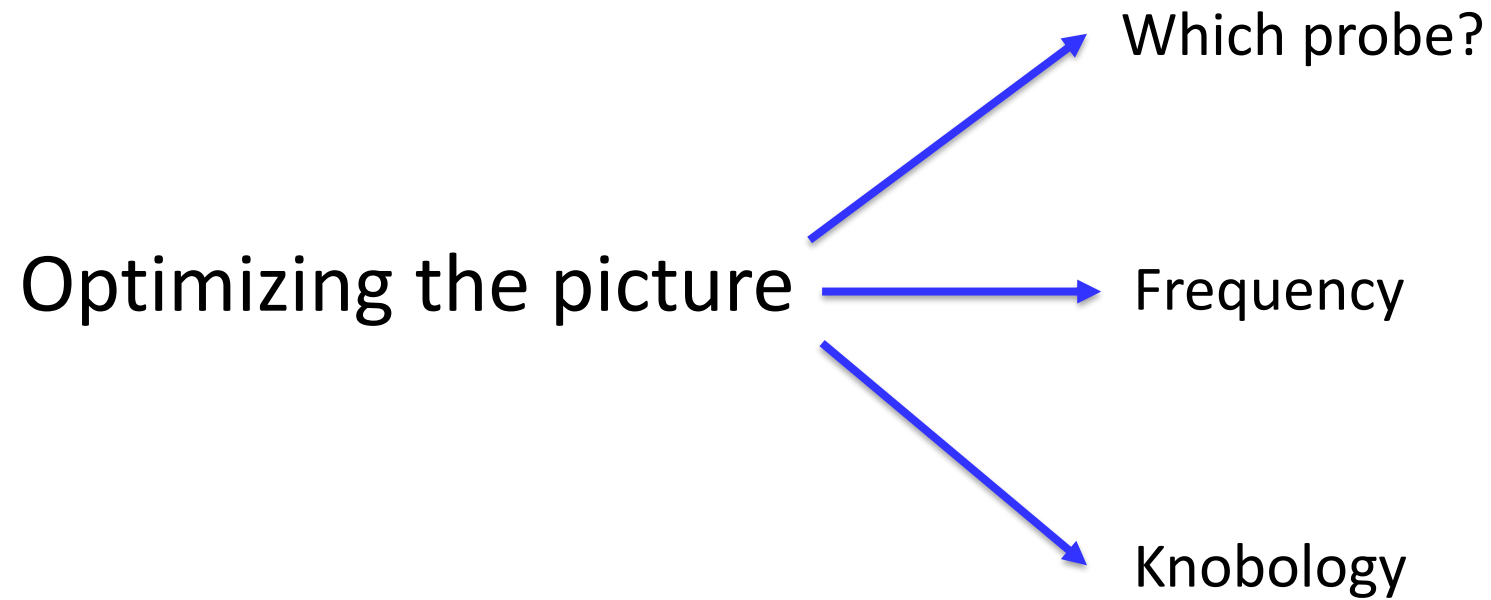
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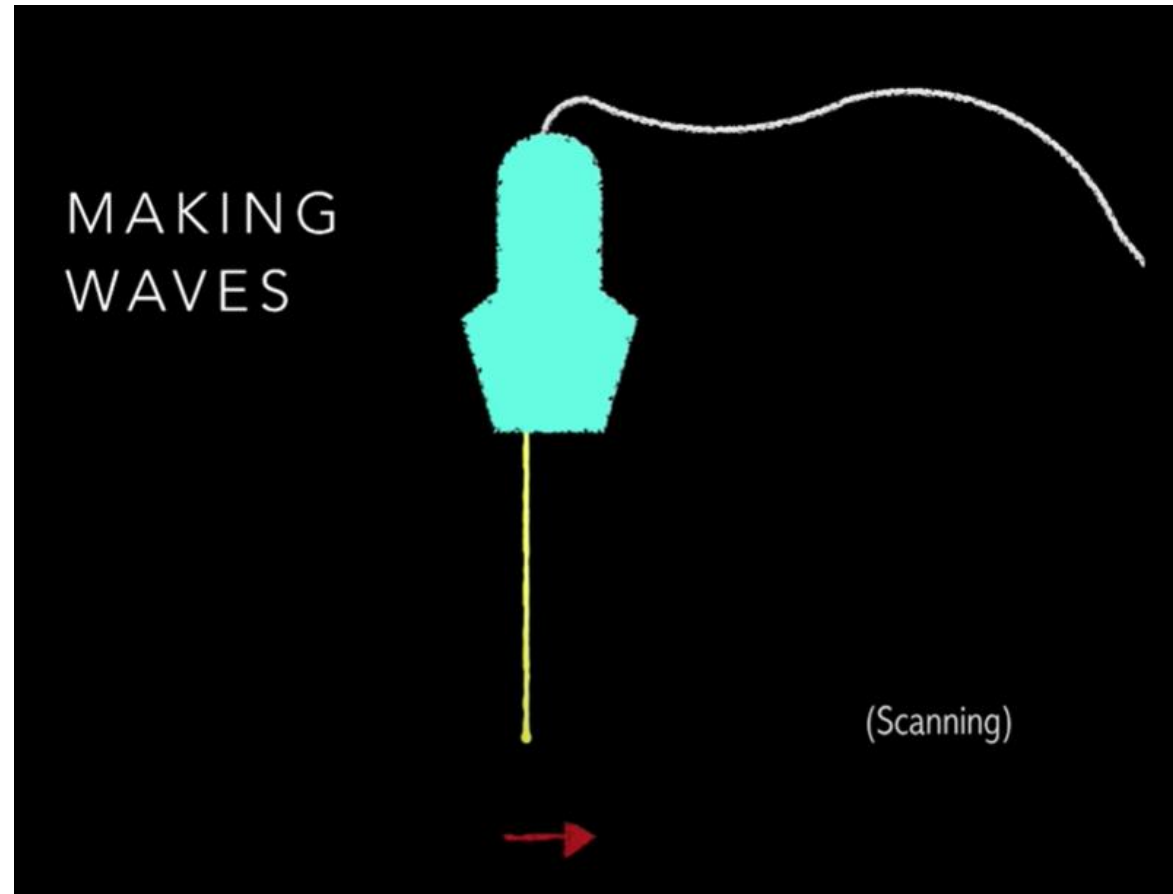
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5. Introduction to ultrasound physics



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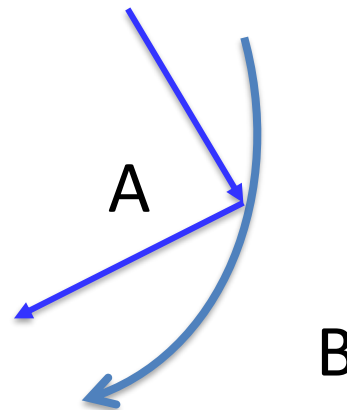
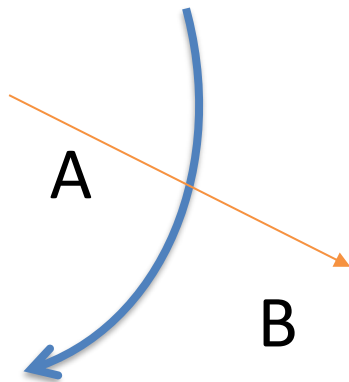
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Difference determines whether sound wave is reflected or transmitted

If two media have the same impedance there will be no reflected ultrasound (no echo)

If two media have very different impedance almost all will be reflected (no transmitted echo)



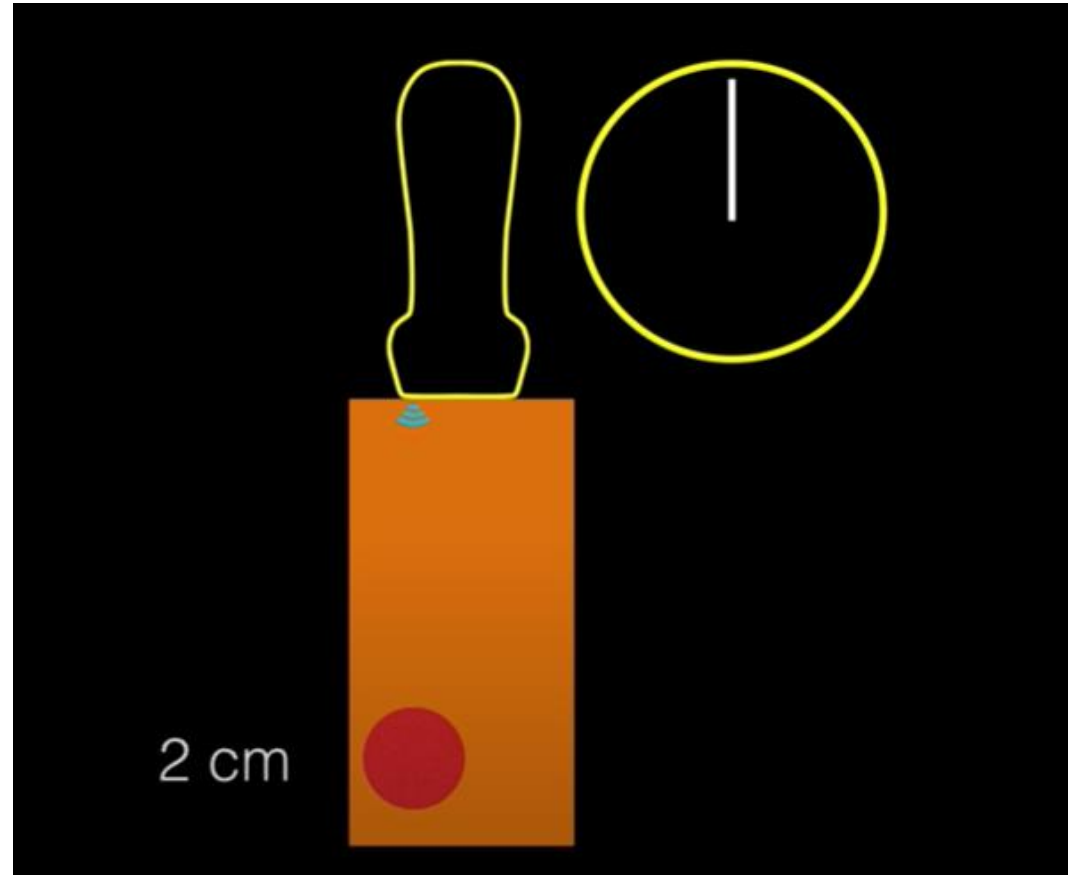
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Round Trip Time

Depth is determined by how long it takes for the echoes to return

There's an echogenic structure at 2 centimeters and a machine sends out a pulse, which is reflected and the round-trip time to the probe is $26 \mu s$



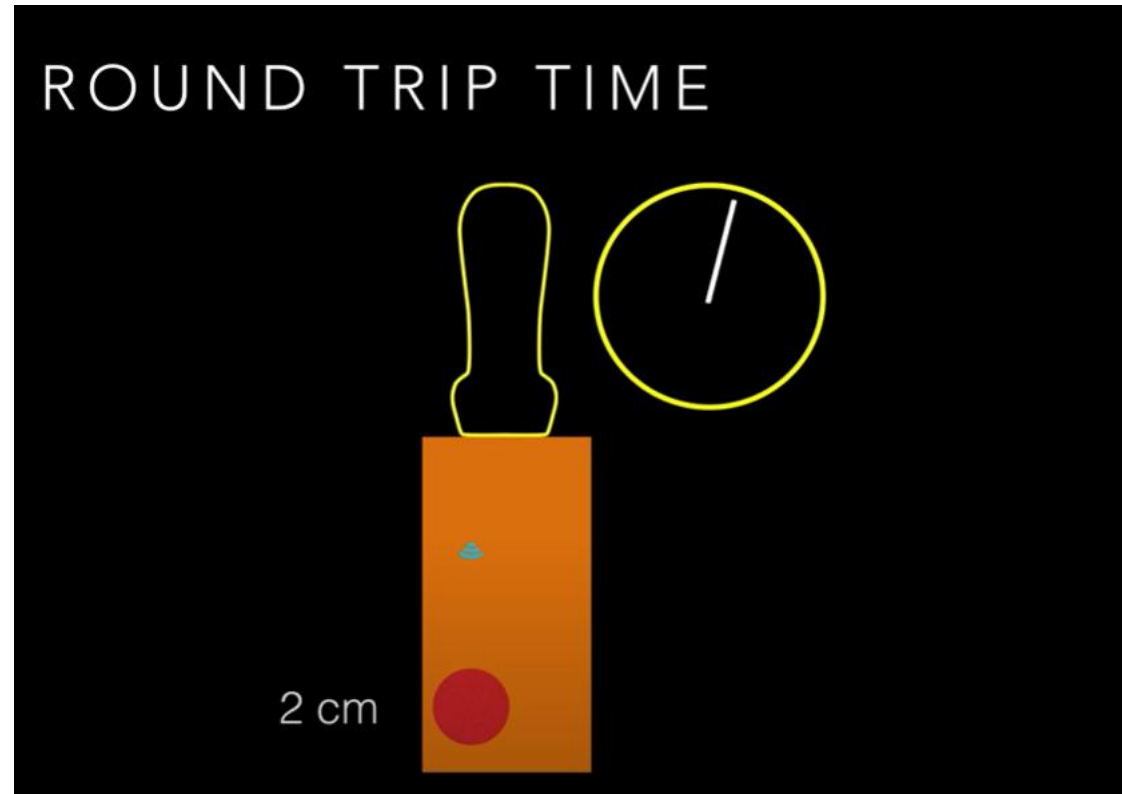
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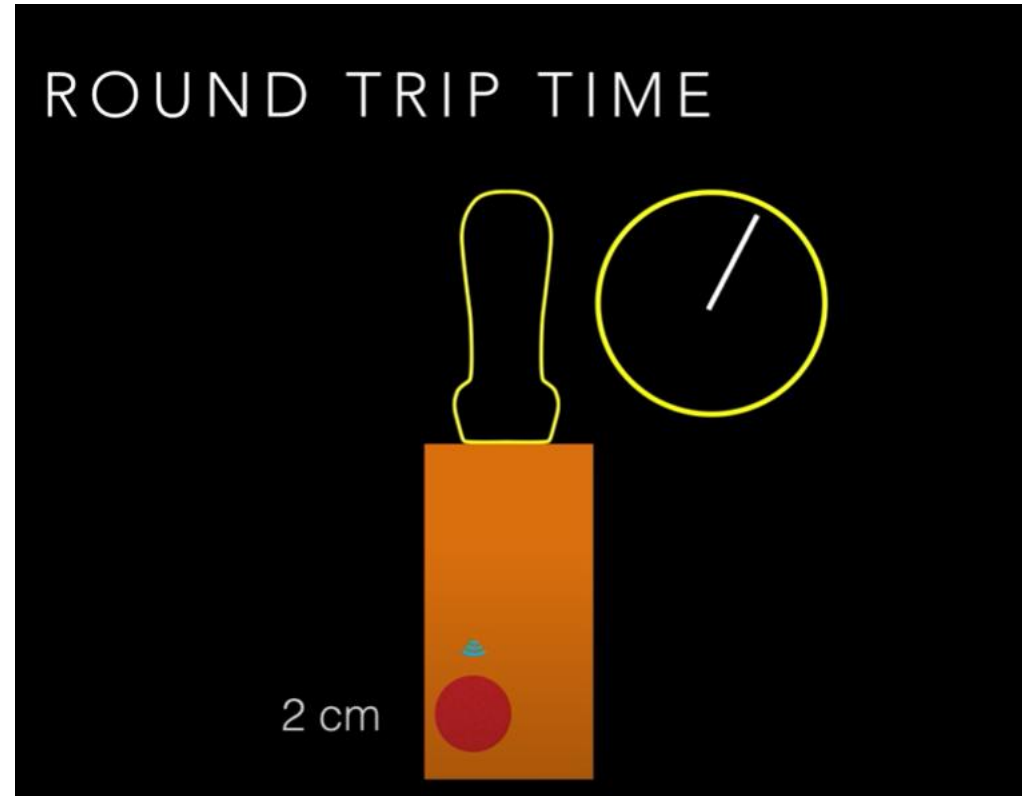
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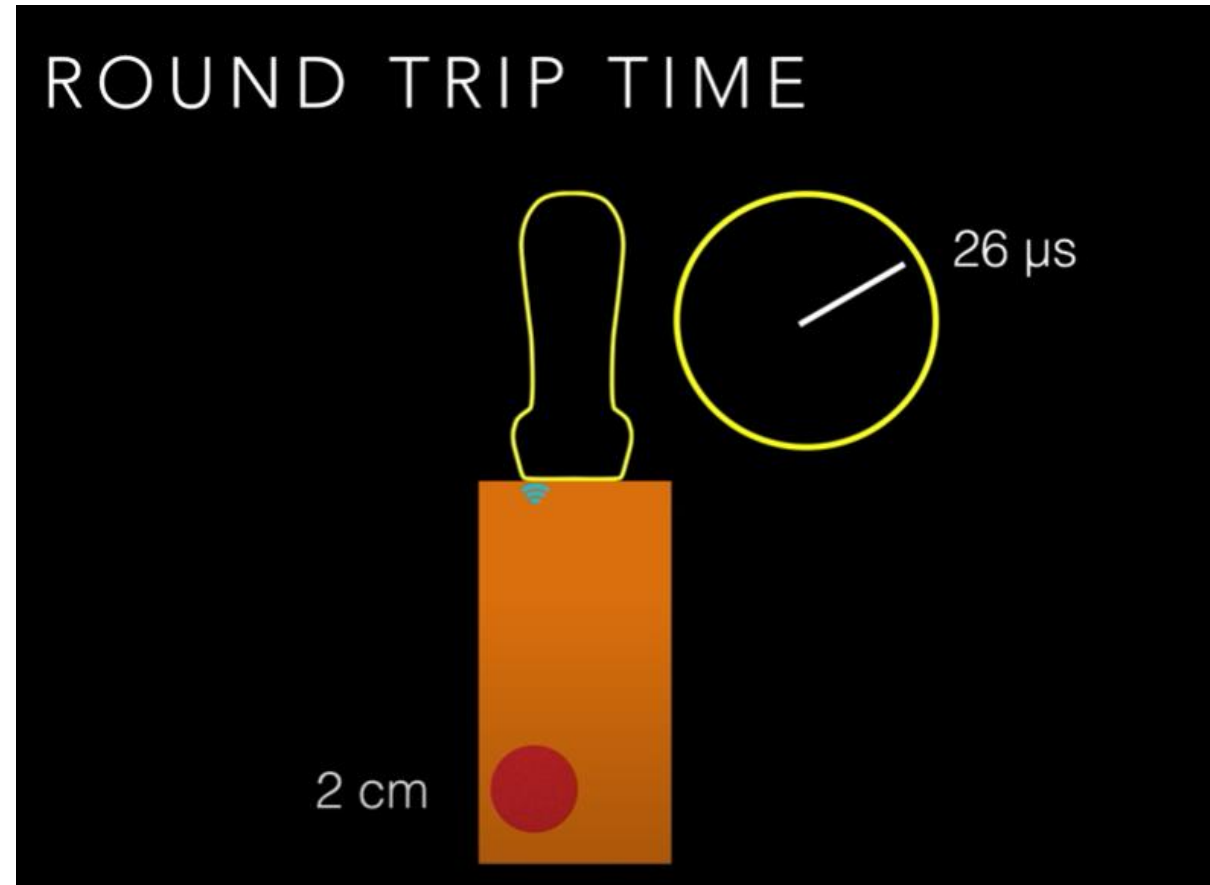
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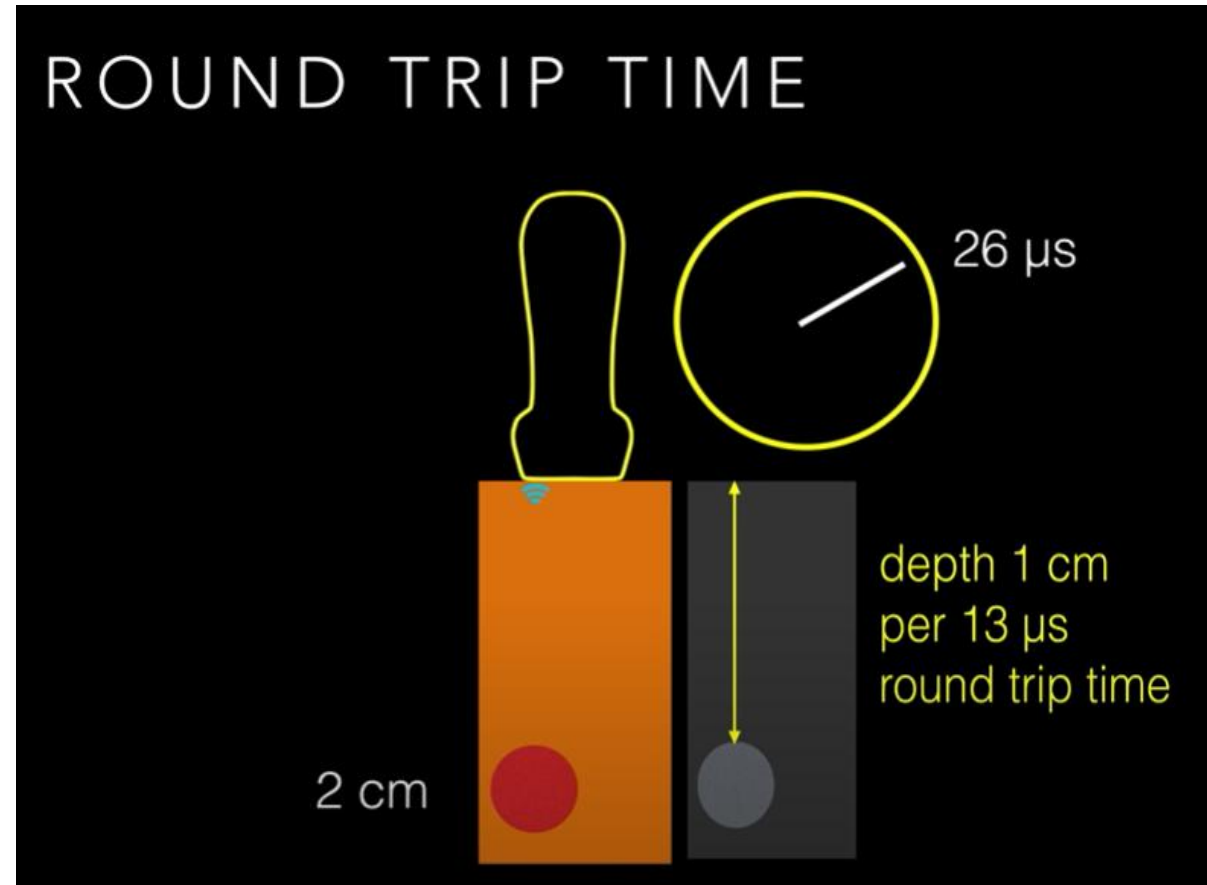
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OVERVIEW OF OPTIMISATION

Probe

Frequency

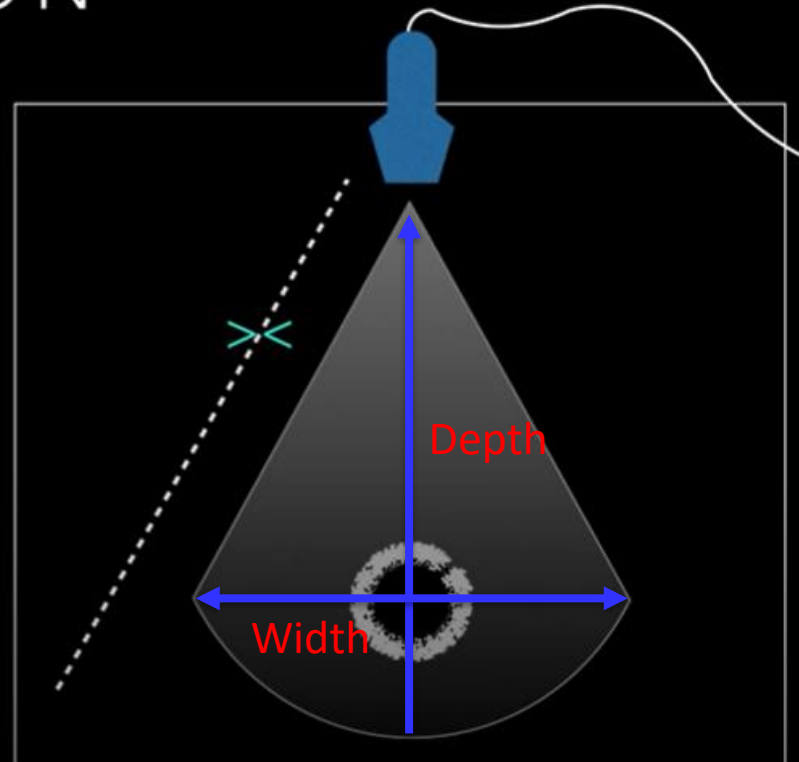
Optimal depth

Narrowed field

Time Gain Compensation

Overall gain

Focussed



Low frequency can image deeper structure.

High frequency give better resolution for superficial structures

Optimize depth and width, adjust the time and gain compensation to get the same brightness.

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WHICH PROBE?

Depends on what
kind of scan you
are planning to do



Phased array



Curvilinear



Linear array

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Phased array

- small footprint
- good penetration
- frequency 1-5MHz
(4-8MHz on paed probes)

Echo/Lung US



Curvilinear

- big footprint
- excellent penetration
- frequency 2-5MHz

Lung/ FAST/ Abdo US



Linear array

- mid sized footprint
- good near field
- frequency 2-13MHz

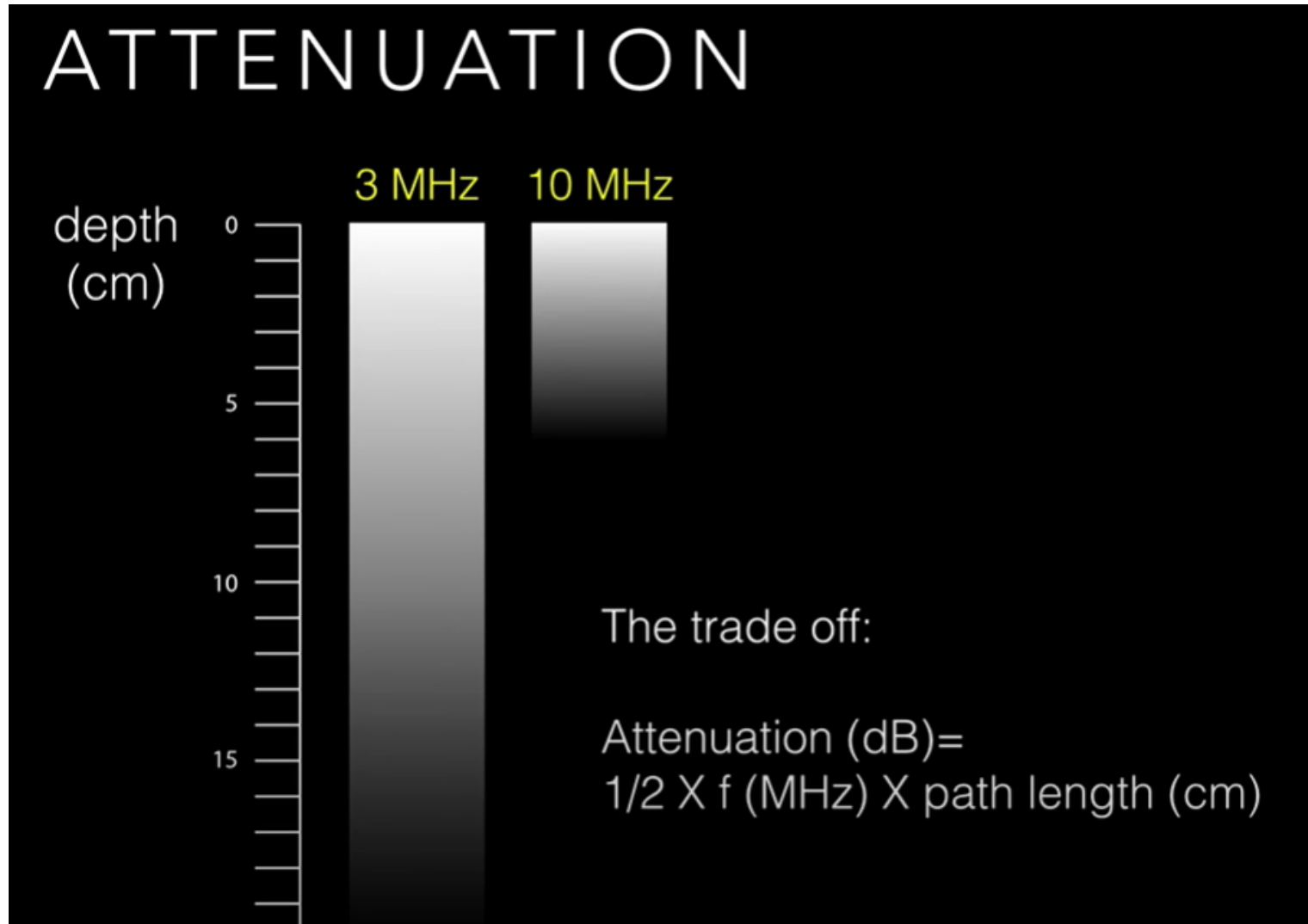
Vascular/procedural US

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Echo imaging with relatively deep structures

Radial art line



Low frequencies penetrate better than high frequencies

High frequencies will lose energy or attenuated rapidly with depth

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[Watch a video](#)



- Multimodal data
- Introduction to dermoscopy
- Introduction to ultrasound
- Deep learning applications